Dual-Directed Algorithm Design for Efficient Pure Exploration

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Abstract

We study pure-exploration problems in adaptive experiments with a finite set of alternatives. The goal is to allocate measurement effort wisely to correctly answer a query about the alternatives with high confidence—one canonical query is to identify a best-performing alternative, commonly known as best-arm identification problems in the machine learning literature, or ranking and selection problems in the simulation literature. For a general query, we formulate its problem complexity measure as a maximin optimization problem and derive the necessary and sufficient conditions for an optimal allocation using the corresponding dual variables. Remarkably, these optimality conditions lead to the extension of top-two algorithm design principle, initially proposed for best-arm identification; moreover, by introducing the dual variables, our analysis yields a simple and effective selection rule, termed information-directed selection, which adaptively identifies and chooses from a candidate set of alternatives based on their current informational value. For canonical Gaussian best-arm identification, we establish that combining with information-directed selection, top-two Thompson sampling achieves asymptotic optimality, resolving a notable open question in the pure-exploration literature. Additionally, our algorithm attains asymptotic optimality for pure-exploration thresholding bandits and ε-best-arm identification (or ranking and selection with a probability of good selection guarantee). Furthermore, our results provide a general principle for adapting Thompson sampling to general pure-exploration problems. Extensive numerical experiments highlight the efficiency of our proposed algorithms compared to existing methods.

1 Introduction

In adaptive experiments, a decision-maker (DM) typically collects noisy observations from a finite set of designs or alternatives. A common objective is to allocate measurement resources wisely to identify the highest-performing option, with as few measurements as possible. This task is widely known as the ranking and selection (R&S) problem, and in the machine learning literature, it is often termed the best-arm identification (BAI) problem. The R&S problem serves as a crucial

abstraction for many applications. For example, it is used to determine optimal hyperparameters for algorithms (Shang et al., 2019). In the context of large language models, it helps in identifying the most effective beam search direction for automatic prompt optimization (Pryzant et al., 2023). More recently, the approach has been applied in brain-computer interface systems to interpret neural signals under a variety of stimulus paradigms (Zhou et al., 2024). For further details on the relationship between BAI and R&S, see Glynn and Juneja (2018).

A distinguishing characteristic of R&S and BAI problems is their emphasis on *end outcomes*. In these settings, the DM is not penalized for losses incurred during the adaptive phase. Instead, the primary focus is on the accuracy of the final decision, which can have substantial real-world consequences. For instance, the final decision might result in the deployment of a "winning" feature, the construction of infrastructure, or the approval of a new drug. The goal is to accurately find the best alternative by the conclusion of the experiment, particularly since resources are then committed to the selected option. In contrast to scenarios that seek immediate gains through exploitation of high-performing alternatives, these problems prioritize exploration to secure a well-informed final decision. Consequently, such sequential decision problems are often referred to as *pure-exploration problems*.

The study of pure-exploration problems dates back to the 1950s in the context of R&S (Bechhofer, 1954). Since then, a wide range of algorithms have been developed to address R&S in both the simulation and statistical learning communities. Examples include elimination algorithms (Kim and Nelson, 2001; Fan et al., 2016; Bubeck et al., 2013), Optimal Computing Budget Allocation algorithms (Chen et al., 2000; Gao and Chen, 2015; Wu and Zhou, 2018b), knowledge-gradient algorithms (Frazier et al., 2008), and UCB-type algorithms (Kalyanakrishnan et al., 2012; Kaufmann and Kalyanakrishnan, 2013). For comprehensive surveys of R&S, we refer readers to Fu and Henderson (2017); Hong et al. (2021); Fan et al. (2024).

1.1 Motivations

A unified approach to general pure-exploration problems. While selecting the best alternative (as in R&S) is a common example, many other problems fall under the umbrella of pure exploration in adaptive experiments. These problems broadly aim to resolve an *exploration query* with high confidence while minimizing sampling resources. For instance, the DM may want to identify alternatives sufficiently close to the best-performing one, often referred to as R&S with a probability of good selection (PGS) guarantee or ε-BAI in the machine learning literature. Other objectives include identifying all alternatives exceeding a specified performance threshold (Locatelli et al., 2016), selecting the best-k alternatives (Chen et al., 2008), or determining whether the mean performance lies within a convex feasible set (Qiao and Tewari, 2024).

Despite various scattered studies in the literature addressing specific exploration tasks, developing a unified and practical algorithmic framework capable of handling these diverse queries remains a significant challenge. To the best of our knowledge, general-purpose algorithms have been discussed only in Wang et al. (2021); Ménard (2019); however, these studies approach the problem

as a generic optimization problem, whereas we explicitly leverage structural properties specific to pure-exploration problems.

Generalizing top-two algorithms. A recent class of algorithms, known as top-two algorithms, initially introduced by Russo (2020) for BAI, has gathered increasing attention. These algorithms employ a two-phase approach for selecting the next alternatives to sample. In the first phase, a pair of top-two candidates—referred to as *leader* and *challenger*—are identified. The second phase, called *selection step*, involves sampling one of these candidates, often based on a potentially biased coin toss.[1] The simplicity and effectiveness of top-two algorithms have led to their widespread adoption, with notable contributions from Qin et al. (2017), Shang et al. (2020), and Jourdan et al. (2022).

However, a notable limitation of these algorithms is their dependence on tuning parameters to achieve optimality. The existence of simple, parameter-free algorithms that are provably optimal remains an open question (see Section 8, Russo 2020). Furthermore, the existing literature on top-two algorithms has predominantly focused on BAI, leaving broader pure-exploration tasks largely unexplored.

Generalizing Thompson sampling. Thompson sampling (TS) has achieved notable success in minimizing regret for stochastic bandits, attracting attention from both academia and industry (Russo et al., 2018). However, its inherent focus on maximizing cumulative rewards leads to excessive exploitation of the estimated best alternative, resulting in insufficient exploration of perceived suboptimal options to confidently verify whether the perceived best alternative is indeed the best. Consequently, modifications to TS are necessary for its effective application to pure exploration. Adapting TS for pure exploration could have an immediate impact in areas where TS is already widely used, such as A/B testing (Graepel et al., 2010), news article recommendation (Chapelle and Li, 2011), and revenue management (Ferreira et al., 2018). Several studies have proposed TS variants for pure exploration, beginning with the top-two Thompson sampling algorithm (Russo, 2020). Other adaptations include the Murphy sampling (Kaufmann et al., 2018), which modifies classical TS to test whether the lowest mean falls below a given threshold.

However, there is a lack of unified guidelines on systematically adapting TS for general pure-exploration queries. We address this gap with the introduction of Algorithm 7.

A unified approach to different problem formulations. Various formulations have been developed to evaluate algorithm performance in pure exploration. The *fixed-budget* problem is often framed as a stochastic dynamic programming problem. However, the complexity of solving such problems exactly has led to a focus on approximate solutions, such as restricting analysis to procedures based on *fixed static allocations*. Building on OCBA (Chen et al., 2000), Glynn and Juneja (2004) provide a large-deviation characterization of the probability of incorrect selection

[1] We use the term "sampling" specifically to refer to the act of picking an alternative to draw a sample from, and reserve the term "selection" to describe the step of determining which candidate should be sampled.

(PICS) for static-allocation procedures, identifying an optimal static allocation that maximizes the exponential decay rate of PICS. This approach introduces a notion of problem complexity as the optimal decay exponent. Subsequent works (Chen and Ryzhov, 2023; Avci et al., 2023) build upon this framework by designing dynamic procedures that converge to the optimal static allocation.

The *fixed-precision* formulation, also referred to as the fixed-confidence setting in the bandit literature, traces its origins to Chernoff (1952)'s seminal work on sequential binary hypothesis testing. R&S and BAI extend this to multiple hypotheses with sequentially chosen experiments. In this setting, the DM determines when to terminate the experiment and propose an estimated top performer. The goal is to minimize the expected sample size while guaranteeing a pre-specified probability of correct selection (PCS). Garivier and Kaufmann (2016) provided matching lower and upper bounds on the sample complexity, framing it as an optimization problem that prescribes the optimal sample allocation ratio.

The *posterior convergence rate* setting (Russo, 2020) maintains a posterior distribution over the unknown parameters and evaluates the algorithms by the posterior PICS. This approach operates within a frequentist framework, where the problem instance is fixed, yet it adopts a Bayesian objective to evaluate performance. Russo (2020) identifies the optimal decay exponent for the posterior PICS as the complexity measure.

Remarkably, for Gaussian models, the problem complexity across the three formulations are identical, see Section 2.5.2. Despite extensive research on individual formulations of R&S and BAI, a unified algorithmic framework capable of addressing all three major formulations simultaneously remains an open challenge.

1.2 Contributions and organization

This paper proposes a unified algorithm design framework that accommodates diverse problem formulations and a wide range of pure-exploration queries. Our main contributions are summarized as follows.

1. *We identify common structural elements across various problem formulations and exploration queries, formulating the problem complexity into a distinctive non-smooth maximin convex optimization problem.* This formulation characterizes the inherent difficulty of each query and serves as the foundation for our unified algorithmic framework. Guided by the shared structure of these problem complexity optimization problems, our framework seamlessly adapts to diverse pure-exploration tasks.

2. *We establish necessary and sufficient conditions for the optimality of problem complexity optimization using dual variables, offering a fresh perspective on existing algorithms and enabling generalizations.* Notably, our use of dual variables uncovers the foundation of top-two algorithms and TS variants in the complementary slackness condition of the Karush-Kuhn-Tucker conditions. This provides a natural generalization of the top-two principle to diverse pure-exploration problems and introduces an adaptive, provably optimal selection step, addressing an open question in Russo (2020).

3. *Our design principle is highly flexible and easily adaptable to practical settings with minimal modifications.* We demonstrate the versatility of our algorithm framework through its application to Gaussian noises with unequal and unknown variances, correlated rewards using common random numbers, and general pure-exploration problems. Additionally, we offer a systematic and accessible approach to tailoring Thompson sampling for general pure-exploration problems in Algorithm 7.

4. *We provide theoretical guarantee for the opimality of our algorithms in various formulations and exploration queries.* In particular, we show that our algorithm is optimal for R&S problems under Gaussian models under all three major formulations—fixed-budget, fixed-precision, and posterior convergence rate. We further establish optimality of our framework in pure-exploration thresholding bandits and ε-BAI.

5. *We conduct extensive numerical studies to showcase the compelling performance of our unified algorithm across various settings.* These experiments encompass a wide range of scenarios, including the fixed-confidence and fixed-budget settings, large-scale problems, applications with common random numbers, and general exploration tasks beyond conventional R&S and BAI. This comprehensive evaluation highlights the performance and robustness of our unified algorithm.

Organization. The paper is organized as follows. In Section 2, we present an illustrative study on indifference-zone-free R&S problems in Gaussian models, demonstrating its optimality in posterior convergence rate and static fixed-budget settings. Section 3 introduces the problem formulation for general pure exploration and provides detailed explanations of our unified algorithm design principle, where we also present optimality for Gaussian BAI in the fixed-confidence setting. In Section 4, we discuss extensions beyond the base cases, including exploration queries with multiple correct answers and Gaussian best-arm identification with unknown variances. Finally, in Section 5, we conduct an extensive numerical study and discuss practical considerations for algorithm implementation.

2 An illustrative example: Indifference-zone-free ranking and selection in Gaussian models

To illustrate our algorithm design, we study the canonical indifference-zone-free R&S problems in Gaussian models with *known but potentially unequal* variances, commonly known as BAI in the machine learning literature. Our analysis leads to optimal algorithms aimed at maximizing the posterior convergence rate (Russo, 2020). The unknown-variance case and additional extensions is discussed further in Section 4.

2.1 Problem formulation

Consider a set of K systems, each with an unknown mean performance. For each system $i \in [K] \triangleq \{1, 2, \ldots, K\}$, the mean performance is characterized by an *unknown* parameter $\theta_i \in \mathbb{R}$. At each time step $t \in \mathbb{N}_0 \triangleq \{0, 1, \ldots\}$, a decision-maker (DM) selects a system $I_t \in [K]$ and observes its random performance: $Y_{t+1,I_t} = \theta_{I_t} + W_{t,I_t}$, with $W_{t,I_t} \sim N(0, \sigma^2_{I_t})$, where W_{t,I_t} is a Gaussian noise with *known variance* $\sigma^2_{I_t}$, independent across time and systems. The true mean vector $\boldsymbol{\theta} \triangleq (\theta_1, \ldots, \theta_K)$ is fixed but *unknown*. We assume there is a unique best system, denoted by $I^*(\boldsymbol{\theta}) \triangleq \operatorname{argmax}_{i\in[K]} \theta_i$, or simply I^* when the dependence on $\boldsymbol{\theta}$ is clear. The DM's goal is to identify this unknown best system I^* by sequentially allocating a finite sampling budget T among the K systems.

Although this problem is set in a *frequentist* framework, with both the mean vector $\boldsymbol{\theta}$ and best system $I^*(\boldsymbol{\theta})$ fixed but unknown, we focus on minimizing the *posterior probability* that a suboptimal system is mistakenly identified as the best, a performance criteria popularized by Russo (2020).

Our performance criteria utilizes a prior distribution Π_0 with density π_0. We focus on *uninformative priors* for the mean vector, which are mutually independent Gaussian distributions with infinite variance. Given the history $\mathcal{H}_t \triangleq (I_0, Y_{1,I_0}, \ldots, I_{t-1}, Y_{t,I_{t-1}})$ collected by any allocation rule, the posterior distribution of each mean θ_i is normal, $N(\theta_{t,i}, \sigma^2_{t,i})$, with posterior mean and variance:

$$\theta_{t,i} \triangleq \frac{1}{N_{t,i}} \sum_{\ell=0}^{t-1} \mathbb{1}(I_\ell = i) Y_{\ell+1,I_\ell} \quad \text{and} \quad \sigma^2_{t,i} \triangleq \frac{\sigma^2_i}{N_{t,i}},$$

where $N_{t,i} \triangleq \sum_{\ell=0}^{t-1} \mathbb{1}(I_\ell = i)$ is the number of samples allocated to system i up to time t.

Define the set of mean vectors for which system I^* is uniquely the best as:

$$\Theta_{I^*} \triangleq \{(\vartheta_1, \ldots, \vartheta_K) \in \mathbb{R}^K : \vartheta_{I^*} > \vartheta_j, \quad \forall j \neq I^*\}.$$

Our goal is to minimize the posterior probability $\Pi_t(\Theta^c_{I^*})$, the probability of misidentifying the best system, where $\Theta^c_{I^*}$ is the complement of Θ_{I^*}. Typically, this probability decays exponentially fast as $t \to \infty$, motivating the asymptotic objective to maximize the exponential decay rate

$$\liminf_{t\to\infty} -\frac{1}{t} \log \Pi_t(\Theta^c_{I^*}).$$

This criterion leads us to define optimality for any fixed problem instance $\boldsymbol{\theta}$ as follows.

Definition 1 (Optimal allocation rules)**.** *An allocation rule* $\mathtt{A}^*$ *is said to be asymptotically optimal for* $\boldsymbol{\theta}$ *if, for every other non-anticipating allocation rule* $\mathtt{A}$*,*

$$\liminf_{t\to\infty} \frac{-\log \Pi^{\mathtt{A}^*}_t(\Theta^c_{I^*})}{-\log \Pi^{\mathtt{A}}_t(\Theta^c_{I^*})} \geq 1 \quad \textit{with probability one,}$$

where the superscript indicates dependence on the allocation rule that determines data collection.

We now present a sufficient condition for asymptotic optimality in terms of long-run allocation proportions. Let $\boldsymbol{p}_t = (p_{t,1}, \ldots, p_{t,K})$ denote the empirical allocation vector, with $p_{t,i}$ being the fraction of samples allocated to system i by time t. The proof of Theorem 1 is collected in Appendix B.

Theorem 1 (A sufficient condition for optimality)**.** *An allocation rule is asymptotically optimal for* $\boldsymbol{\theta}$ *if its empirical allocation vector* $\boldsymbol{p}_t$ *converges almost surely to the unique component-wise strictly positive vector* $\boldsymbol{p}^* = (p_1^*, \ldots, p_K^*)$ *satisfying:*

1. ***Information balance****:*

$$\frac{(\theta_{I^*} - \theta_i)^2}{\sigma_{I^*}^2/p_{I^*}^* + \sigma_i^2/p_i^*} = \frac{(\theta_{I^*} - \theta_j)^2}{\sigma_{I^*}^2/p_{I^*}^* + \sigma_j^2/p_j^*}, \quad \forall i, j \neq I^*, \tag{1}$$

2. ***Stationarity****:*

$$p_{I^*}^* = \sum_{j \neq I^*} h_{I^*}^j(\boldsymbol{p}^*) \cdot \frac{p_j^*}{1 - h_{I^*}^j(\boldsymbol{p}^*)}, \quad \textit{where} \quad h_{I^*}^j(\boldsymbol{p}^*) \triangleq \frac{\sigma_{I^*}^2/p_{I^*}^*}{\sigma_{I^*}^2/p_{I^*}^* + \sigma_j^2/p_j^*}, \quad \forall j \neq I^*. \tag{2}$$

Remark 1 (Equivalent form of stationarity condition)**.** *Plugging* $\{h_{I^*}^j(\boldsymbol{p}^*)\}_{j \neq I^*}$ *into the stationarity condition in* (2) *recovers the conventional* balance-of-sum-of-squares *condition:*

$$p_{I^*}^* = \frac{\sigma_{I^*}^2}{p_{I^*}^*} \sum_{j \neq I^*} \frac{(p_j^*)^2}{\sigma_j^2} \quad \Longleftrightarrow \quad \frac{(p_{I^*}^*)^2}{\sigma_{I^*}^2} = \sum_{j \neq I^*} \frac{(p_j^*)^2}{\sigma_j^2}. \tag{3}$$

This condition has been previously observed in the static fixed-budget setting by Glynn and Juneja (2004, Theorem 1) and Chen and Ryzhov (2023, Lemma 1), and in the fixed-confidence setting by Garivier and Kaufmann (2016, Theorem 5) and Bandyopadhyay et al. (2024, Proposition 2.1). Although these two conditions are mathematically equivalent, the stationarity condition in (2) *explicitly guides our algorithm design principle; see Section 2.3.*

2.2 Top-two design principle as information balance condition

We demonstrate that the information balance condition in (1) naturally leads to the well-known top-two design principle introduced by Russo (2020), where one identifies the two most promising candidates and then carefully selects from these two options. To illustrate this connection, we analyze the *top-two Thompson sampling* (TTTS) algorithm proposed in their work. This condition also underpins various related "top-two" algorithms (Qin et al., 2017; Jourdan et al., 2022; Bandyopadhyay et al., 2024).

2.2.1 Top-two Thompson sampling

The TTTS algorithm maintains a posterior distribution Π_t over the true mean vector $\boldsymbol{\theta}$. In each iteration t, TTTS first generate a sample $\widetilde{\boldsymbol{\theta}} = (\widetilde{\theta}_1, \ldots, \widetilde{\theta}_K)$ from the posterior distribution Π_t and identifies a best system under this sample, denoted by $I_t^{(1)} \in \operatorname{argmax}_{i \in [K]} \widetilde{\theta}_i$. The system $I_t^{(1)}$ is referred to as *leader*. TTTS then repeatedly samples $\widetilde{\boldsymbol{\theta}}' = (\widetilde{\theta}'_1, \ldots, \widetilde{\theta}'_K)$ from Π_t until another system $I_t^{(2)} \in \operatorname{argmax}_{i \in [K]} \widetilde{\theta}'_i$ that differs from $I_t^{(1)}$ emerges. We refer to this system $I_t^{(2)}$ as *challenger*, and the pair $(I_t^{(1)}, I_t^{(2)})$ is referred to as *top-two candidates*.

Algorithm 1 Top-two Thompson sampling (Russo, 2020)

Input: Posterior distribution Π_t.

1: Sample $\widetilde{\boldsymbol{\theta}} = \left(\widetilde{\theta}_1, \ldots, \widetilde{\theta}_K\right) \sim \Pi_t$ and define $I_t^{(1)} \in \operatorname{argmax}_{i\in[K]} \widetilde{\theta}_i$.

2: Repeatedly sample $\widetilde{\boldsymbol{\theta}}' = \left(\widetilde{\theta}'_1, \ldots, \widetilde{\theta}'_K\right) \sim \Pi_t$ until $I_t^{(2)} \in \operatorname{argmax}_{i\in[K]} \widetilde{\theta}'_i$ differs from $I_t^{(1)}$.

3: Return $\left(I_t^{(1)}, I_t^{(2)}\right)$.

Given the top-two candidates $(I_t^{(1)}, I_t^{(2)})$, TTTS selects exactly one candidate to measure using a potentially biased coin toss. Specifically, TTTS measures system $I_t^{(1)}$ with probability β and system $I_t^{(2)}$ with probability $1-\beta$, where β represents the bias of the coin. A common choice is $\beta = 0.5$. The detailed discussion regarding an optimal selection of the coin bias β is provided in Section 2.3.

2.2.2 Top-two algorithms and information balance condition

The idea of TTTS and other top-two algorithms arises naturally from (1). Recall that the key step in TTTS is sampling repeatedly from the posterior to obtain a challenger. We motivate TTTS by heuristically arguing that a system j appears as a challenger, given that a system I^* is the leader, asymptotically only when the following is minimized

$$C_j(\boldsymbol{p}) = C_j(\boldsymbol{p}; \boldsymbol{\theta}) \triangleq \frac{(\theta_{I^*} - \theta_j)^2}{2\left(\sigma_{I^*}^2/p_{I^*} + \sigma_j^2/p_j\right)}, \tag{4}$$

where $\boldsymbol{p} = (p_1, \ldots, p_K)$ is the proportion of samples allocated among the systems. When the context is clear, we will omit the dependence of $C_j(\boldsymbol{p}; \boldsymbol{\theta})$ on the mean vector $\boldsymbol{\theta}$.

To see this, suppose i is the leader, then j appears as the challenger only when the posterior sample satisfies $\widetilde{\theta}'_j > \widetilde{\theta}'_i$. By Russo (2020, Proposition 5), this occurs under the posterior distribution with an asymptotic probability proportional to $\exp\left(-tC_j(\boldsymbol{p})\right)$. The challenger is the system with the highest posterior sample, which corresponds to the system with the smallest $\exp\left(-tC_j(\boldsymbol{p})\right)$ when t is large. Thus, TTTS asymptotically chooses the j with the smallest $C_j(\boldsymbol{p})$ as the challenger. Since $C_j(\boldsymbol{p})$ is a monotonically increasing function in $\boldsymbol{p}$, TTTS satisfies the information balance condition by maintaining equality among the $C_j(\boldsymbol{p})$'s.

2.3 Information-direction selection among top-two via stationarity condition

More importantly, we demonstrate how the stationarity condition in (2) allows us to address a significant open problem identified in Russo (2020): achieving exact optimality rather than merely β-optimality, a form of restricted optimality defined in Russo (2020).

The original TTTS algorithm adopts a non-adaptive tuning parameter β to decide which of the top-two candidates to select in each round. An additional tuning step is required to guarantee that β converges to the unknown optimal β^*, which depends on the unknown mean vector $\boldsymbol{\theta}$.

We propose the *information-directed selection* (IDS) for top-two algorithms under Gaussian distribution. IDS takes the current sample allocation $\boldsymbol{p}_t$ and the current top-two candidates $(I_t^{(1)}, I_t^{(2)})$

as input and returns a selected candidate. IDS is *hyperparameter-free* and guarantees asymptotic optimality.

Algorithm 2 Information-directed selection

Input: Sample allocation $\boldsymbol{p}$ and top-two candidates (i, j).

1: Return i with probability $h_i^j(\boldsymbol{p}) = \frac{\sigma_i^2/p_i}{\sigma_i^2/p_i+\sigma_j^2/p_j}$ and return j with probability $1 - h_i^j(\boldsymbol{p}) = \frac{\sigma_j^2/p_j}{\sigma_i^2/p_i+\sigma_j^2/p_j}$.

We now argue that IDS naturally arises from the stationarity condition in (2). To see this, note that TTTS selects a system using a two-step procedure: it first identifies a pair of candidates and then measures one from the pair. This procedure naturally provides a law of total probability formula by conditioning on the choice of the challenger. In particular, the left-hand side of (2), representing the proportion of samples allocated to the best system I^*, can be decomposed into a summation of terms $h_{I^*}^j(\boldsymbol{p}^*) \cdot \frac{p_j^*}{1-h_{I^*}^j(\boldsymbol{p}^*)}$. By IDS, $h_{I^*}^j(\boldsymbol{p}^*)$ is interpreted as the probability that system I^* is sampled, given that j is proposed as the challenger. Let μ_j^* denote the probability that j is chosen as the challenger. We argue that

$$\mu_j^* = \frac{p_j^*}{1 - h_{I^*}^j(\boldsymbol{p}^*)}, \quad \forall j \neq I^*. \tag{5}$$

Specifically, system j is sampled only if it appears as the challenger (with probability μ_j^*) and is selected by IDS (with probability $1 - h_{I^*}^j(\boldsymbol{p}^*)$). Therefore, the relationship $p_j^* = \mu_j^*(1 - h_{I^*}^j(\boldsymbol{p}^*))$ must hold. As a sanity check, condition (3) ensures that $\boldsymbol{\mu}$ is a proper probability vector: $\mu_j^* > 0$ for all $j \neq I^*$ and

$$\sum_{j\neq I^*} \mu_j^* = \sum_{j\neq I^*} \frac{p_j^*}{1 - h_{I^*}^j(\boldsymbol{p}^*)} = \sum_{j\neq I^*} \frac{p_j^*}{\frac{\sigma_j^2/p_j^*}{\sigma_i^2/p_i^*+\sigma_j^2/p_j^*}} = \sum_{j\neq I^*} p_j^* + p_{I^*}^* \cdot \sum_{j\neq I^*} \frac{(p_j^*)^2/\sigma_j^2}{(p_{I^*}^*)^2/\sigma_{I^*}^2} = \sum_{i\in[K]} p_i = 1,$$

where the second-to-last inequality follows from (3).

Remark 2 (Selection versus tuning)**.** *Our* IDS *algorithm fundamentally differs from the standard tuning rules presented in Russo (2020). In their tuning rules, the same tuning parameter is applied irrespective of the proposed top-two candidates, given the history. In contrast, our* IDS *method explicitly selects between proposed candidates using a candidate-specific selection function* $h_{I_t^{(1)}}^{I_t^{(2)}}(\boldsymbol{p}_t)$*, computed based on the information collected from these candidates.*

2.4 Optimality of top-two algorithms with information-directed selection

Among the top-two algorithms, the analysis of TTTS (Algorithm 1) is arguably the most complex, as both the leader and the challenger are random variables that depend on the history. Therefore, in this section, we focus on the most challenging TTTS-IDS algorithm—specifically, we apply Algorithm 2 to select one of the top-two candidates proposed by Algorithm 1. We prove that TTTS-IDS

is asymptotically optimal, as defined in Definition 1, by verifying that it satisfies the sufficient conditions established in Theorem 1. This resolves an open problem discussed in Russo (2020, Section 8).

Theorem 2. *Suppose that* $\boldsymbol{\theta}$ *satisfies* $\theta_i \neq \theta_j$ *for all* $i \neq j$. *Then,* `TTTS-IDS` *ensures that* $\boldsymbol{p}_t$ *converges almost surely to the optimal allocation* $\boldsymbol{p}^*$, *which verifies the sufficient condition for optimality in Theorem 1.*

This result is a direct consequence of Theorem 5 in Section 3.6, which establishes convergence in a stronger sense than almost sure convergence.

2.5 Discussions

The primary novelty of our algorithm lies in the direct incorporation of dual variables, enabling significant generalizability. Furthermore, our optimality results extend beyond the posterior convergence rate setting.

2.5.1 Novel optimality conditions by incorporating dual variables

The literature has explored optimality conditions such as (1) and (3); see, for instance, Glynn and Juneja (2004, Theorem 1) and Garivier and Kaufmann (2016, Theorem 5). Specifically, these results highlight that the unique optimal allocation vector $\boldsymbol{p}^*$ is indeed the solution to the optimization problem

$$\Gamma^*_{\boldsymbol{\theta}} \triangleq \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{j\neq I^*} C_j(\boldsymbol{p}). \tag{6}$$

Our approach advocates using the condition (2) rather than the equivalent condition (3), as it goes a step further by directly incorporating dual variables into the optimality condition. The variables μ_j^*'s defined in (5) turn out to be the optimal dual variables correspond to the inequality constraints in the following reformulation of the optimal allocation problem (6):

$$\begin{aligned} \Gamma^*_{\boldsymbol{\theta}} = \max_{\phi,\boldsymbol{p}\in\mathcal{S}_K} \quad & \phi, \\ \text{s.t.} \quad & \phi - C_j(\boldsymbol{p}) \leq 0, \quad \forall j \neq I^*. \end{aligned} \tag{7}$$

The conditions in Theorem 1 are derived directly from the Karush-Kuhn-Tucker (KKT) conditions, where the information balance condition (1) corresponds to the complementary slackness condition, and (2) represents the stationarity condition.

The novelty of incorporating dual variables lies in their significant potential for generalization, as we shall demonstrate in Section 3. This is particularly important because generalizing existing approaches (e.g., Chen and Ryzhov 2023, Bandyopadhyay et al. 2024) for general pure-exploration problems is challenging. For example, Zhang et al. (2023, Table 1) notes that the information balance conditions for identifying the best-k alternatives involve a complex combinatorial structure, making explicit generalization of (1) nearly infeasible. You et al. (2023) further illustrates these complexities through a case study of selecting the best two among five alternatives. However, by

leveraging dual variables, we can circumvent these combinatorial difficulties, *implicitly* enforcing information balance through the complementary slackness condition:

$$\mu_j^* \cdot \left(\min_{j \neq I^*} C_j(\boldsymbol{p}^*) - C_j(\boldsymbol{p}^*) \right) = 0, \quad \forall j \neq I^*.$$

This condition applies to a broad range of problems beyond the standard R&S setting; see Theorem 4.

2.5.2 Optimality in alternative performance criteria

The same optimal allocation $\boldsymbol{p}^*$ arises in other performance criteria studied in the literature, and thus our algorithm is also optimal in such settings.

Static-allocation fixed-budget setting. Consider the frequentist PCS, defined as $\mathbb{P}_{\boldsymbol{\theta}}(\widehat{I}_T^* = I^*)$ for a fixed budget T, where the probability is taken over the randomness in both observations and the algorithm and $\widehat{I}_T^*$ is the output of the algorithm at the end of horizon. Due to the dynamic nature of typical algorithms—the decision at later stages depends on earlier observations—analyzing PCS in the fixed-budget setting is challenging (Hong et al., 2021; Komiyama, 2024). To address this, a common framework in the simulation literature, initiated by Glynn and Juneja (2004) and followed by Chen and Ryzhov (2023), focuses on PCS under a *static* algorithm. These static algorithms allocate samples proportionally to a fixed probability vector $\boldsymbol{p} \in \mathcal{S}_K$. The framework characterizes the large deviation rate of the probability of incorrect selection under a static algorithm as $-\lim_{T\to\infty} \frac{1}{T} \log \left(1 - \mathbb{P}_{\boldsymbol{p},\boldsymbol{\theta}}(\widehat{I}_T^* = I^*)\right) = \widetilde{\Gamma}_{\boldsymbol{\theta}}(\boldsymbol{p})$, for some large deviation rate function $\widetilde{\Gamma}_{\boldsymbol{\theta}}(\boldsymbol{p})$, where $\mathbb{P}_{\boldsymbol{p},\boldsymbol{\theta}}$ denotes the probability under system parameter $\boldsymbol{\theta}$ and static allocation $\boldsymbol{p}$. For Gaussian models, the large deviation rate function $\widetilde{\Gamma}_{\boldsymbol{\theta}}(\boldsymbol{p}) = \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p})$. Glynn and Juneja (2004) proceeds to maximize the large deviation rate, leading to the same optimal allocation problem as in (6).[2] A common goal of fixed-budget algorithms is to ensure the dynamic allocation converges almost surely to the optimal static allocation $\boldsymbol{p}^*$. Our algorithm achieves this goal, as guaranteed by Theorem 2.

Fixed-confidence setting. The fixed-confidence formulation for BAI, which leads to the same optimal allocation $\boldsymbol{p}^*$, has been studied in Garivier and Kaufmann (2016, Lemma 4 and Theorem 5). In Section 3, we extend this formulation to general pure-exploration problems and prove that our algorithm achieves optimality for R&S problems in Gaussian models under the fixed-confidence setting.

[2]For general distributions beyond Gaussian, the expression of C_j in the optimal allocation for static fixed-budget and posterior convergence rate settings differs. However, the maximin structure remains unchanged, and our analysis extends to these cases.

3 General pure-exploration problems

We now broaden our focus from traditional R&S problems to address a wider range of pure-exploration queries relevant to decision-making. To align with the common terminology in machine learning literature on pure exploration, we adopt terms such as "arm" instead of "system" and "reward" instead of "performance." We will show that many formulations commonly studied in simulation, such as the PGS and the indifference-zone (IZ) approach, have clear counterparts within the pure-exploration framework.

We consider a stochastic multi-armed bandit problem with K arms, denoted by $i \in [K]$. The rewards from arm i are independent and identically distributed random variables following a distribution F_{θ_i}, parameterized by the mean θ_i and belonging to a *single-parameter exponential family*. We assume that the rewards from different arms are mutually independent. A problem instance is defined by a fixed but unknown vector $\boldsymbol{\theta} \in \mathbb{R}^K$. The DM's goal is to answer an *exploration query* regarding $\boldsymbol{\theta}$ by adaptively allocating the sampling budget across the arms and observing the noisy rewards from the selected arms.

Exploration query. An exploration query specifies a question regarding the unknown mean vector $\boldsymbol{\theta}$. We assume that the query has a *unique* answer, denoted by $\mathcal{I}(\boldsymbol{\theta})$.[3] Several notable exploration queries have been explored in the literature. For the indifference-zone-free R&S problem (Section 2), the query identifies the unique best arm as $\mathcal{I}(\boldsymbol{\theta}) = \operatorname{argmax}_{i\in[K]} \theta_i$. Alternatively, one might seek to identify the exact set of best-k arms, a problem referred to as best-k identification, where $\mathcal{I}(\boldsymbol{\theta}) = \operatorname{argmax}_{\mathcal{S}\subseteq[K],|\mathcal{S}|=k} \left\{\sum_{i\in\mathcal{S}} \theta_i\right\}$. Other exploration queries include R&S under the indifference-zone (IZ) assumption (Bechhofer, 1954), or identifying *all* ε-good arms (Mason et al., 2020), where $\mathcal{I}(\boldsymbol{\theta}) = \{i \in [K] : \theta_i \geq \max_{i\in[K]} \theta_i - \varepsilon\}$. Another example is the pure-exploration thresholding bandit problem (Lai and Liao, 2012; Locatelli et al., 2016), where the goal is to identify all arms with mean rewards exceeding a threshold $\mathfrak{T}$, represented by $\mathcal{I}(\boldsymbol{\theta}) = \{i \in [K] : \theta_i > \mathfrak{T}\}$.

Fixed-confidence performance criteria. We focus on the fixed-confidence setting, which aims to identify the correct answer with a PCS guarantee of at least $1-\delta$ for a fixed $\delta \in (0,1)$[4], while minimizing the sampling budget. In addition to the allocation rule discussed in Section 2, an algorithm requires both stopping and decision rules. At each time step $t \in \mathbb{N}_0$, the DM first determines whether to stop according to a *stopping rule* τ_δ. If stopping occurs at $t = \tau_\delta$, the DM provides an estimate $\widehat{\mathcal{I}}_{\tau_\delta}$ of the answer $\mathcal{I}(\boldsymbol{\theta})$ based on a *decision rule*. If not, she selects an arm I_t according to a *allocation rule* and observe the reward from arm I_t, denoted by Y_{t+1,I_t}. This procedure is summarized in Algorithm 3 where the information that DM has at the beginning of time t is denoted by $\mathcal{H}_t = (I_0, Y_{1,I_0}, \ldots, I_{t-1}, Y_{t,I_{t-1}})$.

We consider problem instances under a mild assumption. Let $\overline{\Theta}$ denote parameters yielding a unique answer to the exploration query, and $\mathfrak{I}$ be the set of all possible correct answers. For

[3] Exploration queries with multiple correct answers require more care, see Section 4.1 for an example.

[4] In simulation literature, the notation δ commonly represents the indifference-zone parameter. In contrast, our work considers an IZ-free formulation, where $1-\delta$ denotes the target PCS level.

Algorithm 3 A template for anytime pure-exploration algorithms

Input: A stopping rule (`stop`), a decision rule (`decision`), an allocation rule (`sample`), and $\mathcal{H}_0 = \varnothing$.

1: **for** $t = 0, 1, \ldots$ **do**
2: **If** $\texttt{stop}(\mathcal{H}_t)$ **return** $\widehat{\mathcal{I}} = \texttt{decision}(\mathcal{H}_t)$.
3: Run allocation rule $I_t = \texttt{sample}(\mathcal{H}_t)$.
4: Select arm I_t, observe reward Y_{t+1,I_t}, and update history $\mathcal{H}_{t+1} = \mathcal{H}_t \cup \{I_t, Y_{t+1,I_t}\}$.
5: **end for**

each $\mathcal{I} \in \mathfrak{I}$, let $\Theta_{\mathcal{I}}$ denote the set of parameters with $\mathcal{I}$ as the correct answer, i.e., $\Theta_{\mathcal{I}} \triangleq \{\boldsymbol{\vartheta} \in \overline{\Theta} : \mathcal{I}(\boldsymbol{\vartheta}) = \mathcal{I}\}$. We assume that each $\Theta_{\mathcal{I}}$ is an *open set*. This rules out statistically unidentifiable scenarios. Our focus is on the parameter space $\Theta \triangleq \cup_{\mathcal{I} \in \mathfrak{I}} \Theta_{\mathcal{I}}$. We note that this assumption is generally unrestrictive; for instance, in R&S, we have $\overline{\Theta} = \Theta$, i.e., meaning we consider all problem instances with a unique solution.

Definition 2 (δ-correct)**.** *A policy is said to be δ-correct if the PCS upon stopping is at least $1-\delta$, i.e.,*

$$\mathbb{P}_{\boldsymbol{\theta}}\left(\tau_\delta < \infty,\ \widehat{\mathcal{I}}_{\tau_\delta} = \mathcal{I}(\boldsymbol{\theta})\right) \geq 1 - \delta, \quad \textit{for any problem instance } \boldsymbol{\theta} \in \Theta.$$

Let $\mathcal{A}$ denote the class of policies that are δ-correct for any $\delta > 0$.

The goal is to design a policy π that minimizes the expected sample complexity, $\mathbb{E}^{\pi}_{\boldsymbol{\theta}}[\tau_\delta]$ over $\mathcal{A}$, where $\tau_\delta \triangleq \min\{t \geq 0 : \texttt{stop}(\mathcal{H}_t) = \texttt{True}\}$ denotes the stopping time when the policy halts. A policy is deemed optimal in the fixed-confidence setting if it achieves universal efficiency, as defined below.

Definition 3 (Universal efficiency in fixed confidence)**.** *A policy π^* is universally efficient if*

1. *π^* is δ-correct (i.e., $\pi^* \in \mathcal{A}$), and*
2. *for any other $\pi \in \mathcal{A}$, we have*

$$\limsup_{\delta \to 0} \frac{\mathbb{E}^{\pi^*}_{\boldsymbol{\theta}}[\tau_\delta]}{\mathbb{E}^{\pi}_{\boldsymbol{\theta}}[\tau_\delta]} \leq 1, \quad \textit{for any problem instance } \boldsymbol{\theta} \in \Theta.$$

Compared with the "pointwise" optimality in Definition 1, Definition 3 is stronger, as it requires optimality to hold universally for all $\boldsymbol{\theta}$ under a δ-correctness constraint that must be satisfied for all δ and all $\boldsymbol{\theta} \in \Theta$.

3.1 Preliminary

We now introduce key concepts and notation that form the foundation of our algorithm design principle.

The culprits. For an algorithm to correctly answer the pure-exploration query, it must distinguish between problem instances that yield different answers. For a given problem instance $\boldsymbol{\theta} \in \Theta$, the intrinsic complexity of the pure-exploration query in the fixed-confidence setting is closely related to the *alternative set*, defined as the set of parameters that yield a correct answer different from $\mathcal{I}(\boldsymbol{\theta})$:

$$\mathrm{Alt}(\boldsymbol{\theta}) \triangleq \{\boldsymbol{\vartheta} \in \Theta : \mathcal{I}(\boldsymbol{\vartheta}) \neq \mathcal{I}(\boldsymbol{\theta})\}.$$

To formulate a maximin problem as in (6), we rely on the following assumption.

Assumption 1 (Wang et al. 2021)**.** *For any fixed problem instance $\boldsymbol{\theta}$, the alternative set $\mathrm{Alt}(\boldsymbol{\theta})$ is a finite union of convex sets. Namely, there exists a finite set $\mathcal{X}(\boldsymbol{\theta})$ and a collection of convex sets $\{\mathrm{Alt}_x(\boldsymbol{\theta}) : x \in \mathcal{X}(\boldsymbol{\theta})\}$ such that $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{x\in\mathcal{X}(\boldsymbol{\theta})}\mathrm{Alt}_x(\boldsymbol{\theta})$.*

This condition is indeed satisfied by commonly encountered exploration queries; see Appendix A.

We refer to the collection $\mathcal{X}(\boldsymbol{\theta})$ as the set of *culprits*. These culprits represent the fundamental causes that lead the correct answer under an alternative problem instance $\boldsymbol{\vartheta}$ to deviate from $\mathcal{I}(\boldsymbol{\theta})$. Thus, the culprit set is determined by both the exploration query and the specific problem instance $\boldsymbol{\theta}$.

Example 1 (R&S)**.** *For R&S, the culprit set is defined as $\mathcal{X}(\boldsymbol{\theta}) = [K]\backslash\{I^*(\boldsymbol{\theta})\}$, where each $j \in \mathcal{X}(\boldsymbol{\theta})$ represents a scenario in which, under an alternative problem instance $\boldsymbol{\vartheta} \in \Theta$, system j outperforms the best system $I^*(\boldsymbol{\theta})$ identified under $\boldsymbol{\theta}$. In such scenarios, $I^*(\boldsymbol{\theta})$ is an incorrect selection due to the existence of the culprit j. The corresponding alternative set for each culprit $j \in \mathcal{X}(\boldsymbol{\theta})$ is $\mathrm{Alt}_j(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \Theta : \vartheta_j \geq \vartheta_{I^*(\boldsymbol{\theta})}\}$.*

Example 2 (Best-k identification)**.** *The goal is to identify the set of k arms with the largest means. The exploration query can be broken down into verifying whether any top arm $i \in \mathcal{I}(\boldsymbol{\theta})$ has a higher mean than any bottom arm $j \in \mathcal{I}^c(\boldsymbol{\theta})$. If there exist pairs $(i, j) \in \mathcal{X}(\boldsymbol{\theta}) \triangleq \mathcal{I}(\boldsymbol{\theta}) \times \mathcal{I}^c(\boldsymbol{\theta})$ such that $\vartheta_j > \vartheta_i$, then either $i \notin \mathcal{I}(\boldsymbol{\vartheta})$ or $j \in \mathcal{I}(\boldsymbol{\vartheta})$, leading to $\mathcal{I}(\boldsymbol{\vartheta}) \neq \mathcal{I}(\boldsymbol{\theta})$. A culprit in this context is specified by a pair of arms $(i, j) \in \mathcal{X}(\boldsymbol{\theta})$, with the alternative set $\mathrm{Alt}_{i,j}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \Theta : \vartheta_j > \vartheta_i\}$.*

Example 3 (Pure-exploration thresholding bandits)**.** *The goal is to identify the set of arms with means above a threshold $\mathfrak{T}$, i.e., $\mathcal{I}(\boldsymbol{\theta}) \triangleq \{i \in [K] : \theta_i > \mathfrak{T}\}$. The exploration query can be broken down into comparing each arm $i \in [K]$ with the threshold $\mathfrak{T}$. A culprit is then specified by an arm $i \in \mathcal{X}(\boldsymbol{\theta}) \triangleq [K]$ with the corresponding alternative set $\mathrm{Alt}_i(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \Theta : (\mathfrak{T} - \vartheta_i)(\mathfrak{T} - \theta_i) < 0\}$. This set indicates that arm i is on opposite sides of $\mathfrak{T}$ under $\boldsymbol{\vartheta}$ and $\boldsymbol{\theta}$, resulting in an incorrect answer.*

The generalized Chernoff information C_x. Let $\theta_{t,i} \triangleq \frac{1}{N_{t,i}}\sum_{\ell=0}^{t-1}\mathbb{1}(I_\ell = i)Y_{\ell+1,I_\ell}$ denotes the sample mean of arm i at time t and $\boldsymbol{\theta}_t = (\theta_{t,1}, \ldots, \theta_{t,K})$. To correctly answer the pure-exploration query, every potential culprit must be ruled out. From a hypothesis testing perspective, this involves testing the hypotheses $H_{0,x} : \boldsymbol{\theta} \in \mathrm{Alt}_x(\boldsymbol{\theta}_t)$ versus $H_{1,x} : \boldsymbol{\theta} \notin \mathrm{Alt}_x(\boldsymbol{\theta}_t)$, for *all* $x \in \mathcal{X}(\boldsymbol{\theta}_t)$, where

$H_{1,x}$ corresponds to the scenario in which x is innocent, i.e., *not* a culprit. For instance, in R&S, rejecting $H_{0,x}$ indicates sufficient evidence to support the desired ordering $\theta_{I^*(\boldsymbol{\theta}_t)} > \theta_x$ for any $x \neq I^*(\boldsymbol{\theta}_t)$.

To this end, we utilize the generalized log-likelihood ratio test (GLRT) statistic, as proposed in Kaufmann and Koolen (2021). Let $\mathcal{S}_K$ denote the probability simplex of dimension $K-1$ and let $\boldsymbol{p} \in \mathcal{S}_K$ represent a generic allocation vector, indicating the proportion of samples allocated to the K arms. If an algorithm allocates according to $\boldsymbol{p}_t$ at time t, then the GLRT statistic given by

$$\ln \frac{\sup_{\boldsymbol{\vartheta} \in \Theta} L(\boldsymbol{\vartheta})}{\sup_{\boldsymbol{\vartheta} \in \mathrm{Alt}_x(\boldsymbol{\theta}_t)} L(\boldsymbol{\vartheta})} = \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}_x(\boldsymbol{\theta}_t)} \ln \frac{L(\boldsymbol{\theta}_t)}{L(\boldsymbol{\vartheta})} = t \cdot \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}_x(\boldsymbol{\theta}_t)} \sum_{i \in [K]} p_i d(\theta_{t,i}, \vartheta_i),$$

where $d(\cdot, \cdot)$ denotes the Kullback-Leibler (KL) divergence. This derivation uses the fact that the unconstrained maximum likelihood estimator for the mean of a single-parameter exponential family is $\boldsymbol{\theta}_t$.

We extend C_j in (4) as the *population version* of the GLRT statistic, normalized by the sample size t,

$$C_x(\boldsymbol{p}) = C_x(\boldsymbol{p}; \boldsymbol{\theta}) \triangleq \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}_x(\boldsymbol{\theta})} \sum_{i \in [K]} p_i d(\theta_i, \vartheta_i), \quad \text{for all } x \in \mathcal{X}(\boldsymbol{\theta}). \tag{8}$$

We refer to C_x as the *generalized Chernoff information*, extending a classical concept in hypothesis testing (Cover and Thomas, 2006). When the context is clear, we will omit the dependence of $C_x(\boldsymbol{p}; \boldsymbol{\theta})$ on the problem instance $\boldsymbol{\theta}$. The following result establishes the basic properties of C_x.

Lemma 1. *Under Assumption 1, for each culprit $x \in \mathcal{X}$ and component-wise strictly positive $\boldsymbol{p} \in \mathcal{S}_K$:*

1. *There exists a unique $\boldsymbol{\vartheta}^x \in \mathrm{Alt}_x(\boldsymbol{\theta})$ that achieves the infimum in (8).*
2. *The functions $C_x(\boldsymbol{p})$ are differentiable concave functions, with $\frac{\partial C_x(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta_i^x)$ for all $i \in [K]$.*
3. *The functions $C_x(\boldsymbol{p})$ satisfy the following partial differential equation*

$$C_x(\boldsymbol{p}) = \sum_{i \in [K]} p_i \frac{\partial C_x(\boldsymbol{p})}{\partial p_i}. \tag{9}$$

When the sampling effort is distributed according to $\boldsymbol{p}$, the function C_x quantifies the average information gain per sample needed to assert $H_{1,x}$, i.e., to acquit culprit x. The partial differential equation (PDE) in (9) shows that this information gain can be effectively decomposed into contributions from the sample allocations to individual arms; see the discussion in Remark 5.

The optimal allocation problem. Recall that $H_{0,x}$ must be rejected for all culprits $x \in \mathcal{X}(\boldsymbol{\theta})$ to correctly answer the pure-exploration query. Furthermore, $C_x(\boldsymbol{p})$ represents the information

gathered to reject $H_{0,x}$ under sample allocation $\boldsymbol{p}$—the larger $C_x(\boldsymbol{p})$ is, the easier it is to reject $H_{0,x}$. By taking the minimum over all culprits $x \in \mathcal{X}(\boldsymbol{\theta})$, we define

$$\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}) \triangleq \min_{x \in \mathcal{X}(\boldsymbol{\theta})} C_x(\boldsymbol{p}),$$

which corresponds to the GLRT statistic for the leading culprit, i.e., the culprit that is hardest to acquit. To maximize information gathering, we should follow an allocation $\boldsymbol{p}$ that maximizes $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p})$. This leads to the following *optimal allocation problem*:

$$\Gamma^*_{\boldsymbol{\theta}} \triangleq \max_{\boldsymbol{p} \in \mathcal{S}_K} \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}) = \max_{\boldsymbol{p} \in \mathcal{S}_K} \min_{x \in \mathcal{X}(\boldsymbol{\theta})} C_x(\boldsymbol{p}), \tag{10}$$

which generalizes (6) in the context of R&S problems in Gaussian models.

Stopping and decision rules. A fixed-confidence algorithm requires three components: stopping, decision, and allocation rules. We follow standard practices in the literature for the stopping and decision rules, allowing us to focus on designing optimal allocation rules.

For the stopping rule, Kaufmann and Koolen (2021) provide the Chernoff stopping rule based on the GLRT for exponential family bandits:

$$\tau_\delta = \inf \left\{ t \in \mathbb{N}_0 \,:\, t \cdot \Gamma_{\boldsymbol{\theta}_t}(\boldsymbol{p}_t) > \gamma(t, \delta) \right\}, \tag{11}$$

where $\gamma(t, \delta)$ is a threshold function. Kaufmann and Koolen (2021, Proposition 15) provide a $\gamma(t, \delta)$ ensuring that, combined with *any* allocation rule, the stopping rule in (11) is δ-correct for any pure-exploration query with a unique correct answer. For completeness, we rephrase their results in Appendix C.1.

We adopt a specific decision rule

$$\texttt{decision}(\mathcal{H}_{\tau_\delta}) = \mathcal{I}(\boldsymbol{\theta}_{\tau_\delta}), \tag{12}$$

where $\mathcal{I}(\boldsymbol{\theta}_{\tau_\delta})$ is the correct answer based on the sample mean $\boldsymbol{\theta}_{\tau_\delta}$ upon stopping.[5]

3.2 A sufficient condition of allocation rules for fixed-confidence optimality

We now present a sufficient condition for an allocation rule to achieve universal efficiency in the fixed-confidence setting. This condition requires the allocation of sampling effort to converge "rapidly" to the optimal solution of (10). We refer to this mode of convergence as $\mathcal{L}$-convergence, which is stronger than almost sure convergence. Specifically, $\mathcal{L}$-convergence ensures that for any fixed neighborhood of the target, the sequence remains within that neighborhood indefinitely after a random time with finite expectation. Let the space $\mathcal{L}$ consist of all real-valued random variables T with $\|T\|_1 < \infty$ where $\|T\|_1 \triangleq \mathbb{E}\left[|T|\right]$.

[5] In cases where multiple correct answers exist under $\boldsymbol{\theta}_{\tau_\delta}$, ties are broken arbitrarily. For simplicity, we abuse the definition $\mathcal{I}(\boldsymbol{\theta}_{\tau_\delta})$ to represent one of the correct answers.

Definition 4 ($\mathcal{L}$-convergence). *For a sequence of real-valued random variables $\{X_t\}_{t\in\mathbb{N}_0}$ and a scalar $x \in \mathbb{R}$, we say $X_t \xrightarrow{\mathcal{L}} x$ if*

$$\text{for all } \epsilon > 0, \quad \text{there exists } T \in \mathcal{L} \text{ such that, for all } t \geq T, \ |X_t - x| \leq \epsilon.$$

For a sequence of random vectors $\{\boldsymbol{X}_t\}_{t\in\mathbb{N}_0}$ taking values in $\mathbb{R}^d$ and a vector $\boldsymbol{x} \in \mathbb{R}^d$ where $\boldsymbol{X}_t = (X_{t,1}, \ldots, X_{t,d})$ and $\boldsymbol{x} = (x_1, \ldots, x_d)$, we say $\boldsymbol{X}_t \xrightarrow{\mathcal{L}} \boldsymbol{x}$ if $X_{t,i} \xrightarrow{\mathcal{L}} x_i$ for all $i \in [d]$.

It is straightforward to observe that $\mathcal{L}$-convergence implies almost sure convergence.

Lemma 2. *If $\boldsymbol{X}_t \xrightarrow{\mathcal{L}} \boldsymbol{x}$, then $\boldsymbol{X}_t \xrightarrow{\text{a.s.}} \boldsymbol{x}$.*

Universal efficiency requires controlling the *expectation* of the stopping time τ_δ in (11). This is achieved by establishing a random time $T_\varepsilon \in \mathcal{L}$ such that $|\Gamma_{\boldsymbol{\theta}_t}(\boldsymbol{p}_t) - \Gamma^*| < \epsilon$ for all $t > T_\epsilon$. Since $T_\varepsilon \in \mathcal{L}$, the expectation $\mathbb{E}^\pi_{\boldsymbol{\theta}}[\tau_\delta]$ is well-controlled. Notably, $\Gamma_{\boldsymbol{\theta}_t}(\boldsymbol{p}_t) \xrightarrow{\mathcal{L}} \Gamma^*$ is implied by $\boldsymbol{p}_t \xrightarrow{\mathcal{L}} \boldsymbol{p}^*$. We formalize this sufficient condition below, with the proof deferred to Appendix C. This condition generalizes the result in Qin et al. (2017), which was specifically tailored to BAI.

Theorem 3 (A sufficient condition for optimality). *Suppose that for any $\boldsymbol{\theta} \in \Theta$, the optimal allocation problem (10) admits a unique, component-wise strictly positive optimal allocation $\boldsymbol{p}^* = \boldsymbol{p}^*(\boldsymbol{\theta})$. A policy π is universally efficient (Definition 3) if the following sufficient conditions are satisfied:*

1. *the stopping rule (11) and decision rule (12) are used, and*
2. *its allocation rule satisfies that*

$$\boldsymbol{p}_t \xrightarrow{\mathcal{L}} \boldsymbol{p}^*, \quad \text{for all } \boldsymbol{\theta} \in \Theta.$$

In particular,

$$\lim_{\delta\to 0} \frac{\mathbb{E}^\pi_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} = (\Gamma^*_{\boldsymbol{\theta}})^{-1}, \quad \text{for all } \boldsymbol{\theta} \in \Theta.$$

Remark 3 (Uniqueness and positivity of optimal solution). *For BAI, Gaussian ε-BAI, pure-exploration thresholding bandits, and best-k identification with single-parameter exponential family distributions, the existence of a unique and componentwise strictly positive optimal solution $\boldsymbol{p}^*(\boldsymbol{\theta}) > 0$ for any $\boldsymbol{\theta} \in \Theta$ is established in Garivier and Kaufmann (2016), Jourdan et al. (2023b), Ménard (2019), and You et al. (2023), respectively.*

Remark 4 (Other performance criteria). *We discussed two alternative performance criteria in Theorem 1 and Section 2.5.2, both focusing on the almost sure convergence of the empirical allocation $\boldsymbol{p}_t \xrightarrow{a.s.} \boldsymbol{p}^*$. For the posterior convergence rate setting (Russo, 2020), the optimal allocation aligns with (10), even for other single-parameter exponential family distributions. For the static-allocation fixed-budget setting (Glynn and Juneja, 2004) under Gaussian distributions, the optimal allocation also coincides with (10) due to the symmetry of the KL divergence. For other distributions, swapping the parameters in the KL divergence in (8) is necessary, but the properties of C_x*

in (9) and the maximin structure of (10) remain valid, leading to a similar sufficient condition. Our sufficient condition in Theorem 3, which requires the $\mathcal{L}$-convergence of the sample allocation, is stronger than almost sure convergence. Consequently, this condition also implies optimality in the posterior convergence rate and the large deviations rate of static allocations.

3.3 Optimality conditions for the optimal allocation problem

We define selection functions that generalize $h_i^j(\boldsymbol{p})$ in Algorithm 2 to general pure-exploration problems:

$$h_i^x(\boldsymbol{p}) = h_i^x(\boldsymbol{p};\boldsymbol{\theta}) \triangleq \frac{p_i \frac{\partial C_x(\boldsymbol{p})}{\partial p_i}}{C_x(\boldsymbol{p})} = \frac{p_i d(\theta_i, \vartheta_i^x)}{\sum_{k\in[K]} p_k d(\theta_k, \vartheta_k^x)}, \quad \text{for } x \in \mathcal{X}(\boldsymbol{\theta}) \text{ and } i \in [K]. \tag{13}$$

Following from the PDE characterization of C_x in Lemma 1, $\boldsymbol{h}^x(\boldsymbol{p}) = \{h_i^x(\boldsymbol{p}) : i \in [K]\} \in \mathcal{S}_K$ forms a valid probability vector for any $x \in \mathcal{X}(\boldsymbol{\theta})$. In fact, $\boldsymbol{h}^x(\boldsymbol{p})$ can be interpreted as the probability distribution over the arms to be sampled when x is identified as the leading culprit.

We derive the following optimality conditions for (10) in the cases where the optimal solution $\boldsymbol{p}^*$ is component-wise strictly positive. These conditions extend those presented in Theorem 1. The detailed proof can be found in Appendix D.

Theorem 4. *Assume that $\boldsymbol{p}^*$ is component-wise strictly positive. A feasible solution $(\phi, \boldsymbol{p})$ to (10) is optimal if and only if $\phi = \min_{x\in\mathcal{X}(\boldsymbol{\theta})} C_x(\boldsymbol{p})$ and there exist dual variables $\boldsymbol{\mu} \in \mathcal{S}_{|\mathcal{X}(\boldsymbol{\theta})|}$ such that*

$$p_i = \sum_{x\in\mathcal{X}(\boldsymbol{\theta})} \mu_x h_i^x, \quad \forall i \in [K], \tag{14a}$$

$$\phi = \min_{x\in\mathcal{X}(\boldsymbol{\theta})} C_x(\boldsymbol{p}), \quad \text{and} \quad \sum_{x\in\mathcal{X}(\boldsymbol{\theta})} \mu_x(\phi - C_x(\boldsymbol{p})) = 0. \tag{14b}$$

The condition (14) is a equivalent form of the KKT condition for optimality, where (14a) is the stationarity condition and (14b) is the complementary slackness condition.

The following result extends the information balance for BAI in (1). It is a direct corollary of (14) and the assumption that $\boldsymbol{p}^*$ is strictly positive. We defer the proof to Appendix G.4.

Corollary 1 (Crude information balance)**.** *Assume that $\boldsymbol{p}^*$ is component-wise strictly positive, then*

$$\Gamma_{\boldsymbol{\theta}}^* = \min_{x\in\mathcal{X}_i} C_x(\boldsymbol{p}^*), \quad \text{for all } i \in [K], \quad \text{where} \quad \mathcal{X}_i \triangleq \left\{ x \in \mathcal{X}(\boldsymbol{\theta}) : \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} > 0 \right\}.$$

Surprisingly, this information balance condition proves to be too crude to guarantee sufficiency for optimality in pure-exploration problems beyond BAI, even when combined with the correct overall balance condition; see You et al. (2023, Example 4) for a counterexample. To address this challenge, we introduce dual variables that implicitly encode a more refined information balance.

3.4 Dual-directed algorithm design

Based on Theorem 4, an allocation rule can be immediately derived by tracking the two sets of conditions in (14). Our general algorithmic design framework follows the same idea as that discussed in Section 2.2 and Section 2.3. In particular, our algorithm takes two steps: it first identifies a leading culprit; and then selects an arm that provides the most information in curbing the identified leading culprit.

We first discuss how the leading culprit is detected. Recall that smaller the value of C_x, the less information we have to rule out x as the root cause of an incorrect answer. Additionally, the dual variable μ_x can be interpreted as the proportion of time x is detected as the leading culprit. In our algorithm, the culprit x with the smallest (empirical estimation of) C_x is detected as the leading culprit x_t. This step is guided by the complementary slackness conditions in (14b), which indicate that only the culprit x with the smallest C_x has $\mu_x > 0$ and therefore requires attention.

Once x_t has been detected as the leading culprit, the algorithm employs an information-directed selection (`IDS`) rule to select an arm for sampling, specified by $\boldsymbol{h}^{x_t}$ defined in (13). This procedure is formalized in Algorithm 4, which extends the `IDS` rule from Algorithm 2 for BAI to general pure-exploration problems. The `IDS` rule is designed to depend on the detected culprit x_t, aiming to maximize the average information gain required to rule out the leading culprit. In particular, this process satisfies the stationarity conditions in (14a), where the sampling probability of arm i is decomposed into a law-of-total-probability formula. Here, h_i^x specifies the probability of sampling i given that x is detected as the leading culprit.

Algorithm 4 Information-directed selection for general pure exploration

Input: Sample allocation $\boldsymbol{p}$, an estimate $\widehat{\boldsymbol{\theta}}$ and the detected leading culprit x.

1: Return $I \in [K]$ sampled from the distribution $\boldsymbol{h}^x(\boldsymbol{p}; \widehat{\boldsymbol{\theta}})$.

Remark 5 (Information-directed selection)**.** *$C_x(\boldsymbol{p})$ is defined as the population version of the GLRT statistic, normalized by the sample size t, for testing $H_{0,x} : \boldsymbol{\theta} \in \mathrm{Alt}_x(\boldsymbol{\theta}_t)$ versus $H_{1,x} : \boldsymbol{\theta} \notin \mathrm{Alt}_x(\boldsymbol{\theta}_t)$. Thus, it quantifies the information gathered per unit sample under the allocation vector $\boldsymbol{p}$ to determine whether the culprit x is responsible for an incorrect answer. The PDE characterization of $C_x(\boldsymbol{p})$ in (9) reveals a decomposition of this information. To interpret, $\frac{\partial C_x(\boldsymbol{p})}{\partial p_i}$ represents the information gain per unit allocation to arm i while $p_i \frac{\partial C_x(\boldsymbol{p})}{\partial p_i}$ quantifies the information gain per unit sample attributed to allocating a p_i proportion of the samples to arm i. The selection function $h_i^x(\boldsymbol{p}) = \frac{p_i \frac{\partial C_x(\boldsymbol{p})}{\partial p_i}}{C_x(\boldsymbol{p})}$ suggests that sampling should be done proportionally to the information gain contributed by each arm when determining the allocation among the candidates. Moreover, the selection functions $h_i^x(\boldsymbol{p})$ identify a set of candidates whose allocation probabilities are positive under a given culprit x, defined as $\mathfrak{C}_x \triangleq \{i \in [K] : \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} > 0\}$. This result is intuitive because only the arms in $\mathfrak{C}_x$ contribute to the information gain relevant to the culprit x. For the specific case of BAI, the cardinality of $\mathfrak{C}_x$ is always two, leading to the well-known class of top-"two" algorithms.*

Remark 6 (The role of the dual variables)**.** *The novelty of our algorithm design framework lies in the incorporation of dual variables. Existing constant-optimal algorithms typically fall into two categories. The first solves the optimal allocation problem using plug-in estimators at each iteration and tracks the resulting solutions, as pioneered by Garivier and Kaufmann (2016) and extended in works such as Kato and Ariu (2024); Al Marjani et al. (2022). The second tracks optimality conditions solely based on the allocation vector* $\boldsymbol{p}$ *(e.g.,* (1) *and* (2)*), as demonstrated in Chen and Ryzhov (2023); Avci et al. (2023); Bandyopadhyay et al. (2024). Our approach is more fundamental and flexible due to the introduction of dual variables, which encode the information balance structure described in Corollary 1. This enables us to avoid explicit characterization of the optimal information balance structure, which is often impractical without knowledge of the true parameter* $\boldsymbol{\theta}$ *(Zhang et al., 2023). Furthermore, dual variables facilitate a law-of-total-probability-type stationarity condition* (14a)*, naturally leading to the* `IDS` *rule that achieves optimality across various pure-exploration tasks.*

3.5 A modular template for proposed algorithms

Our algorithm design principle focuses on adaptively acquiring information that most effectively curbs the leading culprit. Accordingly, we name it Adaptive Culprit Curbing (`ACC`).

Our primary focus is on the `ACC` allocation rules, which determine the arm to be sampled at each time step based on the observed history. These rules are independent of the total budget of samples T or the confidence parameter δ, making them any-time. In the fixed-confidence setting, the `ACC` allocation rule is coupled with a stopping rule to ensure δ-correctness. As detailed in Section 3.1, we adopt standard stopping and decision rules from the literature for this setting. In contrast, the fixed-budget and posterior convergence rate settings terminate at a pre-specified budget T and are evaluated based on their respective PCS measures. This versatility highlights the adaptability of the `ACC` framework across various performance criteria.

We propose the `ACC` allocation rules that follow three steps: first, estimate (`est`) the unknown parameters using $\widehat{\boldsymbol{\theta}}$, which also provides the most plausible answer $\mathcal{I}(\widehat{\boldsymbol{\theta}})$ to the exploration query; next, detect (`det`) the leading culprit x that could make $\mathcal{I}(\widehat{\boldsymbol{\theta}})$ incorrect; and finally, select (`sel`) one arm to sample based on the detected leading culprit x. A template for our allocation rule is summarized in Algorithm 5.

Algorithm 5 Adaptive Culprit Curbing (`ACC`) allocation rule for general pure exploration

Input: Current history $\mathcal{H}$, an estimation rule (`est`), a detection rule (`det`), and a selection rule (`sel`).

1: Run estimation rule to obtain $\widehat{\boldsymbol{\theta}} = \texttt{est}(\mathcal{H})$.
2: Run detection rule to obtain the leading culprit $x = \texttt{det}(\widehat{\boldsymbol{\theta}}, \mathcal{H})$.
3: Return $I = \texttt{sel}\left(x, \widehat{\boldsymbol{\theta}}, \mathcal{H}\right)$.

We will follow the naming convention of `est-det-sel` for our proposed `ACC` algorithms. Below, we elaborate on the options for the three sub-routines required by `ACC`. We provide explicit formulas

of $\mathcal{X}(\boldsymbol{\theta}), C_x(\boldsymbol{p})$ and $\boldsymbol{h}^x(\boldsymbol{p})$ for a wide range of pure-exploration problems in Appendix A.

3.5.1 Estimation (`est`) rules

We consider two estimation rules. The empirical-best (`EB`) rule estimates the means using the empirical mean $\boldsymbol{\theta}_t$ based on the history $\mathcal{H}_t$ at time t. The Thompson-sampling (`TS`) rule estimates the means by drawing a sample $\widetilde{\boldsymbol{\theta}}_t$ from the posterior distribution Π_t at time t.

The estimate $\widehat{\boldsymbol{\theta}}$ returned by the estimation rule determines the perceived best answer $\mathcal{I}(\widehat{\boldsymbol{\theta}})$, culprit set $\mathcal{X}(\widehat{\boldsymbol{\theta}})$ and the decomposed alternative set $\mathrm{Alt}_x(\widehat{\boldsymbol{\theta}})$. Along with the sample allocation $\boldsymbol{p}$, this determines the estimated generalized Chernoff information $C_x(\boldsymbol{p}, \widehat{\boldsymbol{\theta}})$.

3.5.2 Detection (`det`) rules

We propose three detection rules.

1. The `KKT` detection rule (Algorithm 6) explicitly follows the complementary slackness conditions (14b) from our KKT analysis. It determines the leading culprit as the x that minimizes $C_x(\boldsymbol{p}, \widehat{\boldsymbol{\theta}})$.

2. The `TS` detection rule (Algorithm 7) generalizes the `TTTS` algorithm (Algorithm 1) for BAI to more general pure-exploration problems. It identifies an alternative problem instance with a different answer (and hence, a culprit) that is most likely to emerge from the posterior distribution. This framework offers a systematic approach to adapting the classical TS algorithm for pure-exploration problems.

3. The Posterior-Probability-Sampling (`PPS`) detection rule (Algorithm 8) samples the leading culprit x with probability proportional to $\Pi(\mathrm{Alt}_x(\widehat{\boldsymbol{\theta}}))$, where Π denotes the current posterior distribution.

Following the same rationale as in Section 2.2.2, Russo (2020, Proposition 5) establish that the `TS` and `PPS` detection rules are asymptotically equivalent to the `KKT` rule, as $t \to \infty$.

Algorithm 6 `KKT` detection rule for general pure exploration

Input: Current sample allocation $\boldsymbol{p}$ and an estimate $\widehat{\boldsymbol{\theta}}$.

1: Return the leading culprit $x = \mathrm{argmin}_{x \in \mathcal{X}(\widehat{\boldsymbol{\theta}})} C_x(\boldsymbol{p}, \widehat{\boldsymbol{\theta}})$, breaking ties arbitrarily.

Algorithm 7 `TS` detection rule for general pure exploration

Input: Estimation $\widehat{\boldsymbol{\theta}}$ and posterior distribution Π.

1: Repeatedly sample $\widetilde{\boldsymbol{\theta}} \sim \Pi$ until $\widetilde{\boldsymbol{\theta}} \in \mathrm{Alt}(\widehat{\boldsymbol{\theta}})$.

2: Return the leading culprit $x \in \{x' \in \mathcal{X}(\widehat{\boldsymbol{\theta}}) : \widetilde{\boldsymbol{\theta}} \in \mathrm{Alt}_{x'}(\widehat{\boldsymbol{\theta}})\}$, breaking ties arbitrarily.

Algorithm 8 `PPS` detection rule for general pure exploration

Input: Estimation $\widehat{\boldsymbol{\theta}}$ and posterior distribution Π.

1: Return the leading culprit $x \in \mathcal{X}(\widehat{\boldsymbol{\theta}})$ sampled with probability proportional to $\Pi\left(\mathrm{Alt}_x(\widehat{\boldsymbol{\theta}})\right)$.

3.5.3 Selection (`sel`) rule

For the selection rule, we follow the `IDS` rule (Algorithm 4).

Remark 7 (When does the design framework work?)**.** *The derivation of the* `ACC` *algorithm relies only on two ingredients that lead directly to the KKT conditions in Theorem 4: (i) Assumption 1 induces the maximin convex optimization problem central to our framework, which is satisfied by many common exploration queries, as demonstrated in Appendix A; and (ii) by definition (8) using the GLRT, the generalized Chernoff information is homogeneous of degree one, satisfying $C_x(t \cdot \boldsymbol{p}) = t \cdot C_x(\boldsymbol{p})$ for any $t > 0$, which is equivalent to the PDE characterization in (9) by Euler's theorem for homogeneous functions; see Apostol (1969, p. 287) for example.*

Remark 8 (Extensions)**.** *Our analysis extends beyond the base case to accommodate: (i) reward distributions outside the single-parameter exponential family, and (ii) exploration queries with multiple correct answers. We present these extensions in Section 4. Specifically, in Section 4.1, we consider the problem of identifying one of the ε-good alternatives, known as ε-best-arm identification (ε-BAI, Eckman and Henderson 2018; Jourdan et al. 2023b). This problem aligns with R&S under a PGS guarantee. The reason ε-BAI is compatible with our framework is that it can be reformulated as a modified pure-exploration problem with a unique answer. In Section 4.2, we examine the Gaussian case with unknown variances—a scenario in which the reward distribution no longer belongs to a single-parameter exponential family. Similar analyses are possible for other distributions; for instance, Agrawal et al. (2020) provides an analysis for heavy-tailed distributions. Importantly, our algorithm can be readily extended to accommodate these cases.*

3.6 Universal efficiency

In this section, we apply Algorithm 5 to Gaussian BAI (Example 1) and to pure-exploration thresholding bandits (Example 3), demonstrating that two specialized versions of Algorithm 5 are universally optimal for each problem, respectively.

3.6.1 Gaussian BAI.

For BAI presented in Example 1 with Gaussian rewards, we focus on the `TTTS` sub-routine (Algorithm 1) combined with the information-directed selection sub-routine (Algorithm 2). Under the naming convention of Section 3.5, this is the `TS-TS-IDS` algorithm. To honor the original name proposed in Russo (2020), we refer to this algorithm as `TTTS-IDS`. As a theoretical guarantee, we prove that for Gaussian BAI, `TTTS-IDS` satisfies the sufficient condition for optimal allocation rules

in Theorem 3. This addresses an open problem in the pure-exploration literature, as noted by Russo (2020, Section 8). The proof is deferred to Appendix E.

Theorem 5. `TTTS-IDS` *ensures that, for any* $\boldsymbol{\theta}$ *satisfying* $\theta_i \neq \theta_j$ *for all* $i \neq j$*, we have* $\boldsymbol{p}_t \xrightarrow{\mathcal{L}} \boldsymbol{p}^*$*, where the optimal allocation* $\boldsymbol{p}^*$ *is identified in Theorem 1, which verifies the sufficient condition for optimal allocation rules in Theorem 3.*

In conjunction with Theorem 2 on the fixed-budget posterior convergence rate and the discussion of the static fixed-budget setting in Remark 4, the `TTTS-IDS` is optimal across all three settings.

Remark 9 (Fixed-confidence optimality of other top-two algorithms)**.** *The analysis of* `TTTS-IDS` *is arguably the most complex among the top-two algorithms, as both the leader and challenger are random variables conditioned on the history. Our proof strategy can be extended to other top-two algorithms combined with* `IDS`*, such as* `TS-KKT-IDS`*,* `TS-PPS-IDS`*, and those presented in Russo (2020); Qin et al. (2017); Shang et al. (2020); Jourdan et al. (2022).*

3.6.2 Pure-exploration thresholding bandits

For pure-exploration thresholding bandits discussed in Example 3, we prove the universal efficiency of the `EB-KKT-IDS` algorithm. For this problem, we have $\mathcal{X} = [K]$ and the optimal allocation problem simplifies to

$$\Gamma^*_{\boldsymbol{\theta}} = \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{i\in\mathcal{X}} \inf_{\boldsymbol{\vartheta}\in \mathrm{Alt}_i} \sum_i p_i d(\theta_i, \vartheta_i) = \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{i\in\mathcal{X}} \inf_{(\mathfrak{T}-\vartheta_i)(\mathfrak{T}-\theta_i)<0} p_i d(\theta_i, \vartheta_i) = \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{i\in[K]} p_i d(\theta_i, \mathfrak{T}),$$

where the second equality follows from the fact that we can always set $\vartheta_j = \theta_j$ for $j \neq i$, and the last equality follows from the monotonicity of the KL divergence. We apply Lemma 1 to obtain that $\frac{\partial C_i}{\partial p_i} = d(\theta_i, \mathfrak{T})$ and $\frac{\partial C_i}{\partial p_j} = 0$ for any $j \neq i$. Hence, we have $h^i_i(\boldsymbol{p}) = \frac{p_i d(\theta_i, \mathfrak{T})}{C_i(\boldsymbol{p})} = 1$, and $h^i_j(\boldsymbol{p}) = 0$, for all $j \neq i$ and $i \in [K]$.

With the explicit forms of $\mathcal{X}$, C_x, and $\boldsymbol{h}^x$ derived, we can immediately implement `EB-KKT-IDS` algorithm. Theorem 6 establishes that `EB-KKT-IDS` is universally efficient. The proof deferred to Appendix F.

Theorem 6. `EB-KKT-IDS` *ensures that for any* $\boldsymbol{\theta} \in \Theta$*,* $\boldsymbol{p}_t \xrightarrow{\mathcal{L}} \boldsymbol{p}^*$*.*

4 Extensions

In this section, we briefly discuss several important problems not covered by the main framework introduced in Section 3. The first is related to alternative performance criteria, specifically the probability of good selection (PGS), also known as (ϵ, δ)-PAC (probably approximately correct) in the learning literature. The second concerns a practical scenario in which Gaussian distributions have unknown variances.

4.1 Optimal probability of good selection guarantee: Gaussian ε-best-arm identification

This section demonstrates that our framework can be extended to develop optimal algorithms for a variant of the R&S problem, specifically R&S with a PGS guarantee. This formulation is related to, but distinct from, the PCS guarantee in the indifference-zone (IZ) formulation (Perng, 1969; Jennison et al., 1982). For a detailed discussion, see Eckman and Henderson (2018) and Hong et al. (2021). In the bandit literature, this problem is referred to as the ε-best-arm identification problem (Garivier and Kaufmann, 2021; Jourdan et al., 2023b), or (ϵ, δ)-PAC learning.

Consider the same K-armed bandits. Let $\mathcal{I}_\varepsilon(\boldsymbol{\theta}) \triangleq \{i \in [K] : \theta_i \geq \theta_{I^*} - \varepsilon\}$ denote the set of "good enough arms", whose gap from the best arm $I^* = \operatorname{argmax}_{i\in[K]} \theta_i$ (assuming uniqueness) is within a specified tolerance ε. The ε-BAI problem aims to find *any* arm $i \in \mathcal{I}_\varepsilon(\boldsymbol{\theta})$.

To apply our framework, the first step is to establish a lower bound on the problem complexity. Notice that the ε-BAI problem differs significantly from the BAI problem in that the correct answer is non-unique; any arm $i \in \mathcal{I}_\varepsilon$ can serve as a correct answer. When ε exceeds the gap between the best and second-best arm, there may be *multiple correct answers*. For a discussion on pure-exploration problems with multiple correct answers, see Degenne and Koolen (2019).

In this section, we restrict our focus to Gaussian bandits with a homogeneous variance of $\sigma^2 = 1$. For Gaussian bandits, Jourdan et al. (2023b, Lemma 9) establishes the lower bound on problem complexity

$$\lim_{\delta\to 0} \frac{\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \geq (\Gamma^*_{\boldsymbol{\theta},\varepsilon})^{-1},$$

where

$$\Gamma^*_{\boldsymbol{\theta},\varepsilon} = \max_{i\in\mathcal{I}_\varepsilon(\boldsymbol{\theta})} \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{j\neq i} \frac{(\theta_i - \theta_j + \varepsilon)^2}{2\left(1/p_i + 1/p_j\right)} = \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{j\neq I^*} \frac{(\theta_{I^*} - \theta_j + \varepsilon)^2}{2\left(1/p_{I^*} + 1/p_j\right)}$$

To interpret the complexity lower bound, note that finding any ε-good arm can be achieved by asking, for each arm i the question: Is arm i an ε-good arm? This question can be further decomposed into smaller hypothesis tests $H_{0,i,j} : \theta_j - \varepsilon \geq \theta_i$ versus $H_{1,i,j} : \theta_j - \varepsilon < \theta_i$ for any $j \neq i$. To see this, if there exists an arm $i \neq j$ for which $H_{0,i,j}$ holds, then arm i cannot be ε-good, as $\theta_j - \theta_i > \varepsilon$. Therefore, to assert that arm i is ε-good, we need to reject $H_{0,i,j}$ for all $j \neq i$. The term $\frac{(\theta_i-\theta_j+\varepsilon)^2}{2(1/p_i+1/p_j)}$ represents the test statistic, where a higher value increases the likelihood of rejection. Consequently, $\min_{j\neq i} \frac{(\theta_i-\theta_j+\varepsilon)^2}{2(1/p_i+1/p_j)}$ corresponds to the information needed to assert that i is ε-good. The problem complexity depends on maximizing this information across all arms under their optimal allocations. For the last equality, we observe that the best arm I^* is always the easiest to confirm as ε-good.

With the maximin complexity lower bound, the ε-BAI problem falls within the scope of our framework. Specifically, we can adapt the `KKT` detection rule (Algorithm 6) by substituting in

$$C_j(\boldsymbol{p};\boldsymbol{\theta}) = \frac{(\theta_{I^*(\boldsymbol{\theta})} - \theta_j + \varepsilon)^2}{2\left(1/p_{I^*(\boldsymbol{\theta})} + 1/p_j\right)}, \quad \text{and} \quad h^j_i(\boldsymbol{p}) = \frac{p_j}{p_i + p_j}, \quad h^j_j(\boldsymbol{p}) = \frac{p_i}{p_i + p_j}, \quad \text{for } j \neq I^*(\boldsymbol{\theta}).$$

This approach yields the `EB-TC`$_\varepsilon$ algorithm studied in Jourdan et al. (2023b). Following our proof

techniques for Theorem 5, Jourdan et al. (2023b) establishes the fixed-confidence optimality of EB-TC$_\varepsilon$, when combined with IDS and recommending the empirical best arm upon stopping.

Proposition 1 (Lemma 24 in Jourdan et al. 2023b)**.** *Under* EB-TC$_\varepsilon$-IDS*, we have* $\boldsymbol{p}_t \xrightarrow{\mathcal{L}} \boldsymbol{p}^*$.

4.2 Dealing with unknown variances

The Gaussian distributions with unknown variances no longer belong to the single-parameter exponential family. Fortunately, our framework can still be applied with appropriate modifications to the GLRT statistic.

Let $(\boldsymbol{\theta}, \boldsymbol{\sigma}^2)$ denote the unknown mean and variance vectors. The alternative set is given by $\mathrm{Alt}(\boldsymbol{\theta}, \boldsymbol{\sigma}^2) = \{(\boldsymbol{\vartheta}, \boldsymbol{\varsigma}^2) \in \Theta \times \mathbb{R}_+^K : I^*(\boldsymbol{\vartheta}) \neq I^*(\boldsymbol{\theta})\}$, where $I^*(\boldsymbol{\theta})$ is the index of the best arm under $\boldsymbol{\theta}$. This set can be decomposed into a union of convex sets, satisfying Assumption 1, $\mathrm{Alt}(\boldsymbol{\theta}, \boldsymbol{\sigma}^2) = \cup_{x \in \mathcal{X}(\boldsymbol{\theta}, \boldsymbol{\sigma}^2)} \mathrm{Alt}_x(\boldsymbol{\theta}, \boldsymbol{\sigma}^2)$, where $\mathcal{X}(\boldsymbol{\theta}, \boldsymbol{\sigma}^2) \triangleq [K] \backslash \{I^*(\boldsymbol{\theta})\}$ and $\mathrm{Alt}_x(\boldsymbol{\theta}, \boldsymbol{\sigma}^2) = \{(\boldsymbol{\vartheta}, \boldsymbol{\varsigma}^2) \in \Theta \times \mathbb{R}_+^K : \vartheta_x > \theta_x\}$. The generalized Chernoff information is similarly defined as the population version of the GLRT statistic, with the key distinction that the likelihood ratio now involves the KL divergence for a Gaussian distribution with two parameters:

$$C_x(\boldsymbol{p}; \boldsymbol{\theta}, \boldsymbol{\sigma}^2) \triangleq \inf_{(\boldsymbol{\vartheta}, \boldsymbol{\varsigma}^2) \in \mathrm{Alt}_x(\boldsymbol{\theta}, \boldsymbol{\sigma}^2)} \sum_{i \in [K]} p_i d\left((\theta_i, \sigma_i^2), (\vartheta_i, \varsigma_i^2)\right), \quad \text{for all } x \in \mathcal{X}(\boldsymbol{\theta}, \boldsymbol{\sigma}^2), \tag{15}$$

where $d\left((\theta, \sigma^2), (\vartheta, \varsigma^2)\right) = \frac{1}{2}\left(\frac{(\theta-\vartheta)^2}{\varsigma^2} + \frac{\sigma^2}{\varsigma^2} - 1 - \log\left(\frac{\sigma^2}{\varsigma^2}\right)\right)$.

Following Jourdan et al. (2023a, Lemma 11),

$$C_x(\boldsymbol{p}; \boldsymbol{\theta}, \boldsymbol{\sigma}^2) = \inf_{\vartheta \in \left[\theta_x, \theta_{I^*(\boldsymbol{\theta})}\right]} \frac{1}{2}\left[p_{I^*(\boldsymbol{\theta})} \log\left(1 + (\theta_{I^*(\boldsymbol{\theta})} - \vartheta)^2 / \sigma^2_{I^*(\boldsymbol{\theta})}\right) + p_x \log\left(1 + (\theta_x - \vartheta)^2 / \sigma_x^2\right)\right].$$

The added complexity arises from the fact that the minimizer ϑ in the definition cannot be determined in closed form. However, it can be efficiently computed using a bisection search.

Compared to the known-variance case, the generalized Chernoff information takes a distinct form, revealing a fundamental difference between the two problems. Notably, the generalized Chernoff information under unknown variances is *strictly smaller* than that for known variances, and there are problem instances where the ratio between the two can be arbitrarily small (Jourdan et al. 2023a, Lemma 4 and 12). Crucially, substituting empirical variances into the complexity measures derived for known-variance cases, such as those in Glynn and Juneja (2004); Garivier and Kaufmann (2016), does not produce optimal results.

Fortunately, the key components underlying our algorithm framework remain intact. The first component is the maximin formulation of the optimal allocation problem, which now reads:

$$\Gamma^*_{\boldsymbol{\theta}, \boldsymbol{\sigma}^2} = \max_{\boldsymbol{p} \in \mathcal{S}_k} \min_{x \in \mathcal{X}(\boldsymbol{\theta}, \boldsymbol{\sigma}^2)} C_x(\boldsymbol{p}; \boldsymbol{\theta}, \boldsymbol{\sigma}^2).$$

The second component is the PDE characterization of the $C_x(\boldsymbol{p}; \boldsymbol{\theta}, \boldsymbol{\sigma}^2)$ function as in (9), which allows us to formulate the equivalent KKT conditions. The selection function $\boldsymbol{h}^x(\boldsymbol{p}; \boldsymbol{\theta}, \boldsymbol{\sigma}^2)$, defined

by

$$h_i^x(\boldsymbol{p};\boldsymbol{\theta},\boldsymbol{\sigma}^2) \triangleq \frac{p_i \frac{\partial}{\partial p_i} C_x(\boldsymbol{p};\boldsymbol{\theta},\boldsymbol{\sigma}^2)}{C_x(\boldsymbol{p};\boldsymbol{\theta},\boldsymbol{\sigma}^2)}, \quad \text{for } x \in \mathcal{X}(\boldsymbol{\theta}) \text{ and } i \in [K], \tag{16}$$

is still a valid probability vector since $C_x(\boldsymbol{p};\boldsymbol{\theta},\boldsymbol{\sigma}^2)$ is homogeneous of degree one by definition. The KKT condition retains the same form as in Theorem 4, with appropriate substitutions to account for the unknown variances. Specifically, $C_x(\boldsymbol{p};\boldsymbol{\theta},\boldsymbol{\sigma}^2)$ defined in (15) replaces $C_x(\boldsymbol{p};\boldsymbol{\theta})$, and $\mathcal{X}(\boldsymbol{\theta},\boldsymbol{\sigma}^2)$ replaces $\mathcal{X}(\boldsymbol{\theta})$. Additionally, the new `IDS` rule is applied, using $h_i^x(\boldsymbol{p};\boldsymbol{\theta},\boldsymbol{\sigma}^2)$ as defined in (16). With these specifications, a `KKT` detection rule can be implemented directly. Similarly, Bayesian detection rules, such as `TS` and `PPS`, are straightforward to implement when paired with an appropriate conjugate prior.

5 Numerical Experiments

For the numerical study, we focus on the following problem instances of the best-k identification problem. For each case, we examine both Bernoulli and Gaussian bandits (with variance $\sigma^2 = 1$) and adopt an uninformative prior for our algorithms. Additional numerical experiments are collected in Appendix I.

Case ID	K, k	$\boldsymbol{\theta}$
1	$5, 2$	$(0.1, 0.2, 0.3, 0.4, 0.5)$
2	$20, 5$	$\theta_i = 0.05 * i, \forall i \in [20]$
3	$15, 1$	$\theta_i = 0.3, \forall i \in [14]; \theta_{15} = 0.7$
4	$100, 10$	$\theta_i = 0.3, \forall i \in [90]; \theta_i = 0.7, \forall i \in [100] \backslash [90]$
5	$50, 25$	$\theta_i = 0.2, \forall i \in [10]; \theta_i = 0.5, \forall i \in [25] \backslash [10]; \theta_i = 0.8, \forall i \in [50] \backslash [25]$

5.1 Convergence

We established in Theorem 5 that the empirical allocation under our proposed algorithms converges strongly to the optimal allocation for Gaussian BAI. In Figure 1, we present the empirical convergence rate of $O(1/\sqrt{t})$ to the optimal value for our algorithms in Case 1, where $\boldsymbol{\theta} = (0.5, 0.4, 0.3, 0.2, 0.1)$, for identifying the best-2 under both Gaussian and Bernoulli reward distributions. These results suggest that this convergence behavior extends to more general settings. We record the trajectories of $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_t)$ over 10^3 independent replications and plot the mean trajectories, with shaded areas representing the first and third quartiles.

5.2 Fixed-confidence setting

For benchmarks, we compare our proposed algorithms with existing ones, which are briefly summarized in Appendix H. The m-LinGapE and MisLid algorithms, implemented by Tirinzoni and Degenne (2022), are designed for linear bandits and therefore are not implemented for Bernoulli bandits. To ensure fair comparisons, we set the exploration rate as $\beta(t,\delta) = \log\left((\log(t)+1)/\delta\right)$

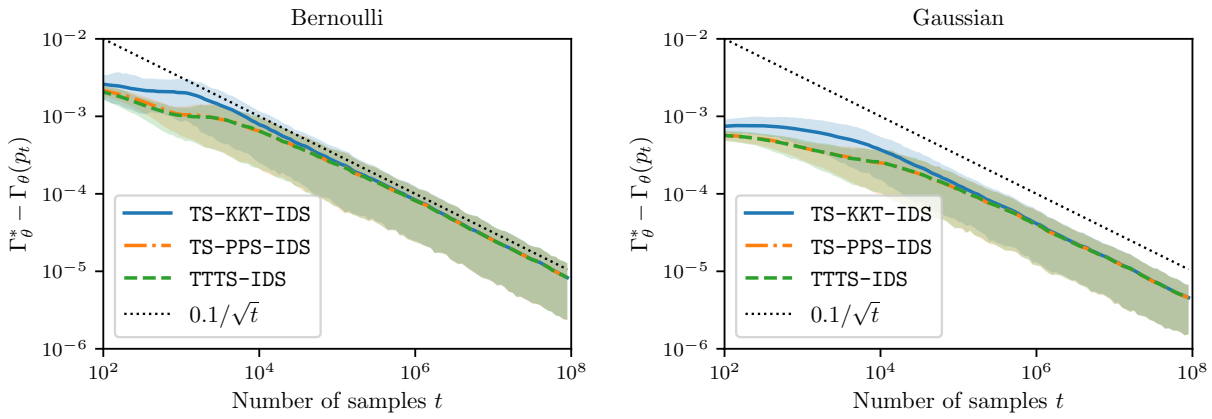


Figure 1: $O(t^{-1/2})$ convergence to the optimal value, with shaded areas representing the first and third quartiles.

in UGapE and KL-LUCB, along with their respective stopping rules. Table 1 presents the fixed-confidence sample complexity, highlighting the superior performance of our proposed algorithms. The FWS algorithm also performs competitively, though with significantly higher computational complexity. Table 1 also reports computation times for selected cases, demonstrating our algorithm's efficiency. Notably, computation time is primarily constrained by the stopping rule calculation, which scales as $O(k(K-k))$; see Section 5.4.3 for further discussion.

5.3 Advantage of IDS over β-tuning.

To isolate the effect of the selection rule, we implement ACC in Algorithm 5 with the Thompson sampling estimation rule and the PPS detection rule, comparing the IDS selection rule in Algorithm 2 with 0.5-tuning, i.e., set $h_i^j = h_j^j = 0.5$. We consider the slippage configuration where $\boldsymbol{\theta}$ is given by $\theta_i = 0.75, i \in [k]$ and $\theta_j = 0.5, j \in [K]\backslash[k]$ for both Gaussian and Bernoulli bandits. Instances are labeled by (K,k)-distribution. Table 2 reports the significant reduction in sample complexity achieved by using IDS.

5.4 Large-scale experiments

The simulation literature has recently seen a surge of interest in large-scale R&S problems, which involve a substantial number of alternative systems. For those interested, we recommend the recent survey by Fan et al. (2024), which provides a comprehensive overview of large-scale simulation optimization. In this section, we evaluate the performance of our proposed algorithms on several large-scale R&S problem instances, considering both fixed-confidence and fixed-budget formulations.

5.4.1 Fixed-confidence

In this experiment, we consider the large-scale slippage configuration studied in Zhong and Hong (2022), where the means of the systems are set as $\mu_0 = 0.1$ and $\mu_i = 0.0$ for $i = 2, \ldots, K$. The

	Case	δ	LB	TTTS-IDS	TS-KKT-IDS	TS-PPS-IDS	FWS	m-LinGapE	MisLid	LMA	KL-LUCB	UGapE	Uniform
Bernoulli	1	0.1	487	838±12	**763**±14	857±13	852±14	—	—	1131±13	1643±26	1639±25	1322±13
		0.01	974	1349±19	**1287**±19	1414±20	1323±20	—	—	1683±18	2687±38	2666±38	2330±17
	2	0.1	1637	3417±27	**2779**±25	3462±26	2964±25	—	—	6329±25	5453±48	5726±50	18465±42
		0.01	3275	5007±38	**4406**±37	5067±38	4565±36	—	—	9314±39	8856±74	8851±73	31236±60
	3	0.1	155	350±1	**282**±1	354±1	284±1	—	—	517±1	533±4	550±2	610±1
		0.01	311	504±2	**435**±2	506±2	443±1	—	—	725±1	843±6	901±2	899±1
	4	0.1	1114	3536±2	**2755**±1	3668±1	2834±3	—	—	7350±1	3859±3	4562±2	8992±1
		0.01	2228	4644±2	**3522**±2	4716±2	3868±3	—	—	9772±1	6048±5	7161±2	11607±1
	5	0.1	934	2868±2	**2121**±2	2951±2	2536±5	—	—	4513±2	3870±48	3419±4	6703±2
		0.01	1869	3759±3	**2869**±3	3820±3	3341±5	—	—	6053±3	5203±5	5566±22	8745±2
Gaussian	1	0.1	2159	3919±58	**3667**±61	3991±61	3684±60	4327±58	4411±65	5809±63	7775±121	7618±117	5975±58
		0.01	4318	6230±85	**5949**±89	6169±85	5974±89	6717±84	6786±91	8431±78	11932±168	11956±167	10162±80
	2	0.1	9459	19424±151	**16371**±146	19282±148	17385±153	24526±162	24534±182	34280±152	32650±284	32828±284	101990±225
		0.01	18918	29214±218	**26274**±215	29685±225	27227±227	33563±215	36828±270	46383±220	52323±436	50977±417	178697±322
	3	0.1	647	1484±5	**1181**±5	1455±5	1240±7	1720±4	1964±7	1959±5	2497±44	2331±7	2617±5
		0.01	1294	2152±7	**1827**±7	2110±7	1904±7	2477±5	2883±9	2768±6	3627±25	3757±10	3933±6
	4	0.1	4605	**14756**±8	19101±161*	15035±6	17251±52	19585±5	25815±15	24967±5	53498±441*	66248±463*	38579±4
		0.01	9210	**19543**±12	20279±87*	19742±8	21669±53	25595±6	32946±16	32672±6	29321±128*	31197±76*	49312±5
	5	0.1	4177	**12994**±14	16492±191*	13113±11	19577±102	15660±11	20502±20	18676±10	55535±552*	56074±480*	31350±8
		0.01	8355	**17226**±15	17340±111*	17291±14	21875±96	20333±14	27519±28	25050±13	25373±138*	25896±88*	40921±9

	Case	δ	(K,k)	Execution time per iteration in microseconds (µs)									
Gaussian	1	0.1	(5,2)	90.3	8.5	7.5	405.8	12.3	17.5	12.9	1.2	3.6	3.2
		0.01	(5,2)	738.7	8.8	8.0	385.6	12.2	16.1	12.2	1.1	3.9	3.1
	4	0.1	(100,10)	279.4	657.5	530.3	31295.7	12998.1	9844.7	4440.2	10.4	153.9	217.1
		0.01	(100,10)	591.4	680.4	531.2	32563.8	13660.1	10323.8	4555.7	10.3	167.6	218.9

Table 1: Fixed-confidence sample complexity, averaged over 1000 replications. Cases where the algorithm did not stop within 10^6 steps for some replications are marked with $*$, with problem complexity recorded as 10^6 in these cases. The 95% confidence interval is provided after the "±" sign.

(K,k)-dist	$(500,1)$-G	$(500,2)$-G	$(500,3)$-G	$(500,5)$-G	$(500,1)$-B	$(500,2)$-B	$(500,3)$-B	$(500,5)$-B
TS-PPS-0.5	285823	285591	283401	283180	62596	61914	62660	62734
TS-PPS-IDS	199374	216668	223860	241869	44640	47044	50436	52767
Reduction	30.2%	24.1%	21.0%	14.6%	28.7%	24.0%	19.5%	15.9%

Table 2: Fixed-confidence sample complexity with $\delta = 0.001$, averaged over 100 replications.

noisy performances of the systems are independent and follows Gaussian distributions with a known variance of $\sigma^2 = 1$. The desired confidence level is set to $\delta = 0.05$.

In Table 3, we compare the fixed-confidence performance of Procedure $\mathcal{KT}_0$ (Zhong and Hong, 2022), Procedure $\mathcal{KN}$ (Kim and Nelson, 2001), and two proposed algorithms, `TS-KKT-IDS` and `TTTS-IDS`. The simulation results for Procedures $\mathcal{KT}_0$ and $\mathcal{KN}$ are taken from Zhong and Hong (2022, Table 2). Both $\mathcal{KT}_0$ and $\mathcal{KN}$ require the specification of an indifference zone parameter, which is set to the (unknown) true gap of 0.1 in this experiment. In contrast, our algorithms are entirely *parameter-free*. For our fixed-confidence algorithms, the stopping rule (11) is applied using a heuristic threshold $\gamma(t, \delta) = \Phi^{-1}(1-\delta)$, i.e., the $1-\delta$ quantile of the standard normal distribution. All results are averaged over 1000 independent replications.

	`TS-KKT-IDS`		`TTTS-IDS`		`EB-TS-IDS`		Procedure $\mathcal{KT}_0$		Procedure $\mathcal{KN}$	
K	PCS	$\mathbb{E}[\tau_\delta]/K$	PCS	$\mathbb{E}[\tau_\delta]/K$	PCS	$\mathbb{E}[\tau_\delta]/K$	PCS	$\mathbb{E}[\tau_\delta]/K$	PCS	$\mathbb{E}[\tau_\delta]/K$
10^2	0.999	616 ± 16	1.0	758 ± 14	0.999	727 ± 12	0.963	1462 ± 3	0.978	884 ± 8
10^3	1.0	764 ± 14	1.0	864 ± 12	1.0	788 ± 7	0.965	1512 ± 1	0.989	1165 ± 8
10^4	1.0	937 ± 13	1.0	1015 ± 12	0.999	898 ± 5	0.956	1527 ± 1	0.994	1419 ± 9
10^5	1.0	1069 ± 14	1.0	1154 ± 13	1.0	1024 ± 3	0.955	1528 ± 1	0.994	1686 ± 8

Table 3: Sample complexity and estimated PCS are compared for the proposed algorithms, Procedure $\mathcal{KT}_0$ and Procedure $\mathcal{KN}$ across different numbers of systems in a slippage configuration with known and equal variances. The 95% confidence interval is provided after the "±" sign, and the estimated PCS is shown in parentheses.

The theoretical stopping rule is designed to ensure PCS for high-confidence levels, i.e., as $\delta \to 0$, but empirically, it tends to be overly conservative for moderate PCS level guarantees. Thus, we implemented a heuristic stopping rule better suited to larger δ. Our results show that this heuristic rule remains conservative, with an estimated PCS exceeding 0.999 in all cases. However, the asymptotically optimal allocation compensates for the conservatism of the stopping rule, achieving smaller sample complexities and higher PCS than competitors. Future work should focus on developing stopping rules that are less conservative for moderate δ. Recent studies, such as Wang et al. (2024) and Tirinzoni and Degenne (2022), propose elimination-based stopping rules that could reduce conservatism and improve computational efficiency. We leave this direction for future research and present this numerical study to illustrate the potential of asymptotically optimized algorithms in large-scale experiments.

5.4.2 Fixed-budget

In this experiment, we evaluate the performance of the proposed algorithms under configurations studied by Li et al. (2025), including the slippage configuration ($\mu_0 = 0.1$ and $\mu_i = 0.0$ for $i = 2, \ldots, K$ with $\sigma_i^2 = 1$), and the equally-spaced configuration with increasing variances ($\mu_0 = 0.1$ and $\mu_i = -i/K$ for $i = 2, \ldots, K$ with $\sigma_i^2 = 1 + i/K$). We assume that all system variances are known.

A key insight from Li et al. (2025) is that allocating samples greedily—sampling the system with

the highest empirical mean—keeps PCS bounded away from zero as the total sampling budget scales linearly with the number of systems, i.e., $B = cK$ for a fixed constant c as $K \to \infty$. This suggests that a degree of greediness is essential in large-scale R&S to promptly eliminate an empirical best system that is not the true best. They introduced the Explore-First Greedy (EFG) procedure to ensure consistency.

Inspired by their work, we test our proposed algorithms with the `EB` estimation rule, i.e., the `EB-*-IDS` algorithms. Figure 2 presents the estimated PCS for the proposed algorithms and benchmarks, including EFG, Greedy, and uniform-allocation procedures. PCS is estimated from 10^4 independent replications, with confidence intervals nearly invisible and thus omitted. As expected, `EB-*-IDS` algorithms appear to be sample-optimal in the sense described by Hong et al. (2022); Li et al. (2025), as the estimated PCS stabilizes at a positive value as $K \to \infty$. This sample optimality is anticipated since the `EB-*-IDS` algorithms allocate sufficient samples to the empirical best. We leave a rigorous investigation of this sample optimality for future work. The advantage of our proposed algorithms becomes more evident as the average budget per system increases, demonstrating that a dynamic allocation strategy informed by asymptotic analysis can enhance performance under moderate to low PCS levels.

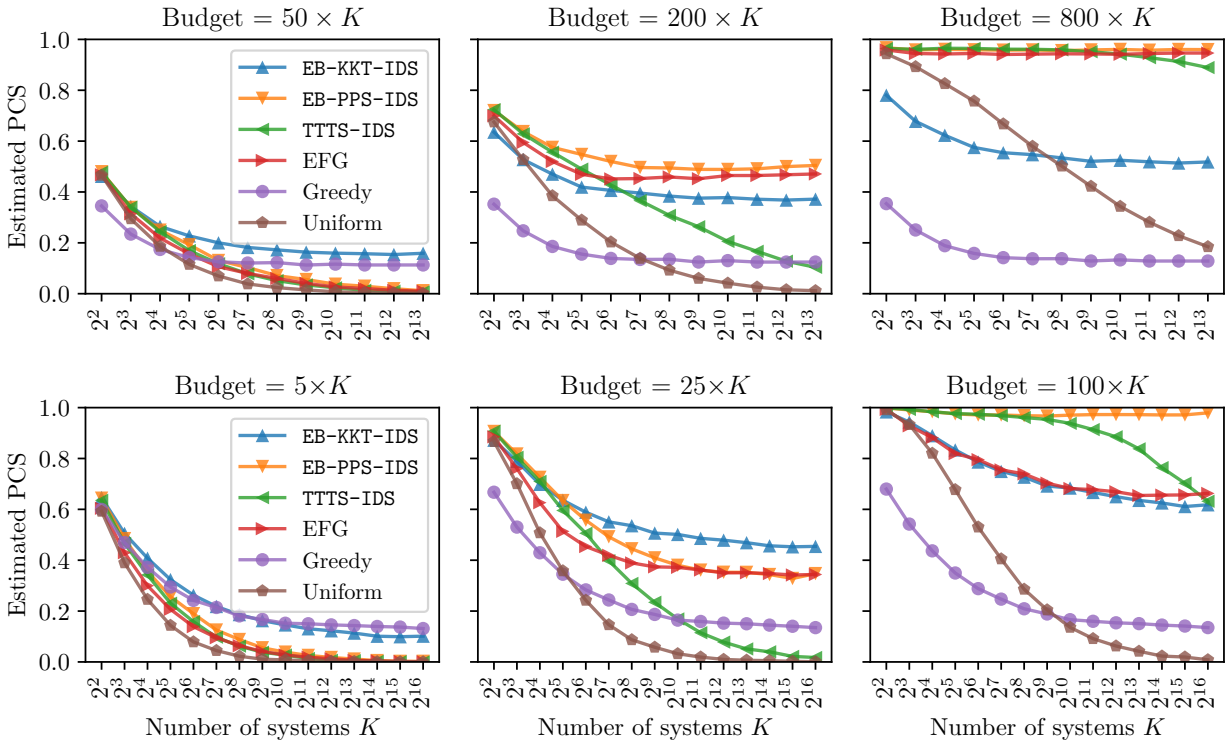


Figure 2: Estimated PCS are compared for the proposed algorithms and benchmarks. Top row is the slippage configuration with equal variances, bottom row is the equally-spaced configuration with increasing variances.

5.4.3 Computation time

To compare the computation time of the proposed algorithms, we consider the slippage configuration, where system means are set as $\mu_0 = 0.1$ and $\mu_i = 0.0$ for $i = 2, \ldots, K$. The noisy performance of each system is independent and follows Gaussian distributions with a known variance of $\sigma^2 = 1$. We vary the number of systems from $K = 10^1$ to 10^4. For each instance, we run the algorithm for $100K$ or $1000K$ time steps and perform 10^3 independent repetitions to estimate the sample mean of the running time along with its 95% confidence interval.

Table 4 reports the execution time of our proposed algorithms, measured in average microseconds (μs) per iteration. We observe that both `TS-KKT-IDS` and `TS-PPS-IDS` are invariant to budget changes, while `TTTS-IDS` slows down as the time period grows. However, this computational inefficiency in `TTTS-IDS` is minimal when PCS is below 0.97. Additionally, the slowdown is less pronounced for larger K as sampling an alternative best system becomes easier with more alternatives available.

All three algorithms scale nearly linearly with the number of systems K, as expected. For instance, in `TS-KKT-IDS`, the bottleneck is searching for the minimum among $K - 1$ C_j functions. Notably, when an algorithm targets *constant optimality*, such scaling is inevitable—after all, even the computation complexity of calculating the problem complexity lower bound scales linearly with K.

	`TTTS-IDS`		`TS-PPS-IDS`		`TS-KKT-IDS`	
Budget	$100K$	$1000K$	$100K$	$1000K$	$100K$	$1000K$
$K = 10^1$	$0.66 \pm 0.01(0.409)$	$5.93 \pm 1.11(0.974)$	$0.84 \pm 0.01(0.411)$	$1.04 \pm 0.01(0.979)$	$0.35 \pm 0.00(0.443)$	$0.39 \pm 0.01(0.971)$
$K = 10^2$	$3.38 \pm 0.12(0.161)$	$7.86 \pm 0.59(0.991)$	$4.75 \pm 0.17(0.179)$	$5.16 \pm 0.18(0.980)$	$1.92 \pm 0.07(0.204)$	$1.99 \pm 0.07(0.988)$
$K = 10^3$	$22.59 \pm 0.98(0.065)$	$25.85 \pm 0.99(0.978)$	$32.94 \pm 1.43(0.059)$	$33.51 \pm 1.44(0.975)$	$13.16 \pm 0.57(0.057)$	$13.13 \pm 0.57(0.983)$
$K = 10^4$	$170.13 \pm 7.96(0.026)$	$172.56 \pm 7.92(0.964)$	$297.96 \pm 14.27(0.018)$	$297.52 \pm 14.22(0.955)$	$98.79 \pm 4.62(0.019)$	$98.62 \pm 4.61(0.964)$

Table 4: Computation time for the proposed algorithms, measured in average microseconds (μs) per iteration. The 95% confidence interval is provided after the "±" sign, and the estimated PCS is shown in parentheses.

5.5 Practical considerations in algorithm implementation

Exploration versus exploitation in pure exploration. The exploration-exploitation trade-off arises even in pure-exploration problems. For instance, the estimation rules `EB` and `TS` aim to identify the best arm. `EB` exploits by selecting the arm with the highest sample mean, while `TS` explores by sampling from the posterior. Asymptotic performance is largely unaffected by the choice of estimation rule, but practical performance can differ under moderate confidence levels or limited budgets. `EB` converges faster when the greedy guess is correct, but it risks poor performance when the guess is wrong, as allocations are based on a misspecified problem instance. In contrast, `TS` explores more to avoid getting stuck with incorrect estimates but requires additional samples when the estimate is already accurate. Thus, `EB-*-IDS` algorithms[6] typically perform better on average

[6]Except for `EB-KKT-IDS`, where both estimation and detection are greedy, leading to significant outliers in slippage configurations.

but are more prone to extreme outliers from inaccurate initial sample means, while `TS-*-IDS` algorithms show slightly worse average performance but fewer outliers. This behavior is discussed in Wu and Zhou (2018a); Jourdan et al. (2022) and conceptually linked to Simchi-Levi et al. (2024). As a side note, Li et al. (2025) observes that `EB` generally stabilizes the PCS at positive levels as the number of alternatives grows, while `TS` does not.

A similar trade-off exists in the choice of the detection rule (`det` in Section 3.5.2). While `KKT` greedily selects the x with the smallest C_x as culprit, `TS` and `PPS` encourage exploration of culprits. For instance, comparing `TTTS-IDS` with `TS-KKT-IDS` in Table 1 and Table 3, we find that `*-KKT-IDS` algorithms perform better on average but are more prone to extreme outliers (e.g., Table 1, Case 4 and 5, Gaussian). In contrast, `*-TS-IDS` algorithms exhibit slightly worse average performance but no outliers. However, in most cases, exploration in the estimation rule alone suffices to mitigate outliers.

Common Random Numbers (CRN). In simulation studies, the performances of different systems are often simulated using common random numbers (CRN), introducing positive correlations between the noisy performances. This typically reduces sampling complexity by decreasing the variances of the differences in sample means, see Nelson and Matejcik (1995); Zhong and Hong (2022).

The main challenge in fully leveraging CRN lies in the need to allow noisy performances of different systems to be mutually correlated. For simplicity of exposition, we introduced our framework under the assumption that the rewards are i.i.d. random variables from a single-parameter exponential family. However, our framework can be extended to handle correlated observations.

To demonstrate, consider the case where performances follow a multivariate normal distribution with means μ_i, variances σ_i^2, and pairwise correlation ρ, where both σ_i^2 and ρ are known.[7] The generalized Chernoff information is modified to account for the correlation as:

$$C_j(\boldsymbol{p};\boldsymbol{\theta}) = \frac{(\theta_{I^*}-\theta_j)^2}{2\left(\sigma_{I^*}^2/p_{I^*} + \sigma_j^2/p_j - 2\rho\sigma_{I^*}\sigma_j/\sqrt{p_{I^*}p_j}\right)}, \quad \text{for } j \in [K]\backslash\{I^*\}. \tag{17}$$

It can be verified that (9) holds, along with Theorem 4, allowing the immediate application of an `ACC` algorithm. For the fixed-confidence setting, the modified C_j must replace the original in the stopping rule. Although the universal efficiency of `ACC` under CRN remains an open question, numerical evidence in Table 5 demonstrates its effectiveness.

6 Conclusion

This paper introduces a unified algorithm design principle for general pure-exploration problems by linking asymptotic lower bounds on problem complexities to maximin optimal allocation problems. This connection yields reformulated Karush-Kuhn-Tucker conditions involving dual variables,

[7] Extensions to unknown covariance matrices are possible using the multiparameter exponential family, as in Section 4.2.

Case	K	TS-KKT-IDS			
		$\rho = 0.0$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$
A	10^2	72±4(0.999)	62±4(0.988)	52±3(0.979)	35±2(0.957)
	10^3	87±3(1.000)	80±3(0.999)	72±3(0.997)	55±2(0.984)
	10^4	112±1(1.000)	106±2(1.000)	94±2(1.000)	75±2(0.995)
B	10^2	616±16(1.000)	537±16(0.998)	423±17(0.992)	254±10(0.965)
	10^3	743±14(1.000)	670±16(1.000)	554±17(0.996)	336±12(0.992)
	10^4	937±13(1.000)	832±17(1.000)	693±18(1.000)	455±14(0.998)

Table 5: Sample complexity per arm and estimated PCS are compared for the `TS-KKT-IDS` algorithm with varying ρ. Case A corresponds to the slippage configuration with equal variances (Section 5.4.1), and Case B to the equally-spaced configuration with increasing variances (Section 5.4.2). The 95% confidence interval is shown after "±" signs, followed by the estimated PCS.

offering a novel interpretation of the top-two algorithm principle as a method for tracking complementary slackness conditions. Furthermore, the derived stationarity conditions motivate the information-directed selection rule, which, when integrated with the `TTTS` algorithm, achieves optimal performance for Gaussian BAI in posterior convergence, fixed-confidence, and fixed-budget settings.

Our framework demonstrates broad applicability, optimally extending to settings such as ε-BAI and pure-exploration thresholding bandits, notably without relying on forced exploration. We propose practical modifications to Thompson sampling, simplifying its implementation for practitioners addressing various pure-exploration problems. Experimental results confirm our algorithms' superior performance, highlighting improvements in both fixed-budget probability of incorrect selection and fixed-confidence sample complexity, with robust results across diverse and large-scale problem configurations.

Future research directions include refining stopping rules to enhance efficiency in finite-sample or moderate-confidence settings (Simchowitz et al., 2017) and extending theoretical results to broader settings, such as heavy-tailed distributions (Agrawal et al., 2020). Another promising avenue is developing parallelizable algorithm variants suited for large-scale simulations (Avci et al., 2023) and incorporating elimination procedures to reduce parallelization overhead (Zhong and Hong, 2022).

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A Examples of proposed algorithms for pure-exploration problems

In this section, we collect a wide variety of examples of pure-exploration queries that fall within our algorithmic framework and provide explicit formulas for implementing our algorithms.

To implement the `KKT` detection rule (Algorithm 6) and the `IDS` selection rule (Algorithm 4), we require expressions for the culprit set $\mathcal{X}(\boldsymbol{\theta})$, the generalized Chernoff information $C_x(\boldsymbol{p})$, and the selection functions $h_i^x(\boldsymbol{p})$. To implement the `TS` detection rule (Algorithm 7), we require an explicit formula for the culprit set $\mathcal{X}(\boldsymbol{\theta})$, the alternative set $\mathrm{Alt}(\boldsymbol{\theta})$, and an oracle to sample from the posterior distribution (this is usually straightforward in the case of a single-parameter exponential family). Moreover, to implement the `PPS` detection rule (Algorithm 8), we need to calculate the posterior probability of the alternative set, which is also made possible by the explicit formulas for the alternative sets.

To enable direct implementation of our algorithms, we explicitly derive the following components for each pure-exploration problem:

1. Correct answer $\mathcal{I}(\boldsymbol{\theta})$.
2. The alternative set $\mathrm{Alt}(\boldsymbol{\theta})$, the culprit set $\mathcal{X}(\boldsymbol{\theta})$, and the decomposition of $\mathrm{Alt}(\boldsymbol{\theta})$.
3. The unique solution $\boldsymbol{\vartheta}^x$ in Lemma 1.
4. The generalized Chernoff information $C_x(\boldsymbol{p})$.
5. The partial derivative of $C_x(\boldsymbol{p})$, from which the selection functions $h_i^x(\boldsymbol{p})$ can be derived.

A.1 Variants of BAI

BAI or IZ-free R&S. This is studied in detail in Section 2. The goal is to find the arm with the highest mean.

1. Correct answer: $\mathcal{I}(\boldsymbol{\theta}) = \{I^*(\boldsymbol{\theta})\}$, where $I^*(\boldsymbol{\theta}) = \mathrm{argmax}_i\{\theta_i\}$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{j\in\mathcal{X}(\boldsymbol{\theta})}\mathrm{Alt}_j(\boldsymbol{\theta})$ with $\mathcal{X}(\boldsymbol{\theta}) = [K]\backslash\mathcal{I}(\boldsymbol{\theta})$ and $\mathrm{Alt}_j(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} : \vartheta_j > \vartheta_{I^*(\boldsymbol{\theta})}\}$.
3. $\boldsymbol{\vartheta}^j$ is given by $\vartheta^j_{I^*(\boldsymbol{\theta})} = \vartheta^j_j = \frac{p_{I^*(\boldsymbol{\theta})}\theta_{I^*(\boldsymbol{\theta})} + p_j\theta_j}{p_{I^*(\boldsymbol{\theta})} + p_j}$ and $\vartheta^j_k = \theta_k$ for all $j \in \mathcal{X}(\boldsymbol{\theta})$ and $k \neq I^*(\boldsymbol{\theta}), j$.[8]
4. $C_j(\boldsymbol{p}) = p_{I^*(\boldsymbol{\theta})}d(\theta_{I^*(\boldsymbol{\theta})}, \vartheta^j_{I^*(\boldsymbol{\theta})}) + p_j d(\theta_j, \vartheta^j_j)$ for all $j \in \mathcal{X}(\boldsymbol{\theta})$.
5. $\frac{\partial C_j(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta^j_i)$ for all $j \in \mathcal{X}(\boldsymbol{\theta})$, most of them are zero, except for $i = I^*(\boldsymbol{\theta})$ or j.

[8] Our algorithm relies on the vector $\boldsymbol{\vartheta}^j$ as defined in the function $\boldsymbol{h}^x(\boldsymbol{p})$ used in `IDS`. Note that $\boldsymbol{\vartheta}^j$ is undefined when $p_{I^*} = p_j = 0$. In these cases, we assign $\boldsymbol{h}^x(\boldsymbol{p}) = \frac{1}{K}\mathbf{1}$, which corresponds to the uniform vector. This choice is made without loss of generality, since beginning with one allocation per arm ensures that the empirical allocations will never become exactly zero, no matter how small the allocation may get. A similar argument applies to the best-k identification, pure-exploration thresholding bandit (selecting the arm closest to a threshold), and all-ε-best-arm identification scenarios, so we do not repeat this discussion for those cases.

Best-k identification. The goal is to find k arms with the highest means.

1. Correct answer: $\mathcal{I}(\boldsymbol{\theta}) = \operatorname{argmax}_{\mathcal{S}\subseteq[K],|\mathcal{S}|=k} \left\{\sum_{i\in\mathcal{S}} \theta_i\right\}$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{(i,j)\in\mathcal{X}(\boldsymbol{\theta})} \mathrm{Alt}_{i,j}(\boldsymbol{\theta})$ with $\mathcal{X}(\boldsymbol{\theta}) = \mathcal{I}(\boldsymbol{\theta}) \times \mathcal{I}(\boldsymbol{\theta})^c$ and $\mathrm{Alt}_{i,j}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} : \vartheta_j > \vartheta_i\}$.
3. $\boldsymbol{\vartheta}^{(i,j)}$ is given by $\vartheta_i^{(i,j)} = \vartheta_j^{(i,j)} = \frac{p_i\theta_i + p_j\theta_j}{p_i+p_j}$ and $\vartheta_k^{(i,j)} = \theta_k$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$ and $k \neq i,j$.
4. $C_{i,j}(\boldsymbol{p}) = p_i d(\theta_i, \vartheta_i^{(i,j)}) + p_j d(\theta_j, \vartheta_j^{(i,j)})$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$.
5. $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_k} = d(\theta_k, \vartheta_k^{(i,j)})$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$, most of them are zero, except for $k = i$ or j.

Pairwise pure exploration. The BAI and best-k identification problems can be further generalized to the so-called pairwise pure exploration (Wu et al., 2025). Let $\mathcal{R} \subset \{(i,j) : i \neq j\}$ be any subset of (ordered) pairs.[9] A pairwise pure exploration query partitions the permutation group of the arm set $[K]$ into mutually exclusive and collectively exhaustive subsets $\{\mathcal{R}_1, \ldots, \mathcal{R}_M\}$, and the goal is to determine which of the subsets $\mathcal{R}_m$ is consistent with the true ranking of $\boldsymbol{\theta}$.

1. Correct answer: $\mathcal{I}(\boldsymbol{\theta}) = \{m^*(\boldsymbol{\theta})\}$, where $m^*(\boldsymbol{\theta})$ is the unique index such that $\boldsymbol{\theta}$ satisfies all constraints in $\mathcal{R}_{m^*(\boldsymbol{\theta})}$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{(i,j)\in\mathcal{X}(\boldsymbol{\theta})} \mathrm{Alt}_{i,j}(\boldsymbol{\theta})$ with $\mathcal{X}(\boldsymbol{\theta}) = \mathcal{R}_{m^*(\boldsymbol{\theta})}$ and $\mathrm{Alt}_{i,j}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} : \vartheta_j > \vartheta_i\}$.
3. $\boldsymbol{\vartheta}^{(i,j)}$ is given by $\vartheta_i^{(i,j)} = \vartheta_j^{(i,j)} = \frac{p_i\theta_i + p_j\theta_j}{p_i+p_j}$ and $\vartheta_k^{(i,j)} = \theta_k$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$ and $k \neq i,j$.
4. $C_{i,j}(\boldsymbol{p}) = p_i d(\theta_i, \vartheta_i^{(i,j)}) + p_j d(\theta_j, \vartheta_j^{(i,j)})$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$.
5. $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_k} = d(\theta_k, \vartheta_k^{(i,j)})$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$, most of them are zero, except for $k = i$ or j.

All-ε-best-arm identification. The goal is to find *all* arms with means larger than $\max_{j\in[K]} \theta_j - \varepsilon$; see Mason et al. (2020).

1. Correct answer: $\mathcal{I}(\boldsymbol{\theta}) = \{i : \theta_i \geq \max_{j\in[K]} \theta_j - \varepsilon\}$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{(i,j)\in\mathcal{X}} \mathrm{Alt}_{i,j}(\boldsymbol{\theta})$, where $\mathcal{X}(\boldsymbol{\theta}) = \{(i,j) \in [K]\times[K] : i \in \mathcal{I}(\boldsymbol{\theta}), j \neq i\}$, and for any $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$, $\mathrm{Alt}_{i,j}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} : \vartheta_j - \varepsilon > \vartheta_i\}$. To interpret, if there exists an arm $j \neq i$ with $\theta_j - \theta_i > \epsilon$, then arm i cannot be ε-good. Notice the subtle difference between the culprit set $\mathcal{X}(\boldsymbol{\theta})$ here and that for the best-k identification problem.
3. $\boldsymbol{\vartheta}^{(i,j)}$ is given by $\vartheta_i^{(i,j)} = \frac{p_i(\theta_i+\varepsilon)+p_j\theta_j}{p_i+p_j} - \varepsilon$, $\vartheta_j^{(i,j)} = \frac{p_i(\theta_i+\varepsilon)+p_j\theta_j}{p_i+p_j}$ and $\vartheta_k^{(i,j)} = \theta_k$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$ and $k \neq i,j$.
4. $C_{i,j}(\boldsymbol{p}) = p_i d(\theta_i, \vartheta_i^{(i,j)}) + p_j d(\theta_j, \vartheta_j^{(i,j)})$ for all $(i,j) \in \mathcal{X}(\boldsymbol{\theta})$.
5. $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_k} = d(\theta_k, \vartheta_k^{(i,j)})$ for all $(i,j) \in \mathcal{X}$, most of them are zero, except for $k = i$ or j.

[9] The set $\mathcal{R}$ must be admissible in the sense that (i,j) and (j,i) are not simultaneously in $\mathcal{R}$.

A.2 Variants of pure-exploration thresholding bandits

Pure-exploration thresholding bandits. This is studied in Section 3.6.2. The goal is to find the set of *all* arms whose mean rewards are above a given threshold $\mathfrak{T} \in \mathbb{R}$.

1. Correct answer: $\mathcal{I}(\boldsymbol{\theta}) = \{k \in [K] : \theta_k > \mathfrak{T}\}$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{j \in \mathcal{X}} \mathrm{Alt}_j(\boldsymbol{\theta})$ where $\mathcal{X} = [K]$ and $\mathrm{Alt}_j(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} : (\mathfrak{T} - \vartheta_j)(\mathfrak{T} - \theta_j) < 0\}$.
3. $\boldsymbol{\vartheta}^j$ is given by $\vartheta^j_j = \mathfrak{T}$ and $\vartheta^j_k = \theta_k$ for all $j \in \mathcal{X}$ and $k \neq j$.
4. $C_j(\boldsymbol{p}) = \sum_{i \in [K]} p_i d(\theta_i, \vartheta^j_i) = p_j d(\theta_j, \mathfrak{T})$.
5. $\frac{\partial C_j(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta^j_i)$ for all $j \in \mathcal{X}$, most of them are zero, except for $i = j$.

Pure-exploration thresholding bandit (selecting one arm closest to a threshold). The goal is to find *the* arm whose mean reward is the closest to a given threshold $\mathfrak{T} \in \mathbb{R}$.

1. Correct answer: $\mathcal{I}(\boldsymbol{\theta}) = \{I^*(\boldsymbol{\theta})\}$, where $I^*(\boldsymbol{\theta}) = \mathrm{argmin}_{i \in [K]} |\theta_i - \mathfrak{T}|$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \cup_{j \in \mathcal{X}} \mathrm{Alt}_j(\boldsymbol{\theta})$, where $\mathcal{X}(\boldsymbol{\theta}) = [K] \backslash \mathcal{I}(\boldsymbol{\theta})$ and $\mathrm{Alt}_j(\boldsymbol{\theta}) = \{\boldsymbol{\theta} : |\theta_j - \mathfrak{T}| < |\theta_{I^*(\boldsymbol{\theta})} - \mathfrak{T}|\}$.
3. $\boldsymbol{\vartheta}^j$ is given by $\vartheta^j_{I^*(\boldsymbol{\theta})} = \vartheta^j_j = \frac{p_{I^*(\boldsymbol{\theta})} \theta_{I^*(\boldsymbol{\theta})} + p_j \theta_j}{p_{I^*(\boldsymbol{\theta})} + p_j}$, and $\vartheta^j_k = \theta_k$ for all $k \neq I^*(\boldsymbol{\theta}), j$.
4. $C_j(\boldsymbol{p}) = \sum_{i \in [K]} p_i d(\theta_i, \vartheta^j_i)$ for all $j \in \mathcal{X}$.
5. $\frac{\partial C_j(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta^j_i)$ for all $j \in \mathcal{X}$, most of them are zero, except for $i = I^*(\boldsymbol{\theta})$ and j.

Signed bandits. This is an variant of pure-exploration thresholding bandit problem. The goal is to identified the whether the arm means are all above or all below a given threshold $\mathfrak{T} \in \mathbb{R}$. We assume that either all arm means are above the threshold $\mathfrak{T}$ or below it—let this restricted parameter space be denoted as $\widetilde{\Theta}$; see Ménard (2019).

1. Correct answers: $\mathcal{I}(\boldsymbol{\theta}) = \{+\}$ if $\min_{i \in [K]} \theta_i > \mathfrak{T}$ and $\mathcal{I}(\boldsymbol{\theta}) = \{-\}$ if $\max_{i \in [K]} \theta_i < \mathfrak{T}$.
2. $\mathrm{Alt}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \widetilde{\Theta} : \max_{i \in [K]} \vartheta_i < \mathfrak{T}\}$ if $\mathcal{I}(\boldsymbol{\theta}) = \{+\}$ and $\mathrm{Alt}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \widetilde{\Theta} : \min_{i \in [K]} \vartheta_i > \mathfrak{T}\}$ if $\mathcal{I}(\boldsymbol{\theta}) = \{-\}$. In both cases, the culprit sets are singleton sets and no decomposition is needed. We remark that this satisfies Assumption 1.
3. For both $\mathcal{I}(\boldsymbol{\theta}) = \{+\}$ and $\mathcal{I}(\boldsymbol{\theta}) = \{-\}$, we have $\boldsymbol{\vartheta} = \mathfrak{T}\mathbf{1}$, i.e., a constant vector of $\mathfrak{T}$.
4. $C(\boldsymbol{p}) = \sum_{i \in [K]} p_i d(\theta_i, \mathfrak{T})$.
5. $\frac{\partial C(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \mathfrak{T})$ for all $i \in [K]$.

Murphy sampling. Consider yet another variant of pure-exploration thresholding bandit, where the goal is to determine whether the smallest mean reward is lower than a given threshold $\mathfrak{T}$ or not. If the smallest mean is lower than $\mathfrak{T}$, we say that the answer is "feasible," and we say that the answer is "infeasible" if otherwise; see Kaufmann et al. (2018); Juneja and Krishnasamy (2019).

1. Correct answers: $\mathcal{I}(\boldsymbol{\theta}) = \{\text{feasible}\}$ if $\min_{i\in[K]} \theta_i < \mathfrak{T}$ and $\mathcal{I}(\boldsymbol{\theta}) = \{\text{infeasible}\}$ if $\min_{i\in[K]} \theta_i > \mathfrak{T}$.

2. $\text{Alt}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \widetilde{\Theta} : \min_{i\in[K]} \vartheta_i > \mathfrak{T}\}$ if $\mathcal{I}(\boldsymbol{\theta}) = \{\text{feasible}\}$ (so that the culprit set is a singleton set and no decomposition is needed) and $\text{Alt}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \widetilde{\Theta} : \min_{i\in[K]} \vartheta_i < \mathfrak{T}\} = \cup_{j\in\mathcal{X}} \text{Alt}_j(\boldsymbol{\theta}) = \cup_{j\in\mathcal{X}} \{\boldsymbol{\vartheta} : \vartheta_j < \mathfrak{T}\}$ if $\mathcal{I}(\boldsymbol{\theta}) = \{\text{infeasible}\}$, where $\mathcal{X}(\boldsymbol{\theta}) = [K]$.

3. If $\mathcal{I}(\boldsymbol{\theta}) = \{\text{feasible}\}$, then $\boldsymbol{\vartheta} = \mathfrak{T}\mathbf{1}$, i.e., a constant vector of $\mathfrak{T}$. If $\mathcal{I}(\boldsymbol{\theta}) = \{\text{infeasible}\}$, then $\boldsymbol{\vartheta}^j$ is given by $\vartheta^j_j = \mathfrak{T}$ and $\vartheta^j_k = \theta_k$ for all $j \in \mathcal{X}$ and $k \neq j$.

4. $C(\boldsymbol{p}) = \sum_{i\in[K]} p_i d(\theta_i, \mathfrak{T})$ if $\mathcal{I}(\boldsymbol{\theta}) = \{\text{feasible}\}$, and $C_j(\boldsymbol{p}) = \sum_{i\in[K]} p_i d(\theta_i, \vartheta^j_i) = p_j d(\theta_j, \mathfrak{T})$ if $\mathcal{I}(\boldsymbol{\theta}) = \{\text{infeasible}\}$.

5. $\frac{\partial C(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \mathfrak{T})$ for all $i \in [K]$, if $\mathcal{I}(\boldsymbol{\theta}) = \{\text{feasible}\}$. $\frac{\partial C_j(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta^j_i)$ for all $j \in \mathcal{X}$, if $\mathcal{I}(\boldsymbol{\theta}) = \{\text{infeasible}\}$.

A.3 Partition identification

In Juneja and Krishnasamy (2019), the authors identify a class of pure-exploration problems, termed *partition identification*, which generalizes the variants of pure-exploration thresholding bandits discussed above. These problems, where the goal is to decide in which partition of the parameter space the true mean vector $\boldsymbol{\theta}$ lies, fall naturally within our framework.

Half-space problem. The goal is to check whether the mean vector lies on one side or the other of the half-space, i.e., $\{\boldsymbol{\theta} \in \mathbb{R}^K : \sum_{i=1}^K a_i\theta_i > b\}$ or $\{\boldsymbol{\theta} \in \mathbb{R}^K : \sum_{i=1}^K a_i\theta_i < b\}$, for a specified $(a_1, \ldots, a_K, b) \in \mathbb{R}^{K+1}$; see Juneja and Krishnasamy (2019, Theorem 5).

1. Correct answers: $\mathcal{I}(\boldsymbol{\theta}) = \{+\}$ if $\sum_{i=1}^K a_i\theta_i > b$ and $\mathcal{I}(\boldsymbol{\theta}) = \{-\}$ if $\sum_{i=1}^K a_i\theta_i < b$.

2. If $\mathcal{I}(\boldsymbol{\theta}) = \{+\}$, then $\text{Alt}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \mathbb{R}^K : \sum_{i=1}^K a_i\vartheta_i < b\}$. If $\mathcal{I}(\boldsymbol{\theta}) = \{-\}$, then $\text{Alt}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta} \in \mathbb{R}^K : \sum_{i=1}^K a_i\vartheta_i > b\}$. In both cases, the culprit set is a singleton and no decomposition is needed.

3. For both cases, define the unique optimizer $\boldsymbol{\vartheta}^* = \text{argmin}_{\boldsymbol{\vartheta}\in\mathbb{R}^K : \sum_{i=1}^K a_i\vartheta_i = b} \sum_{i=1}^K p_i\, d(\theta_i, \vartheta_i)$. The first-order optimality conditions read $\frac{\partial}{\partial\vartheta_i}(p_i\, d(\theta_i, \vartheta^*_i)) = -\lambda\, a_i$, for $i = 1, \ldots, K$, with the Lagrange multiplier λ chosen such that $\sum_{i=1}^K a_i\vartheta^*_i = b$.

4. For both cases, $C(\boldsymbol{p}) = \sum_{i=1}^K p_i\, d(\theta_i, \vartheta^*_i)$.

5. For both cases, $\frac{\partial C(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta^*_i)$ for all $i \in [K]$.

Convex-set feasibility problem. We note that the half-space problem naturally extends to verifying whether the mean vector $\boldsymbol{\theta}$ lies within a given convex feasible set; see, e.g., Juneja and Krishnasamy (2019, Section 3.3). In particular, if the complement of the convex feasible set can be expressed as a finite union of convex sets, then our Assumption 1 is satisfied. For example, when the feasibility set is a convex polytope (Gangrade et al., 2024), its complement can be written as a finite union of half-spaces, and hence the assumption holds. Due to the wide variety of potential feasible sets and the associated technical complexity, we omit detailed derivations.

B Proof of Theorem 1

We first formally establish the existence and uniqueness of a probability vector that satisfies the information balance condition (1) and stationarity condition (2) in Theorem 1, as well as its properties, which will be needed in the proof of the sufficient condition in Theorem 1.

Lemma 3 (Existence and properties of $\boldsymbol{p}^*$)**.** *There exists a unique and component-wise strictly positive probability vector $\boldsymbol{p}^*$ satisfying information balance condition (1) and stationarity condition (2). Furthermore, $\boldsymbol{p}^*$ is the unique solution to the maximin optimization problem in (6), i.e.,*

$$\boldsymbol{p}^* = \operatorname*{argmax}_{\boldsymbol{p}\in\mathcal{S}_K} \min_{j\neq I^*} C_j(\boldsymbol{p}).$$

Proof. The proof of this result directly follows from the proofs of Garivier and Kaufmann (2016, Lemma 4 and Theorem 5). Although the statements of Garivier and Kaufmann (2016, Lemma 4 and Theorem 5) are given for the case of equal variances, their proofs can be extended straightforwardly to the case of unequal variances, establishing that:

1. the solution to maximin optimization problem in (6) is unique, and
2. it is the only probability vector that satisfies both the information balance condition (1) and the sum-of-squares balance condition (3).

As shown in Remark 1, the sum-of-squares balance condition (3) is equivalent to the stationarity condition (2). Hence, we have $\boldsymbol{p}^* = \operatorname{argmax}_{\boldsymbol{p}\in\mathcal{S}_K} \min_{j\neq I^*} C_j(\boldsymbol{p})$. The strict positivity of each component of $\boldsymbol{p}^*$ follows from Garivier and Kaufmann (2016, Theorem 5) and its proof. □

The sufficient condition in Theorem 1 is directly led by the next result, which shows that

1. the optimal value of the maximin optimization problem in (6), denoted by $\Gamma^*_{\boldsymbol{\theta}}$, represents the optimal rate of posterior convergence, and
2. an allocation rule with asymptotic allocation converging to the unique solution of this maximin optimization problem, denoted by $\boldsymbol{p}^*$, is indeed optimal.

Theorem 7 (Optimal rate of posterior convergence—A stronger result that implies Theorem 1)**.** *For any allocation rule* $\mathtt{A}$*, with probability one,*

$$\limsup_{t\to\infty} -\frac{1}{t}\log \Pi_t^{\mathtt{A}}(\Theta_{I^*}^c) \le \Gamma_{\boldsymbol{\theta}}^*,$$

where $\Gamma_{\boldsymbol{\theta}}^*$ *is the optimal value of the maximin optimization problem in* (6)*.*

Furthermore, if an allocation rule $\mathtt{A}^*$ *ensures* $\boldsymbol{p}_t \overset{a.s.}{\to} \boldsymbol{p}^*$ *for the unique* $\boldsymbol{p}^*$ *defined in Theorem 1, then with probability one,*

$$\lim_{n\to\infty} -\frac{1}{t}\log \Pi_t^{\mathtt{A}^*}(\Theta_{I^*}^c) = \Gamma_{\boldsymbol{\theta}}^*.$$

Proof. The proof of this result directly follows from that of Qin et al. (2017, Theorem 2). Although the statement of Qin et al. (2017, Theorem 2) is given for the case of equal variances, its proof can be extended straightforwardly to the case of unequal variances. □

C A threshold for the stopping rule and the Proof of Theorem 3

C.1 A threshold for the stopping rule to ensure δ-correctness

Let $h(u) = u - \log u$ and define for any $z \in [1, e]$ and $x \ge 0$

$$\widetilde{h}_z(x) = \begin{cases} e^{1/h^{-1}(x)} h^{-1}(x), & \text{if } x \ge h(1/\log z), \\ z(x - \log\log z), & \text{otherwise.} \end{cases}$$

Furthermore, define

$$\mathcal{C}_{\exp}(x) = 2\widetilde{h}_{3/2}\left(\frac{h^{-1}(1+x) + \log(2\zeta(2))}{2}\right),$$

where $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Riemann zeta function. One can easily verify that $\mathcal{C}_{\exp}(x) \sim x$ when $x \to \infty$.

Proposition 2 (Kaufmann and Koolen 2021, Proposition 15)**.** *The stopping rule* (11) *with the sequence of thresholds*

$$\gamma(t,\delta) = 3\sum_{i=1}^{K} \log(1 + \log N_{t,i}) + K\mathcal{C}_{\exp}\left(\frac{\log(1/\delta)}{K}\right) \tag{18}$$

is such that, for every allocation rule and any $\delta > 0$*,*

$$\mathbb{P}_{\boldsymbol{\theta}}\left(\tau_\delta < \infty,\ \mathcal{I}(\boldsymbol{\theta}_{\tau_\delta}) = \mathcal{I}(\boldsymbol{\theta})\right) \ge 1 - \delta, \quad \text{for any problem instance } \boldsymbol{\theta} \in \Theta,$$

where $\boldsymbol{\theta}_t$ *is the sample mean of the arms at time* t*.*

C.2 Proof of Theorem 3

We rely on the following technical finding implied by Proposition 14 in Qin and Russo (2024).

Proposition 3 (Proposition 14 in Qin and Russo 2024)**.** *If an allocation rule π satisfies that there exist $\rho > 0$, $\alpha > 0$ and $T \in \mathcal{L}$ such that for any $t \geq T$, $\min_{i\in[k]} N_{t,i} \geq \rho \cdot t^{\alpha}$, then $\boldsymbol{\theta}_t \xrightarrow{\mathcal{L}} \boldsymbol{\theta}$.*

Proof of Theorem 3. The first condition in Theorem 3 ensures that the policy π is δ-correct (Definition 2).

Now we fix $\boldsymbol{\theta} \in \Theta$ and write $\boldsymbol{p}^* = \boldsymbol{p}^*(\boldsymbol{\theta})$ for notational convenience. Since $\boldsymbol{p}^* > \mathbf{0}$ componentwisely, the second condition $\boldsymbol{p}_t \xrightarrow{\mathcal{L}} \boldsymbol{p}^*$ in Theorem 3 ensures that the condition in Proposition 3 is satisfied, and thus we have $\boldsymbol{\theta}_t \xrightarrow{\mathcal{L}} \boldsymbol{\theta}$. Then by the continuous mapping theorem,

$$\min_{x\in\mathcal{X}(\boldsymbol{\theta}_t)} C_x(\boldsymbol{p}_t;\boldsymbol{\theta}_t) \xrightarrow{\mathcal{L}} \min_{x\in\mathcal{X}(\boldsymbol{\theta})} C_x(\boldsymbol{p}^*;\boldsymbol{\theta}) = \Gamma^*_{\boldsymbol{\theta}},$$

where the equality follows from that $\boldsymbol{p}^*$ is the maximizer of (10). For any $\epsilon > 0$, there exists a random time $T_{\epsilon,1} \in \mathcal{L}$ such that for any $t \geq T_{\epsilon,1}$,

$$\min_{x\in\mathcal{X}(\boldsymbol{\theta}_t)} C_x(\boldsymbol{p}_t;\boldsymbol{\theta}_t) \geq \Gamma^*_{\boldsymbol{\theta}} - \epsilon.$$

On the other hand, there exists a deterministic time $T_{\epsilon,2}$ such that for any $t \geq T_{\epsilon,2}$,

$$\gamma(t,\delta) \leq 3K\log(1+\log t) + K\mathcal{C}_{\exp}\left(\frac{\log(1/\delta)}{K}\right) \leq \epsilon t + K\mathcal{C}_{\exp}\left(\frac{\log(1/\delta)}{K}\right).$$

Hence, for any $t \geq T_\epsilon \triangleq \max\{T_{\epsilon,1}, T_{\epsilon,2}\} \in \mathcal{L}$,

$$t \cdot \min_{x\in\mathcal{X}(\boldsymbol{\theta}_t)} C_x(\boldsymbol{p}_t;\boldsymbol{\theta}_t) - \gamma(t,\delta) \geq (\Gamma^*_{\boldsymbol{\theta}} - 2\epsilon)\, t - K\mathcal{C}_{\exp}\left(\frac{\log(1/\delta)}{K}\right),$$

which implies

$$\tau_\delta \leq \max\left\{T_\epsilon, \frac{K\mathcal{C}_{\exp}\left(\frac{\log(1/\delta)}{K}\right)}{\Gamma^*_{\boldsymbol{\theta}} - 2\epsilon}\right\} \leq T_\epsilon + \frac{K\mathcal{C}_{\exp}\left(\frac{\log(1/\delta)}{K}\right)}{\Gamma^*_{\boldsymbol{\theta}} - 2\epsilon}.$$

Since $T_\epsilon = \max\{T_{\epsilon,1}, T_{\epsilon,2}\} \in \mathcal{L}$, we have $\mathbb{E}[T_\epsilon] < \infty$, and then

$$\limsup_{\delta\to 0} \frac{\mathbb{E}^\pi_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{\Gamma^*_{\boldsymbol{\theta}} - 2\epsilon},$$

where we also use the fact that $\mathcal{C}_{\exp}(x) \sim x$ when $x \to \infty$. Since $\epsilon > 0$ can be arbitrarily small, we have $\limsup_{\delta\to 0} \frac{\mathbb{E}^\pi_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{\Gamma^*_{\boldsymbol{\theta}}}$. We note that a matching lower bound on the sample complexity has been established in Wang et al. (2021, Appendix B), i.e., for any other δ-correct algorithm π',

$$\liminf_{\delta\to 0} \frac{\mathbb{E}^{\pi'}_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \geq \frac{1}{\Gamma^*_{\boldsymbol{\theta}}}, \quad \text{and thus we have} \quad \limsup_{\delta\to 0} \frac{\mathbb{E}^{\pi}_{\boldsymbol{\theta}}[\tau_\delta]}{\mathbb{E}^{\pi'}_{\boldsymbol{\theta}}[\tau_\delta]} \leq 1.$$

The above inequality holds for any $\boldsymbol{\theta} \in \Theta$, which implies that π is universal efficient (Definition 3). □

D Proof of Theorem 4

Proof. Note that the optimal allocation problem (10) is equivalent to

$$\Gamma^* = \max_{\phi, \boldsymbol{p}} \quad \phi \tag{19a}$$

$$\text{s.t.} \quad \sum_{i\in[K]} p_i - 1 = 0, \tag{19b}$$

$$p_i \geq 0, \quad \forall i \in [K], \tag{19c}$$

$$\phi - C_x(\boldsymbol{p}) \leq 0, \quad \forall x \in \mathcal{X}. \tag{19d}$$

The corresponding Lagrangian is given by

$$\mathcal{L}(\phi, \boldsymbol{p}, \lambda, \boldsymbol{\mu}) = \phi - \lambda \left(\sum_{i\in[K]} p_i - 1 \right) - \sum_{x\in\mathcal{X}} \mu_x (\phi - C_x(\boldsymbol{p})) + \sum_{i\in[K]} \iota_i p_i. \tag{20}$$

Slater's condition holds since $\boldsymbol{p} = \mathbf{1}/K$ and $\phi = 0$ is a feasible solution such that all inequality constraints hold strictly, where $\mathbf{1}$ is a vector of 1's. Hence, the KKT conditions are necessary and sufficient for global optimality, which are then given by

$$-1 + \sum_{x\in\mathcal{X}} \mu_x = 0, \tag{21a}$$

$$\lambda - \sum_{x\in\mathcal{X}} \mu_x \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} - \iota_i = 0, \quad \forall i \in [K], \tag{21b}$$

$$\sum_{i\in[K]} p_i - 1 = 0, \quad p_i \geq 0, \quad \forall i \in [K], \tag{21c}$$

$$\phi - C_x(\boldsymbol{p}) \leq 0, \quad \forall x \in \mathcal{X}, \tag{21d}$$

$$\mu_x \geq 0, \quad \forall x \in \mathcal{X}, \quad \iota_i \geq 0 \quad \forall i \in [K], \tag{21e}$$

$$\sum_{x\in\mathcal{X}} \mu_x (\phi - C_x(\boldsymbol{p})) = 0, \quad \sum_{i\in[K]} \iota_i p_i = 0. \tag{21f}$$

We first show that the KKT conditions (21) implies (14). To this end, we reformulate the stationarity condition (21b), leaving the feasibility and complementary slackness conditions unchanged. Define the selection functions h_i^x as in (13), which we repeat here for convenience:

$$h_i^x \triangleq h_i^x(\boldsymbol{\theta}, \boldsymbol{p}) = \frac{p_i d(\theta_i, \vartheta_i^x)}{C_x(\boldsymbol{p})}, \quad \forall (x, i) \in \mathcal{X} \times [K]. \tag{22}$$

Now, we can rewrite the KKT conditions by eliminating the dual variables λ and ι.

$$\begin{aligned}
& & \lambda - \sum_{x\in\mathcal{X}} \mu_x \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} - \iota_i = 0, \quad \forall i \in [K], \\
\text{C-S for } \boldsymbol{\mu} \quad &\Rightarrow \quad \frac{\lambda}{\phi} - \sum_{x\in\mathcal{X}} \mu_x \frac{\frac{\partial C_x(\boldsymbol{p})}{\partial p_i}}{C_x(\boldsymbol{p})} - \frac{\iota_i}{\phi} = 0, \quad \forall i \in [K], \qquad (23) \\
\text{Multiply } p_i \text{ and use C-S for } \boldsymbol{\iota} \quad &\Rightarrow \quad \frac{\lambda}{\phi} p_i - \sum_{x\in\mathcal{X}} \mu_x \frac{p_i \frac{\partial C_x(\boldsymbol{p})}{\partial p_i}}{C_x(\boldsymbol{p})} = 0, \quad \forall i \in [K], \qquad (24) \\
& \Rightarrow \quad \frac{\lambda}{\phi} p_i - \sum_{x\in\mathcal{X}} \mu_x h_i^x = 0, \quad \forall i \in [K], \\
\text{Sum over } i \quad &\Rightarrow \quad \frac{\lambda}{\phi} - \sum_{x\in\mathcal{X}} \mu_x \sum_{i\in[K]} h_i^x = 0, \quad \forall i \in [K], \\
\text{Primal and dual feasibility} \quad &\Rightarrow \quad \frac{\lambda}{\phi} = 1, \quad \forall i \in [K].
\end{aligned}$$

Substituting $\lambda = \phi$ into (24) yields the desired result. Note that (24) is equivalent to (23) if and only if $\boldsymbol{p}^*$ is strictly positive.

Next, we show that (14) implies the KKT conditions in (21) when the optimal solution is strictly positive. First, when $\boldsymbol{p}^*$ is strictly positive, we have $\boldsymbol{\iota} = \mathbf{0}$. We have that (21a), (21d), (21e), and (21f) hold automatically. Let $\lambda = \phi$, we have (21b) using (24), (21f) and $\boldsymbol{\iota} = \mathbf{0}$. Finally, (21c) is obtained by summing (14a) over all $i \in [K]$ and noting that $\boldsymbol{\mu} \in \mathcal{S}_{|\mathcal{X}|}$. This completes the proof. □

E Proof of Theorem 5

We prove Theorem 5 in this appendix. For clarity of exposition, our analysis focuses on the case of a common variance. However, the argument readily generalizes to the heterogeneous variance setting by replacing the common variance σ^2 with either $\sigma^2_{\min} = \min_{i\in[K]} \sigma_i^2$ or $\sigma^2_{\max} = \max_{i\in[K]} \sigma_i^2$, depending on the direction of the inequalities.

E.1 Preliminary

E.1.1 Notation

For Gaussian BAI with reward distributions $N(\theta_i, \sigma^2)$ for $i \in [K]$, let $C_{t,j}$ denote the empirical versions of $C_j(\cdot)$ obtained by plugging in the sample proportion $\boldsymbol{p}_t$ and the sample mean $\boldsymbol{\theta}_t$, i.e.,

$$C_{t,j} \triangleq C_j(\boldsymbol{p}_t; \boldsymbol{\theta}_t),$$

for any culprit $j \in \mathcal{X}(\boldsymbol{\theta}) = [K] \backslash \{I^*(\boldsymbol{\theta})\}$.

Minimum and maximum gaps. We define the minimum and maximum values between the expected rewards of two different arms:

$$\Delta_{\min} \triangleq \min_{i\neq j} |\theta_i - \theta_j| \quad \text{and} \quad \Delta_{\max} \triangleq \max_{i,j\in[K]} (\theta_i - \theta_j).$$

By the assumption that the arm means are unique, we have $\Delta_{\max} \geq \Delta_{\min} > 0$.

Another measure of cumulative effort and average effort. We have introduced one measure of cumulative effort and average effort: the number of samples and the proportion of total samples allocated allocated to arm $i \in [K]$ before time $t \in \mathbb{N}_0$ are denoted as

$$N_{t,i} = \sum_{\ell=0}^{t-1} \mathbb{1}(I_\ell = i) \quad \text{and} \quad p_{t,i} = \frac{N_{t,i}}{t}.$$

To analyze randomized algorithms (such as `TTTS`), we introduce the probability of sampling arm i at time t as

$$\psi_{t,i} \triangleq \mathbb{P}(I_t = i | \mathcal{H}_t),$$

where the history $\mathcal{H}_t$ is the sigma-algebra generated by the observations $\left(I_0, Y_{1,I_0}, \ldots, I_{t-1}, Y_{t,I_{t-1}}\right)$ available up to time t. Then we define another measure of cumulative effort and average effort:

$$\Psi_{t,i} \triangleq \sum_{\ell=0}^{t-1} \psi_{\ell,i} \quad \text{and} \quad w_{t,i} \triangleq \frac{\Psi_{t,i}}{t}.$$

Posterior means and variances under independent uninformative priors. Under the independent uninformative prior, the posterior mean and variance of the unknown mean of arm $i \in [K]$ are

$$\theta_{t,i} = \frac{1}{N_{t,i}} \sum_{\ell=0}^{t-1} \mathbb{1}(I_\ell = i) Y_{\ell+1,I_\ell} \quad \text{and} \quad \sigma^2_{t,i} = \frac{\sigma^2}{N_{t,i}}.$$

Posterior probability of being the best arm. To analyze the `TTTS` algorithm, we need the posterior probability that arm i emerges as the best arm, defined as

$$\alpha_{t,i} \triangleq \mathbb{P}_{\widetilde{\boldsymbol{\theta}} \sim \Pi_t} \left(\widetilde{\theta}_i = \max_{j \in [k]} \widetilde{\theta}_j \right).$$

E.1.2 Definition and properties of strong convergence

The proof of our Theorem 5 requires bounding time until quantities reaches their limits. Following Qin and Russo (2022), we first introduce a class of random variables that are "light-tailed." Let the space $\mathcal{L}_p$ consist of all measurable T with $\|T\|_p < \infty$ where $\|T\|_p \triangleq \left(\mathbb{E}\left[|T|^p\right]\right)^{1/p}$ is the p-norm.

Definition 5 (Definition 1 in Qin and Russo 2022)**.** *For a real valued random variable T, we say $T \in \mathbb{M}$ if and only if $\|T\|_p < \infty$ for all $p \geq 1$. Equivalently, $\mathbb{M} = \cap_{p \geq 1} \mathcal{L}_p$.*

Then we introduce the so-called *strong convergence* for random variables, which is a stronger notion of convergence than almost sure convergence.

Definition 6 ($\mathbb{M}$-convergence, Definition 2 in Qin and Russo 2022)**.** *For a sequence of real valued random variables $\{X_t\}_{t \in \mathbb{N}_0}$ and a scalar $x \in \mathbb{R}$, we say $X_t \xrightarrow{\mathbb{M}} x$ if*

$$\textit{for all } \epsilon > 0 \textit{ there exists } T \in \mathbb{M} \textit{ such that for all } t \geq T, \ |X_t - x| \leq \epsilon.$$

We say $X_t \xrightarrow{\mathbb{M}} \infty$ *if*

$$\textit{for all } c > 0 \textit{ there exists } T \in \mathbb{M} \textit{ such that for all } t \geq T,\ X_t \geq c,$$

and similarly, we say $X_t \xrightarrow{\mathbb{M}} -\infty$ *if* $-X_t \xrightarrow{\mathbb{M}} \infty$.
For a sequence of random vectors $\{\boldsymbol{X}_t\}_{t\in\mathbb{N}_0}$ *taking values in* $\mathbb{R}^d$ *and a vector* $\boldsymbol{x} \in \mathbb{R}^d$ *where* $\boldsymbol{X}_t = (X_{t,1}, \ldots, X_{t,d})$ *and* $\boldsymbol{x} = (x_1, \ldots, x_d)$, *we say* $\boldsymbol{X}_t \xrightarrow{\mathbb{M}} \boldsymbol{x}$ *if* $X_{t,i} \xrightarrow{\mathbb{M}} x_i$ *for all* $i \in [d]$.

Corollary 2. *If* $\boldsymbol{X}_t \xrightarrow{\mathbb{M}} \boldsymbol{x}$, *then* $\boldsymbol{X}_t \xrightarrow{\mathcal{L}} \boldsymbol{x}$ *and* $\boldsymbol{X}_t \xrightarrow{\text{a.s.}} \boldsymbol{x}$.

Note that the sufficient condition for optimality in Theorem 3 requires only $\mathcal{L}$-convergence, which is implied by the stronger $\mathbb{M}$-convergence. In the proofs of the fixed-confidence optimality (Theorem 5 and Theorem 6), we establish the stronger $\mathbb{M}$-convergence.

Qin and Russo (2022) also introduces a number of properties of the class $\mathbb{M}$ (Definition 5) and strong convergence (Definition 6).

Lemma 4 (Lemma 1 in Qin and Russo 2022: Closedness of $\mathbb{M}$)**.** *Let* X *and* Y *be non-negative random variables in* $\mathbb{M}$.

1. $aX + bY \in \mathbb{M}$ *for any scalars* $a, b \in \mathbb{R}$.
2. $XY \in \mathbb{M}$.
3. $\max\{X, Y\} \in \mathbb{M}$.
4. $X^q \in \mathbb{M}$, *for any* $q \geq 0$.
5. *If* $g : \mathbb{R} \to \mathbb{R}$ *satisfies* $\sup_{x\in-[c,c]} |g(x)| < \infty$ *for all* $c \geq 0$ *and* $|g(x)| = O(|x|^q)$ *as* $|x| \to \infty$ *for some* $q \geq 0$, *then* $g(X) \in \mathbb{M}$.
6. *If* $g : \mathbb{R} \to \mathbb{R}$ *is continuous and* $|g(x)| = O(|x|^q)$ *as* $|x| \to \infty$ *for some* $q \geq 0$, *then* $g(X) \in \mathbb{M}$.

As mentioned in Qin and Russo (2022), many of the above properties can be extended to any finite collection of random variables in $\mathbb{M}$. A notable one is that if $X_i \in \mathbb{M}$ for each $i \in [d]$ then

$$X_1 + \cdots + X_d \in \mathbb{M} \quad \text{and} \quad \max\{X_1, \ldots, X_d\} \in \mathbb{M}.$$

Qin and Russo (2022) also shows an equivalence between pointwise convergence and convergence in the maximum norm and a continuous mapping theorem under strong convergence (Definition 6).

Lemma 5 (Lemma 2 in Qin and Russo 2022)**.** *For a sequence of random vectors* $\{X_n\}_{n\in\mathbb{N}}$ *taking values in* $\mathbb{R}^d$ *and a vector* $x \in \mathbb{R}^d$, *if* $X_n \xrightarrow{\mathbb{M}} x$, *then* $\|X_n - x\|_\infty \xrightarrow{\mathbb{M}} 0$. *Equivalently, if* $X_{n,i} \xrightarrow{\mathbb{M}} x_i$ *for all* $i \in [d]$, *then for all* $\epsilon > 0$, *there exists* $N \in \mathbb{M}$ *such that* $n \geq N$ *implies* $|X_{n,i} - x_i| \leq \epsilon$ *for all* $i \in [d]$.

Lemma 6 (Lemma 3 in Qin and Russo 2022: Continuous Mapping)**.** *For any* x *taking values in a normed vector space and any random sequence* $\{X_n\}_{n\in\mathbb{N}}$*:*

1. *If* g *is continuous at* x, *then* $X_n \xrightarrow{\mathbb{M}} x$ *implies* $g(X_n) \xrightarrow{\mathbb{M}} g(x)$.
2. *If the range of function* g *belongs to* $\mathbb{R}$ *and* $g(y) \to \infty$ *as* $y \to \infty$, *then* $X_n \xrightarrow{\mathbb{M}} \infty$ *implies* $g(X_n) \xrightarrow{\mathbb{M}} \infty$.

E.1.3 Maximal inequalities

Following Qin et al. (2017); Shang et al. (2020), we introduce the following path-dependent random variable to control the impact of observation noises:

$$W_1 = \sup_{(t,i)\in\mathbb{N}_0\times[K]} \sqrt{\frac{N_{t,i}+1}{\log(N_{t,i}+e)}}\frac{|\theta_{t,i}-\theta_i|}{\sigma}. \tag{25}$$

In addition, to control the impact of algorithmic randomness, we introduce the other path-dependent random variable:

$$W_2 = \sup_{(t,i)\in\mathbb{N}_0\times[K]} \frac{|N_{t,i}-\Psi_{t,i}|}{\sqrt{(t+1)\log(t+e^2)}}. \tag{26}$$

The next lemma ensures that these maximal deviations have light tails.

Lemma 7 (Lemma 6 of Qin et al. 2017 and Lemma 4 of Shang et al. 2020)**.** *For any $\lambda > 0$,*

$$\mathbb{E}[e^{\lambda W_1}] < \infty \quad \textit{and} \quad \mathbb{E}[e^{\lambda W_2}] < \infty.$$

The definition of W_2 in Equation (26) implies the following corollary for sufficiently large t.

Corollary 3. *There exists a deterministic T such that for any $t \geq T$,*

$$|N_{t,i} - \Psi_{t,i}| \leq W_2 t^{0.6}, \quad \forall i \in [K].$$

The exponent 0.6 is not essential in the sense that it can be any constant in $(0.5, 1)$. Corollary 3 implies

$$p_{t,i} - w_{t,i} \xrightarrow{\mathbb{M}} 0, \quad \forall i \in [K].$$

E.2 Proof outline

Sufficient conditions for optimality in fixed-confidence setting. Theorem 3 shows a sufficient condition of allocation rules that ensures the optimality in fixed-confidence setting. It is easier to study the following alternative but equivalent sufficient conditions to prove optimality of allocation rules.

Proposition 4 (Alternative sufficient conditions)**.** *The following conditions are equivalent:*

1. $\boldsymbol{p}_t \xrightarrow{\mathbb{M}} \boldsymbol{p}^*$;
2. $\frac{p_{t,j}}{p_{t,I^*}} \xrightarrow{\mathbb{M}} \frac{p^*_j}{p^*_{I^*}}$ *for any* $j \neq I^*$;
3. $\boldsymbol{w}_t \xrightarrow{\mathbb{M}} \boldsymbol{p}^*$; *and*
4. $\frac{w_{t,j}}{w_{t,I^*}} \xrightarrow{\mathbb{M}} \frac{p^*_j}{p^*_{I^*}}$ *for any* $j \neq I^*$.

Proof. By Corollary 3, conditions 1 and 3 are equivalent.

Now we are going to show conditions 1 and 2 are equivalent. By Lemma 4, condition 1 implies condition 2. On the other hand, applying Lemma 4 to condition 3 gives

$$\frac{1-p_{t,I^*}}{p_{t,I^*}} = \sum_{j\neq I^*} \frac{p_{t,j}}{p_{t,I^*}} \xrightarrow{\mathbb{M}} \sum_{j\neq I^*} \frac{p^*_j}{p^*_{I^*}} = \frac{1-p^*_{I^*}}{p^*_{I^*}},$$

Using Lemma 4 again yields $p_{t,I^*} \xrightarrow{\mathbb{M}} p^*_{I^*}$, and then condition 2 gives $p_{t,j} \xrightarrow{\mathbb{M}} p^*_j$ for any $j \neq I^*$ Hence, condition 2 implies condition 1, and thus they are equivalent. Similarly, conditions 2 and 4 are equivalent. This completes the proof. □

To prove Theorem 5, we show that `TTTS-IDS` satisfies the last sufficient condition in Proposition 4:

Proposition 5. *Under* `TTTS-IDS`*,*

$$\frac{w_{t,j}}{w_{t,I^*}} \xrightarrow{\mathbb{M}} \frac{p^*_j}{p^*_{I^*}}, \quad \forall j \neq I^*.$$

The rest of this appendix is devoted to proving Proposition 5.

E.3 Empirical overall balance under `TTTS-IDS`

Previous papers (Qin et al., 2017; Shang et al., 2020; Jourdan et al., 2022) establishes β-optimality by proving either $w_{t,I^*} \xrightarrow{\mathbb{M}} \beta$ or $p_{t,I^*} \xrightarrow{\mathbb{M}} \beta$ where β is a fixed tuning parameter taken as an input to the algorithms, while the optimal $\beta^* =$ (i.e., $p^*_{I^*}$) is unknown a priori. In our information-directed selection rule, at each time step, h_t is adaptive to the top-two candidates, so novel proof techniques are required. In this subsection, we first show that the "empirical" version of overall balance (3) holds.

Proposition 6. *Under* `TTTS-IDS`*,*

$$w^2_{t,I^*} - \sum_{j\neq I^*} w^2_{t,j} \xrightarrow{\mathbb{M}} 0 \quad \textit{and} \quad p^2_{t,I^*} - \sum_{j\neq I^*} p^2_{t,j} \xrightarrow{\mathbb{M}} 0.$$

To prove this proposition, we need the following supporting result.

Lemma 8. *Under* `TTTS-IDS`*, for any* $t \in \mathbb{N}_0$*,*

$$\alpha_{t,I^*}\left(\sum_{\ell\neq I^*} \frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}} \frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) \leq \psi_{t,I^*} \leq \alpha_{t,I^*}\left(\sum_{\ell\neq I^*} \frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}} \frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + (1-\alpha_{t,I^*}),$$

and for any $j \neq I^*$*,*

$$\alpha_{t,I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}}{p_{t,I^*}+p_{t,j}} \leq \psi_{t,j} \leq \alpha_{t,I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}}{p_{t,I^*}+p_{t,j}} + (1-\alpha_{t,I^*}).$$

Proof. Fix $t \in \mathbb{N}_0$. Recall that $\alpha_{t,i}$ is the posterior probability at time t that arm i appears as the best arm. For a fixed $j \in [K]$ and any $\ell \neq j$, TTTS obtains the top-two candidates

$$\left(I_t^{(1)}, I_t^{(2)}\right) = \begin{cases} (j,\ell), & \text{with probability } \alpha_{t,j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}, \\ (\ell,j), & \text{with probability } \alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}, \end{cases}$$

and when TTTS is applied with IDS,

$$\psi_{t,j} = \alpha_{t,j}\left(\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}\right) + \sum_{\ell\neq j}\alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}.$$

For $j = I^*$, we have

$$\psi_{t,I^*} = \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + \sum_{\ell\neq I^*}\alpha_{t,\ell}\frac{\alpha_{t,I^*}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}},$$

so

$$\psi_{t,I^*} \geq \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right)$$

and

$$\begin{aligned}\psi_{t,I^*} &\leq \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + \sum_{\ell\neq I^*}\alpha_{t,\ell} \\ &= \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + (1-\alpha_{t,I^*}),\end{aligned}$$

where the inequality uses that for any $\ell \neq I^*$, $\frac{\alpha_{t,I^*}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}} \leq \frac{\alpha_{t,I^*}}{1-\alpha_{t,\ell}} = \frac{\alpha_{t,I^*}}{\sum_{\ell'\neq\ell}\alpha_{t,\ell'}} \leq 1$.

For $j \neq I^*$, we have

$$\begin{aligned}\psi_{t,j} &= \sum_{\ell\neq j}\alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}} + \alpha_{t,j}\left(\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}\right) \\ &= \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}} + \sum_{\ell\neq j,I^*}\alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,\ell}+p_{t,j}} + \alpha_{t,j}\left(\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}\right),\end{aligned}$$

so

$$\psi_{t,j} \geq \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}}$$

and

$$\psi_{t,j} \leq \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}} + \left(\sum_{\ell\neq j,I^*}\alpha_{t,\ell}\right) + \alpha_{t,j} = \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}} + (1-\alpha_{t,I^*}),$$

where the inequality uses that

1. for any $\ell \neq j, I^*$, $\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}} \frac{p_{t,\ell}}{p_{t,\ell}+p_{t,j}} \leq \frac{\alpha_{t,j}}{1-\alpha_{t,\ell}} = \frac{\alpha_{t,j}}{\sum_{\ell' \neq \ell} \alpha_{t,\ell'}} \leq 1$; and
2. $\sum_{\ell \neq j} \frac{\alpha_{t,\ell}}{1-\alpha_{t,j}} \frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}} \leq \sum_{\ell \neq j} \frac{\alpha_{t,\ell}}{1-\alpha_{t,j}} = 1$.

This completes the proof. □

Now we are ready to complete the proof of Proposition 6.

Proof of Proposition 6. By Corollary 3, it suffices to only show

$$w_{t,I^*}^2 - \sum_{j \neq I^*} w_{t,j}^2 \xrightarrow{\mathbb{M}} 0.$$

For any $t \in \mathbb{N}_0$, we let

$$G_t \triangleq \Psi_{t,I^*}^2 - \sum_{j \neq I^*} \Psi_{t,j}^2,$$

and we are going to show $\frac{G_t}{t^2} \xrightarrow{\mathbb{M}} 0$.

We argue that it suffices to show that there exists $T \in \mathbb{M}$ such that for any $t \geq T$,

$$|G_{t+1} - G_t| \leq 2W_2 t^{0.6} + 2. \tag{27}$$

The reason is that given (27), for any $t \geq T$

$$\begin{aligned}
|G_t| = \left| G_T + \sum_{\ell=T}^{t-1} (G_{\ell+1} - G_\ell) \right| &\leq |G_T| + \sum_{\ell=T}^{t-1} |G_{\ell+1} - G_\ell| \\
&\leq |G_T| + \sum_{\ell=T}^{t-1} \left(2W_2 \ell^{0.6} + 2\right) \\
&\leq |G_T| + 2(t-T) + 2W_2 \int_{T+1}^{t+1} x^{0.6} \mathrm{d}x \\
&= |G_T| + 2(t-T) + \frac{2W_2}{1.6} \left[(t+1)^{1.6} - (T+1)^{1.6}\right] \\
&\leq T^2 + 2(t-T) + \frac{2W_2}{1.6} \left[(t+1)^{1.6} - (T+1)^{1.6}\right],
\end{aligned}$$

where in the last inequality, $|G_T| \leq T^2$ holds since $G_T \geq -\sum_{j \neq I^*} \Psi_{T,j}^2 \geq -\left(\sum_{j \neq I^*} \Psi_{T,j}\right)^2 \geq -T^2$, and $G_T \leq \Psi_{T,I^*}^2 \leq T^2$. This gives $\frac{|G_t|}{t^2} \xrightarrow{\mathbb{M}} 0$.

Now we are going to prove the sufficient condition (27). We can calculate

$$\begin{aligned}
G_{t+1} - G_t &= \left[(\Psi_{t,I^*} + \psi_{t,I^*})^2 - \Psi_{t,I^*}^2\right] - \sum_{j \neq I^*} \left[(\Psi_{t,j} + \psi_{t,j})^2 - \Psi_{t,j}^2\right] \\
&= 2\left(\psi_{t,I^*} \Psi_{t,I^*} - \sum_{j \neq I^*} \psi_{t,j} \Psi_{t,j}\right) + \left(\psi_{t,I^*}^2 - \sum_{j \neq I^*} \psi_{t,j}^2\right).
\end{aligned}$$

Then by Lemma 8, we have

$$\begin{aligned}
&G_{t+1} - G_t \\
\geq& 2\left[\alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,j}\Psi_{t,I^*}}{p_{t,I^*}+p_{t,j}} - \alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}} - (1-\alpha_{t,I^*}) \sum_{j\neq I^*} \Psi_{t,j}\right] - 1 \\
\geq& 2\alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}} - 2(1-\alpha_{t,I^*})t - 1,
\end{aligned}$$

where the last inequality uses $\sum_{j\neq I^*} \Psi_{t,j} \leq t$, and

$$\begin{aligned}
&G_{t+1} - G_t \\
\leq& 2\left[\alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,j}\Psi_{t,I^*}}{p_{t,I^*}+p_{t,j}} + (1-\alpha_{t,I^*})\Psi_{t,I^*} - \alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right] + 1 \\
\leq& 2\alpha_{t,I^*} \left(\sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right) + 2(1-\alpha_{t,I^*})t + 1,
\end{aligned}$$

where the last inequality uses $\Psi_{t,I^*} \leq t$. Hence,

$$|G_{t+1} - G_t| \leq 2\alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \left|\frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right| + 2(1-\alpha_{t,I^*})t + 1.$$

By Lemma 12, there exists $T_1 \in \mathbb{M}$ such that for any $t \geq T_1$, $(1-\alpha_{t,I^*})t \leq 1/2$, and thus

$$\begin{aligned}
|G_{t+1} - G_t| &\leq 2\alpha_{t,I^*} \sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \max_{j\neq I^*} \left|\frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right| + 2 \\
&\leq 2\max_{j\neq I^*} \left|\frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right| + 2 \\
&= 2\max_{j\neq I^*} \left|\frac{N_{t,j}\Psi_{t,I^*} - N_{t,I^*}\Psi_{t,j}}{N_{t,I^*}+N_{t,j}}\right| + 2,
\end{aligned}$$

where the last inequality applies $\alpha_{t,I^*} \leq 1$ and $\sum_{j\neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} = 1$. By Corollary 3, there exists $T_2 \in \mathbb{M}$ such that for any $t \geq T_2$ and $i \in [K]$, $|N_{t,i} - \Psi_{t,i}| \leq W_2 t^{0.6}$, and thus for $j \neq I^*$

$$\left|\frac{N_{t,j}\Psi_{t,I^*} - N_{t,I^*}\Psi_{t,j}}{N_{t,I^*}+N_{t,j}}\right| \leq \left|\frac{N_{t,j}\left(N_{t,I^*} + W_2 t^{0.6}\right) - N_{t,I^*}\left(N_{t,j} - W_2 t^{0.6}\right)}{N_{t,I^*}+N_{t,j}}\right| \leq W_2 t^{0.6}.$$

Hence, for $t \geq T \triangleq \max\{T_1, T_2\}$, $|G_{t+1} - G_t| \leq 2W_2 t^{0.6} + 2$. Note that $T \in \mathbb{M}$ by Lemma 4. This completes the proof of the sufficient condition (27). □

E.3.1 Implication of empirical overall balance

Here we present some results that are implied by empirical overall balance (Proposition 6) and needed for proving the sufficient condition for optimality in fixed-confidence setting (Proposition 5). The first one shows that when t is large enough, two measures of average effort are bounded away from 0 and 1.

Lemma 9. *Let* $b_1 \triangleq \frac{1}{\sqrt{32(K-1)}}$ *and* $b_2 \triangleq \frac{3}{4}$. *Under* TTTS-IDS, *there exists a random time* $T \in \mathbb{M}$ *such that for any* $t \geq T$,

$$b_1 \leq w_{t,I^*} \leq b_2 \quad \textit{and} \quad b_1 \leq p_{t,I^*} \leq b_2.$$

Since w_{t,I^*} and p_{t,I^*} are bounded away from 0 when t is large enough, we obtain the following corollary by dividing the equations in Proposition 6 by w_{t,I^*} and p_{t,I^*}, respectively,

Corollary 4. *Under* TTTS-IDS,

$$\sum_{j \neq I^*} \frac{w_{t,j}^2}{w_{t,I^*}^2} \xrightarrow{\mathbb{M}} 1 \quad \textit{and} \quad \sum_{j \neq I^*} \frac{p_{t,j}^2}{p_{t,I^*}^2} \xrightarrow{\mathbb{M}} 1.$$

Proof of Lemma 9. We only show the upper and lower bounds for w_{t,I^*} since proving those for p_{t,I^*} is the same.

By Proposition 6, there exists $T_1 \in \mathbb{M}$ such that for any $t \geq T_1$,

$$w_{t,I^*}^2 - \sum_{j \neq I^*} w_{t,j}^2 \leq \frac{1}{2}.$$

This implies

$$\frac{1}{2} \geq w_{t,I^*}^2 - \left(\sum_{j \neq I^*} w_{t,j} \right)^2 = w_{t,I^*}^2 - (1 - w_{t,I^*})^2 = 2w_{t,I^*} - 1,$$

which gives $w_{t,I^*} \leq b_2 \triangleq \frac{3}{4}$.

Using Proposition 6 again, there exists $T_2 \in \mathbb{M}$ such that for any $t \geq T_2$,

$$w_{t,I^*}^2 - \sum_{j \neq I^*} w_{t,j}^2 \geq -\frac{1}{32(K-1)}.$$

By Cauchy-Schwarz inequality,

$$\begin{aligned} w_{t,I^*}^2 \geq -\frac{1}{32(K-1)} + \frac{\left(\sum_{j \neq I^*} w_{t,j}\right)^2}{K-1} &= -\frac{1}{32(K-1)} + \frac{(1 - w_{t,I^*})^2}{K-1} \\ &\geq -\frac{1}{32(K-1)} + \frac{1}{16(K-1)} = \frac{1}{32(K-1)}, \end{aligned}$$

where the last inequality uses $w_{t,I^*} \leq b_2 = \frac{3}{4}$. This gives $w_{t,I^*} \geq b_1 \triangleq \frac{1}{\sqrt{32(K-1)}}$. Taking $T \triangleq \max\{T_1, T_2\}$ completes the proof since $T \in \mathbb{M}$ by Lemma 4. □

E.4 Sufficient exploration under TTTS-IDS

In this subsection, we show that TTTS-IDS sufficiently explores all arms:

Proposition 7. *Under* TTTS-IDS, *there exists* $T \in \mathbb{M}$ *such that for any* $t \geq T$,

$$\min_{i \in [K]} N_{t,i} \geq \sqrt{t/K}.$$

The previous analyses in Qin et al. (2017); Shang et al. (2020); Jourdan et al. (2022) only work for any sequence of tuning parameters $\{h_t\}_{t\in\mathbb{N}_0}$ that is *uniformly strictly bounded*, i.e., there exists $h_{\min} > 0$ such that with probability one,

$$\inf_{t\in\mathbb{N}_0} \min\{h_t, 1-h_t\} \geq h_{\min}.$$

Our `IDS` is adaptive to the algorithmic randomness in `TTTS` in the sense that it depends on the algorithmic randomness in picking top-two candidates, so it does not enjoy the property above. We need some novel analysis to prove the sufficient exploration of `TTTS-IDS`.

Following Shang et al. (2020) of analyzing `TTTS`, we define the following two "most promising arms":

$$J_t^{(1)} \in \operatorname*{argmax}_{i\in[K]} \alpha_{t,i} \quad \text{and} \quad J_t^{(2)} \in \operatorname*{argmax}_{i\neq J_t^{(1)}} \alpha_{t,i}. \tag{28}$$

We further define the one that is less sampled as:

$$J_t \triangleq \operatorname*{argmin}_{i\in\left\{J_t^{(1)}, J_t^{(2)}\right\}} N_{t,i} = \operatorname*{argmin}_{i\in\left\{J_t^{(1)}, J_t^{(2)}\right\}} p_{t,i}.$$

Note that the identity of J_t can change over time. We show that `IDS` allocates a decent effort to J_t.

Lemma 10. *Under* `TTTS-IDS`*, for any* $t \in \mathbb{N}_0$*,*

$$\psi_{t,J_t} \geq \frac{1}{2K(K-1)}.$$

Proof. Fix $t \in \mathbb{N}_0$. We have

$$\left(I_t^{(1)}, I_t^{(2)}\right) = \left(J_t^{(1)}, J_t^{(2)}\right) \quad \text{with probability} \quad \alpha_{t,J_t^{(1)}} \frac{\alpha_{t,J_t^{(2)}}}{1-\alpha_{t,J_t^{(1)}}} \geq \frac{1}{K(K-1)},$$

where the inequality follows from the definition of $J_t^{(1)}$ and $J_t^{(2)}$. Note that

$$h^{J_t^{(2)}}_{J_t^{(1)}}(\boldsymbol{p}_t;\boldsymbol{\theta}_t) = \frac{p_{t,J_t^{(2)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}} \quad \text{and} \quad h^{J_t^{(2)}}_{J_t^{(2)}}(\boldsymbol{p}_t;\boldsymbol{\theta}_t) = \frac{p_{t,J_t^{(1)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}}.$$

If $J_t = \operatorname{argmin}_{i\in\left\{J_t^{(1)}, J_t^{(2)}\right\}} p_{t,i} = J_t^{(1)}$,

$$\psi_{t,J_t} \geq \frac{1}{K(K-1)} \frac{p_{t,J_t^{(2)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}} \geq \frac{1}{2K(K-1)}.$$

Otherwise, $J_t = \operatorname{argmin}_{i\in\left\{J_t^{(1)}, J_t^{(2)}\right\}} p_{t,i} = J_t^{(2)}$ and

$$\psi_{t,J_t} \geq \frac{1}{K(K-1)} \frac{p_{t,J_t^{(1)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}} \geq \frac{1}{2K(K-1)}.$$

This completes the proof. □

Following Qin et al. (2017), we define an insufficiently sampled set for any $t \in \mathbb{N}_0$ and $s \geq 0$:

$$U_t^s \triangleq \{i \in [K] \,:\, N_{t,i} < s^{1/2}\}.$$

Proposition 7 states that there exists $T \in \mathbb{M}$ such that for all $t \geq T$, $U_t^{t/K}$ is empty. Lemma 11 in Shang et al. (2020) shows a nice property of `TTTS`: if `TTTS` with a selection or tuning rule allocates a decent amount of effort to J_ℓ at any time ℓ, the insufficiently sampled set U_t^s is empty when t is large enough.

Lemma 11 (Lemmas 9 and 11 in Shang et al. 2020)**.** *Under* `TTTS` *with any selection rule (e.g.,* `IDS`*) such that*

$$\exists \psi_{\min} > 0 \quad \forall t \in \mathbb{N}_0 \quad \psi_{t,J_t} \geq \psi_{\min},$$

there exists $S \in \mathbb{M}$ such that for any $s \geq S$, $U^s_{\lfloor Ks \rfloor} = \emptyset$.

We remark that, while the original result focuses on β-tuning, the same proof applies to any selection rule that satisfies the condition.

Now we can complete the proof of Proposition 7.

Proof of Proposition 7. By Lemma 10, `IDS` satisfies the condition in Lemma 11. Take the corresponding S in Lemma 11 for `IDS`, and let $T \triangleq KS$. For any $t \geq T$, we let $s = t/K \geq S$, and then by Lemma 11, we have $U^s_{\lfloor Ks \rfloor} = U_t^{t/K}$ is empty. □

E.4.1 Implication of sufficient exploration

Here we present some results that are implied by sufficient exploration (Proposition 7) and needed for proving the sufficient condition for optimality in fixed-confidence setting (Proposition 5). With sufficient exploration, the posterior means strongly converge to the unknown true means, and the probability of any sub-optimal arm being the best decays exponentially.

Lemma 12 (Lemmas 6 and 12 in Shang et al. 2020)**.** *Under* `TTTS` *with any selection rule that ensures sufficient exploration (e.g.,* `IDS`*), i.e.,*

$$\exists \widetilde{T} \in \mathbb{M} \quad \forall t \geq \widetilde{T} \quad \min_{i \in [K]} N_{t,i} \geq \sqrt{t/K},$$

the following properties hold:

1. $\boldsymbol{\theta}_t \xrightarrow{\mathbb{M}} \boldsymbol{\theta}$*; and*
2. *there exists $T \in \mathbb{M}$ such that for any $t \geq T$ and $j \neq I^*$,*

$$\alpha_{t,j} \leq \exp\left(-ct^{1/2}\right), \quad \textit{where} \quad c \triangleq \frac{\Delta_{\min}^2}{16\sigma^2 K^{1/2}}.$$

We remark that, while the original result focuses on β-tuning, the same proof applies to any selection rule that ensures sufficient exploration.

E.5 Strong convergence to optimal proportions: completing the proof of Proposition 5

In this subsection, we are going to complete the proof of Proposition 5, the sufficient condition for optimality for fixed-confidence setting, which requires a sequence of results.

Lemma 13. *Under* TTTS-IDS*, for any arm* $j \neq I^*$*,*

$$\frac{C_{t,j}}{p_{t,I^*} \cdot f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right)} \xrightarrow{\mathbb{M}} 1, \quad \text{where} \quad f_j(x) \triangleq \frac{(\theta_{I^*} - \theta_j)^2}{2\sigma^2\left(1 + \frac{1}{x}\right)} \quad \text{and} \quad C_{t,j} = \frac{(\theta_{t,I^*} - \theta_{t,j})^2}{2\sigma^2\left(\frac{1}{p_{t,I^*}} + \frac{1}{p_{t,j}}\right)}.$$

Proof. Fix $j \neq I^*$. We have

$$p_{t,I^*} f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right) = p_{t,I^*} \frac{(\theta_{I^*} - \theta_j)^2}{2\sigma^2\left(1 + \frac{p_{t,I^*}}{p_{t,j}}\right)} = \frac{(\theta_{I^*} - \theta_j)^2}{2\sigma^2\left(\frac{1}{p_{t,I^*}} + \frac{1}{p_{t,j}}\right)}.$$

By Proposition 7, TTTS-IDS ensures sufficient exploration, and thus by Lemma 12, $\boldsymbol{\theta}_t \xrightarrow{\mathbb{M}} \boldsymbol{\theta}$. Then applying the continuous mapping theorem (Lemma 6), we have the desired convergence. □

Lemma 14. *Under* TTTS-IDS*, for any* $\epsilon > 0$*, there exists a deterministic constant* $c_\epsilon > 0$ *and a random time* $T_\epsilon \in \mathbb{M}$ *such that for any* $t \geq T_\epsilon$ *and* $j \neq I^*$*,*

$$\frac{w_{t,j}}{w_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \psi_{t,j} \leq \exp\left(-c_\epsilon t\right) + (K-1)\exp\left(-ct^{1/2}\right),$$

where c *is defined in Lemma 12.*

Proof. Fix $\epsilon > 0$. It suffices to show that for any $j \neq I^*$, there exist a deterministic constant $c_\epsilon^{(j)} > 0$ and a random time $T_\epsilon^{(j)} \in \mathbb{M}$ such that for any $t \geq T_\epsilon^{(j)}$,

$$\frac{w_{t,j}}{w_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \psi_{t,j} \leq \exp\left(-c_\epsilon^{(j)} t\right) + (K-1)\exp\left(-ct^{1/2}\right),$$

since this completes the proof by taking $T_\epsilon \triangleq \max_{j \neq I^*} T_\epsilon^{(j)}$ and $c_\epsilon \triangleq \min_{j \neq I^*} c_\epsilon^{(j)}$.

From now on, we fix $j \neq I^*$. By Corollary 3, there exists $T_{\epsilon,1}^{(j)} \in \mathbb{M}$ such that for any $t \geq T_{\epsilon,1}^{(j)}$,

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \frac{p_{t,j}}{p_{t,I^*}} = \frac{N_{t,j}}{N_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*}.$$

Then by Corollary 4, there exists $T_{\epsilon,2}^{(j)} \in \mathbb{M}$ such that for any $t \geq T_{\epsilon,2}^{(j)}$,

$$\frac{p_{t,j}}{p_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*} \quad \Longrightarrow \quad \exists A_t \neq I^* \ : \ \frac{p_{t,A_t}}{p_{t,I^*}} \leq \frac{p_{A_t}^*}{p_{I^*}^*}.$$

From now on, we consider $t \geq \max\left\{T_{\epsilon,1}^{(j)}, T_{\epsilon,2}^{(j)}\right\}$ and we have

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \frac{p_{t,j}}{p_{t,I^*}} = \frac{N_{t,j}}{N_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*} \quad \text{and} \quad \exists A_t \neq I^* \ : \ \frac{p_{t,A_t}}{p_{t,I^*}} \leq \frac{p_{A_t}^*}{p_{I^*}^*}. \tag{29}$$

By Lemmas 8 and 12,

$$\psi_{t,j} \le \alpha_{t,I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}}{p_{t,I^*}+p_{t,j}} + (1-\alpha_{t,I^*}) \le \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} + (K-1)\exp\left(-ct^{1/2}\right).$$

We can upper bound the numerator $\alpha_{t,j}$ as follows,

$$\alpha_{t,j} \le \mathbb{P}_t\left(\widetilde{\theta}_j \ge \widetilde{\theta}_{I^*}\right) = \mathbb{P}\left(\frac{\widetilde{\theta}_{I^*} - \widetilde{\theta}_j - (\theta_{t,I^*} - \theta_{t,j})}{\sigma\sqrt{1/N_{t,I^*} + 1/N_{t,j}}} \le \frac{-(\theta_{t,I^*} - \theta_{t,j})}{\sigma\sqrt{1/N_{t,I^*} + 1/N_{t,j}}}\right) = \Phi\left(-\sqrt{2tC_{t,j}}\right),$$

where $\widetilde{\boldsymbol{\theta}} = \left(\widetilde{\theta}_1, \ldots, \widetilde{\theta}_K\right)$ is a sample drawn from the posterior distribution Π_t and $\Phi(\cdot)$ is Gaussian cumulative distribution function. Similarly, we can lower bound the denominator $1-\alpha_{t,I^*}$ as follows,

$$1-\alpha_{t,I^*} = \mathbb{P}_t\left(\exists a \ne I^* : \widetilde{\theta}_a \ge \widetilde{\theta}_{I^*}\right) \ge \mathbb{P}_t\left(\widetilde{\theta}_{A_t} \ge \widetilde{\theta}_{I^*}\right) = \Phi\left(-\sqrt{2tC_{t,A_t}}\right).$$

Hence,

$$\psi_{t,j} \le \frac{\Phi\left(-\sqrt{2tC_{t,j}}\right)}{\Phi\left(-\sqrt{2tC_{t,A_t}}\right)} + (K-1)\exp\left(-ct^{1/2}\right).$$

The remaining task is to show that there exist a deterministic constant $c_\epsilon^{(j)} > 0$ and a random time $T_{\epsilon,3}^{(j)} \in \mathbb{M}$ such that for any $t \ge T_{\epsilon,3}^{(j)}$,

$$\frac{\Phi\left(-\sqrt{2tC_{t,j}}\right)}{\Phi\left(-\sqrt{2tC_{t,A_t}}\right)} \le \exp\left(-c_\epsilon^{(j)} t\right). \tag{30}$$

By Lemma 13, for any $\delta > 0$,

$$\begin{aligned}\frac{\Phi\left(-\sqrt{2tC_{t,j}}\right)}{\Phi\left(-\sqrt{2tC_{t,A_t}}\right)} &\le \frac{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right)(1-\delta)}\right)}{\Phi\left(-\sqrt{2tp_{t,I^*}f_{A_t}\left(\frac{p_{t,A_t}}{p_{t,I^*}}\right)(1+\delta)}\right)} \le \frac{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_j^*+\epsilon/2}{p_{I^*}^*}\right)(1-\delta)}\right)}{\Phi\left(-\sqrt{2tp_{t,I^*}f_{A_t}\left(\frac{p_{A_t}^*}{p_{I^*}^*}\right)(1+\delta)}\right)} \\ &= \frac{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_j^*+\epsilon/2}{p_{I^*}^*}\right)(1-\delta)}\right)}{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_j^*}{p_{I^*}^*}\right)(1+\delta)}\right)},\end{aligned}$$

where the second inequality follows from (29) and the monotonicity of f_j and f_{A_t}, while the final equality uses the information balance (1).

We can pick a sufficiently small δ as a function of ϵ such that

$$c_1 \triangleq f_j\left(\frac{p_j^*+\epsilon/2}{p_{I^*}^*}\right)(1-\delta) > f_j\left(\frac{p_j^*}{p_{I^*}^*}\right)(1+\delta) \triangleq c_2.$$

Then (30) follows from Gaussian tail upper and lower bounds (or $\frac{1}{t}\log\Phi\left(-\sqrt{t}x\right) \to -x^2/2$ as $t \to \infty$) and that p_{t,I^*} is bounded away from 0 and 1 when t is large (Lemma 9). This completes the proof. □

Lemma 15. *Under* TTTS-IDS*, for any $\epsilon > 0$, there exists $T_\epsilon \in \mathbb{M}$ such that for any $t \geq T_\epsilon$,*

$$\frac{w_{t,j}}{w_{t,I^*}} \leq \frac{p_j^* + \epsilon}{p_{I^*}^*}, \quad \forall j \neq I^*.$$

Proof. Fix $\epsilon > 0$ and $j \neq I^*$. By Lemma 14, there exists $\widetilde{T}_\epsilon \in \mathbb{M}$ such that for any $t \geq \widetilde{T}_\epsilon$ and $j \neq I^*$,

$$\frac{w_{t,j}}{w_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*} \quad \Longrightarrow \quad \psi_{t,j} \leq \exp\left(-c_\epsilon t\right) + (K-1)\exp\left(-ct^{1/2}\right).$$

Define

$$\kappa_\epsilon \triangleq \sum_{\ell=0}^{\infty}\left[\exp\left(-c_\epsilon \ell\right) + (K-1)\exp\left(-c\ell^{1/2}\right)\right] < \infty.$$

From now on, we consider $t \geq \widetilde{T}_\epsilon$. We consider two cases, and provide upper bounds on $\frac{w_{t,j}}{w_{t,I^*}}$ for each case.

1. The first case supposes that

$$\forall \ell \in \left\{\widetilde{T}_\epsilon, \widetilde{T}_\epsilon + 1, \ldots, t-1\right\}: \quad \frac{w_{\ell,j}}{w_{\ell,I^*}} \geq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}.$$

   We have

$$\begin{aligned}\Psi_{t,j} = \Psi_{\widetilde{T}_\epsilon,j} + \sum_{\ell=\widetilde{T}_\epsilon}^{t-1}\psi_{\ell,j} &= \Psi_{\widetilde{T}_\epsilon,j} + \sum_{\ell=\widetilde{T}_\epsilon}^{t-1}\psi_{\ell,j}\mathbf{1}\left(\frac{w_{\ell,j}}{w_{\ell,I^*}} \geq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\right) \\ &\leq \Psi_{\widetilde{T}_\epsilon,j} + \sum_{\ell=\widetilde{T}_\epsilon}^{t-1}\left[\exp\left(-c_\epsilon \ell\right) + (K-1)\exp\left(-c\ell^{1/2}\right)\right] \leq \Psi_{\widetilde{T}_\epsilon,j} + \kappa_\epsilon.\end{aligned}$$

   Take $T \in \mathbb{M}$ in Lemma 9, so for any $t \geq T$, $\Psi_{t,I^*} \geq b_1 t$, and thus

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} \leq \frac{\Psi_{\widetilde{T}_\epsilon,j} + \kappa_\epsilon}{\Psi_{t,I^*}} \leq \frac{\widetilde{T}_\epsilon + \kappa_\epsilon}{b_1 t}. \tag{31}$$

2. The alternative case supposes that

$$\exists \ell \in \left\{\widetilde{T}_\epsilon, \widetilde{T}_\epsilon + 1, \ldots, t-1\right\}: \quad \frac{w_{\ell,j}}{w_{\ell,I^*}} < \frac{p_j^* + \epsilon/2}{p_{I^*}^*}.$$

   We define

$$L_t \triangleq \max\left\{\ell \in \left\{\widetilde{T}_\epsilon, \widetilde{T}_\epsilon + 1, \ldots, t-1\right\} : \frac{w_{\ell,j}}{w_{\ell,I^*}} < \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\right\}.$$

   We have

$$\begin{aligned}\Psi_{t,j} &= \Psi_{L_t,j} + \psi_{L_t,j} + \sum_{\ell=L_t+1}^{t-1}\psi_{\ell,j} = \Psi_{L_t,j} + \psi_{L_t,j} + \sum_{\ell=L_t+1}^{t-1}\psi_{\ell,j}\mathbf{1}\left(\frac{w_{\ell,j}}{w_{\ell,I^*}} \geq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\right) \\ &\leq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\Psi_{L_t,I^*} + 1 + \sum_{\ell=L_t+1}^{t-1}\left[\exp\left(-c_\epsilon \ell\right) + (K-1)\exp\left(-c\ell^{1/2}\right)\right] \\ &\leq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\Psi_{L_t,I^*} + (\kappa_\epsilon + 1),\end{aligned}$$

where the first inequality uses $\frac{\Psi_{L_t,j}}{\Psi_{L_t,I^*}} = \frac{w_{L_t,j}}{w_{L_t,I^*}} < \frac{p_j^*+\epsilon/2}{p_{I^*}^*}$. Then by Lemma 9, for any $t \geq T$,

$$\frac{w_{t,j}}{w_{t,I^*}} \leq \frac{p_j^*+\epsilon/2}{p_{I^*}^*}\frac{\Psi_{L_t,I^*}}{\Psi_{t,I^*}} + \frac{1+\kappa_\epsilon}{\Psi_{t,I^*}} \leq \frac{p_j^*+\epsilon/2}{p_{I^*}^*} + \frac{1+\kappa_\epsilon}{b_1 t}. \tag{32}$$

Putting (31) and (32) together, we find that for any $t \geq T_1$,

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} \leq \max\left\{\frac{\widetilde{T}_\epsilon+\kappa_\epsilon}{b_1 t}, \frac{p_j^*+\epsilon/2}{p_{I^*}^*} + \frac{\kappa_\epsilon+1}{b_1 t}\right\} \leq \frac{p_j^*+\epsilon/2}{p_{I^*}^*} + \frac{\widetilde{T}_\epsilon+2\kappa_\epsilon+1}{t}.$$

Then we have

$$t \geq \frac{2\left(\widetilde{T}_\epsilon+2\kappa_\epsilon+1\right)}{\epsilon} \quad \Longrightarrow \quad \frac{w_{t,j}}{w_{t,I^*}} \leq \frac{p_j^*+\epsilon}{p_{I^*}^*}.$$

Taking $T_\epsilon \triangleq \max\left\{\widetilde{T}_\epsilon, T, \frac{2\left(\widetilde{T}_\epsilon+2\kappa_\epsilon+1\right)}{\epsilon}\right\}$ completes the proof since Lemma 4 implies $T_\epsilon \in \mathbb{M}$. □

Now we are ready to complete the proof of Proposition 5.

Proof of Proposition 5. Corollary 4 and Lemma 15 give Proposition 5. □

F Proof of Theorem 6

We overload the notation $\Delta_{\min}$ and $\Delta_{\max}$ for pure-exploration thresholding bandits as follows,

$$\Delta_{\min} \triangleq \min_{i\in[K]} |\theta_i - \mathfrak{T}| \quad \text{and} \quad \Delta_{\max} \triangleq \max_{i\in[K]} |\theta_i - \mathfrak{T}|.$$

F.1 Sufficient exploration

Proposition 8. *There exists $T \in \mathbb{M}$ such that for any $t \geq T$, $\min_{i\in[K]} N_{t,i} \geq \sqrt{t/K}$.*

Following Qin et al. (2017); Shang et al. (2020), we define the following sets for any $t \in \mathbb{N}_0$ and $s \geq 0$:

$$U_t^s \triangleq \{i \in [K] : N_{t,i} < s^{1/2}\} \quad \text{and} \quad V_t^s \triangleq \{i \in [K] : N_{t,i} < s^{3/4}\}.$$

We let $\overline{U}_t^s = [K] \setminus U_t^s$ and $\overline{V}_t^s = [K] \setminus V_t^s$.

Lemma 16. *There exists $S \in \mathbb{M}$ such that for any $s \geq S$, if U_t^s is nonempty, then $I_t \in V_t^s$.*

Proof. By Lemma 7, there exists $S_1 \in \mathbb{M}$ such that for any $s \geq S_1$ and any $t \in \mathbb{N}_0$, if $i \in \overline{V}_t^s$, then $|\theta_{t,i} - \mathfrak{T}| \geq \Delta_{\min}/2$. On the other hand, if $j \in U_t^s$, then $|\theta_{t,j} - \mathfrak{T}| \leq \Delta_{\max} + \sigma W_1$ by the definition of W_1. Hence, there exists $S_2 \in \mathbb{M}$ such that for $s \geq S_1$, $p_{t,j} d(\theta_{t,j}, \mathfrak{T}) < p_{t,i} d(\theta_{t,i}, \mathfrak{T})$, which implies $I_t \notin \overline{V}_t^s$. □

The proof of Proposition 8 follows from the above lemma and a variant of Lemma 11.

F.2 Strong convergence to optimal proportions

For notational convenience, let $w_{t,i} \triangleq \frac{\Psi_{t,i}}{t}$ for any $(t,i) \in \mathbb{N}_0 \times [K]$. Now we define the over-sampled sets as follows: for any $t \in \mathbb{N}_0$ and $\epsilon > 0$,

$$O_{t,\epsilon} \triangleq \{j \in [K] \,:\, w_{t,i} > p_i^* + \epsilon\}\,.$$

Lemma 17. *For any arm $i \in [K]$,*

$$\frac{C_{t,i}}{D_{t,i}} \xrightarrow{\mathbb{M}} 1 \quad \textit{where} \quad D_{t,i} \triangleq p_{t,i} d(\theta_i, \mathfrak{T}) = \frac{p_{t,i}\,(\theta_i - \mathfrak{T})^2}{2\sigma^2}.$$

Lemma 18. *For any $\epsilon > 0$, there exists a random time $T_\epsilon \in \mathbb{M}$ such that for any $t \geq T_\epsilon$ and $i \in [K]$,*

$$i \in O_{t,\epsilon} \quad \Longrightarrow \quad \psi_{t,i} = 0.$$

Proof. There exists $T_1 \in \mathbb{M}$ such that for any $t \geq T_1$ and $i \in [K]$,

$$i \in O_{t,\epsilon} \quad \Longrightarrow \quad p_{t,i} > p_i^* + \epsilon/2 \quad \text{and} \quad \exists J_t \neq I^* \text{ s.t. } p_{t,J_t} \geq p_{J_t}.$$

$$\begin{aligned}\frac{C_{t,i}}{C_{t,J_t}} = \frac{p_{t,i}\,(\theta_{t,i} - \mathfrak{T})^2}{p_{t,J_t}\,(\theta_{t,J_t} - \mathfrak{T})^2} &\geq \frac{p_{t,i}\,(\theta_i - \mathfrak{T})^2\,(1-\delta)}{p_{t,J_t}\,(\theta_{J_t} - \mathfrak{T})^2\,(1+\delta)} \\ &\geq \frac{\left(p_i + \frac{\epsilon}{2}\right)(\theta_i - \mathfrak{T})^2\,(1-\delta)}{p^*_{J_t}\,(\theta_{J_t} - \mathfrak{T})^2\,(1+\delta)} > 1,\end{aligned}$$

where the first inequality uses the above lemma and picks a sufficient small δ as a function of ϵ such that the last inequality holds. □

Lemma 19. *For any $\epsilon > 0$, there exists $T_\epsilon \in \mathbb{M}$ such that for any $t \geq T_\epsilon$,*

$$\forall i \in [K], \quad i \notin O_{t,\epsilon} \quad \textit{i.e.} \quad w_{t,i} \leq p_i^* + \epsilon.$$

Proof. Since K is finite, it suffices to consider a fixed $i \in [K]$. There exists $\widetilde{T}_\epsilon \in \mathbb{M}$ such that for any $t \geq \widetilde{T}_\epsilon$,

$$i \in O_{t,\epsilon} \quad \Longrightarrow \quad \psi_{t,i} = 0.$$

We need to consider two cases.

1. $\exists \ell \in \left\{\widetilde{T}_\epsilon, \widetilde{T}_\epsilon + 1, \ldots, t-1\right\}, w_{\ell,i} \leq p_i^* + \epsilon$.

   Define

   $$L_t \triangleq \max\left\{\ell \in \left\{\widetilde{T}_\epsilon, \widetilde{T}_\epsilon + 1, \ldots, t-1\right\} \,:\, w_{\ell,i} \leq p_i^* + \epsilon\right\}.$$

   Then

   $$w_{t,i} = \frac{\Psi_{t,i}}{t} = \frac{\Psi_{L_t,i}}{t} \leq \frac{\Psi_{L_t,i}}{L_t} = w_{L_t,i} \leq p_i^* + \epsilon.$$

2. $\forall \ell \in \left\{\widetilde{T}_\epsilon, \widetilde{T}_\epsilon + 1, \ldots, t-1\right\}, w_{\ell,i} > p_i^* + \epsilon/2$.

We have

$$w_{t,i} = \frac{\Psi_{t,i}}{t} = \frac{\Psi_{\widetilde{T}_\epsilon,i}}{t}.$$

Letting $T_{\epsilon,i} \triangleq \max\left\{\widetilde{T}_\epsilon, \frac{\Psi_{\widetilde{T}_\epsilon,i}}{p_i^*+\epsilon}\right\}$ completes the proof. □

G Miscellaneous results and proofs

In this appendix, we collect auxiliary proofs and useful tools used in our discussion.

G.1 Facts about the exponential family

We restrict our attention to reward distribution following a one-dimensional natural exponential family. In particular, a natural exponential family has identity natural statistic in canonical form, i.e.,

$$p(y|\eta) = b(y)\exp\{\eta y - A(\eta)\}, \quad \eta \in \mathcal{T} = \left\{\eta : \int b(y)e^{\eta y}\mathrm{d}y < \infty\right\} \tag{33}$$

where η is the natural parameter and $A(\eta)$ is assumed to be twice differentiable. The distribution is called non-singular if $\mathrm{Var}_\eta(Y_{t,i}) > 0$ for all $\eta \in \mathcal{T}^o$, where $\mathcal{T}^o$ denotes the interior of the parameter space $\mathcal{T}$. Denote by θ the mean reward $\theta(\eta) = \int yp(y|\eta)dy$. By the properties of the exponential family, we have, for any $\eta \in \mathcal{T}^o$,

$$\theta(\eta) = \mathbb{E}_\eta[Y_{t,i}] = A'(\eta), \quad \text{and} \quad A''(\eta) = \mathrm{Var}_\eta(Y_{t,i}) > 0.$$

We immediately have the following lemma.

Lemma 20. *At any $\eta \in \mathcal{T}^o$, $\theta(\eta)$ is strictly increasing and $A(\eta)$ is strictly convex.*

Lemma 20 implies that $\theta(\cdot)$ is a one-to-one mapping. Consequently, we can use θ to denote the distribution $p(\cdot|\eta)$ for convenience. Let θ_1 and θ_2 be the mean reward of the distribution $p(\cdot|\eta_1)$ and $p(\cdot|\eta_2)$. Denote by $d(\theta_1, \theta_2)$ the Kullback-Leibler (KL) divergence from $p(\cdot|\eta_1)$ to $p(\cdot|\eta_2)$. We have convenient closed-form expressions of $d(\cdot,\cdot)$ and its partial derivative.

Lemma 21. *The partial derivatives of the KL divergense of a one-dimensional natural exponential family satisfies*

$$d(\theta_1, \theta_2) = A(\eta_2) - A(\eta_1) - \theta_1(\eta_2 - \eta_1), \tag{34}$$

$$\frac{\partial d(\theta_1,\theta_2)}{\partial \theta_1} = \eta_1 - \eta_2, \quad \frac{\partial d(\theta_1,\theta_2)}{\partial \theta_2} = (\theta_2 - \theta_1)\frac{\mathrm{d}\eta(\theta_2)}{\mathrm{d}\theta_2}. \tag{35}$$

G.2 Proof of Lemma 1

For easy reference, we restate Lemma 1 here.

Lemma 1. *Under Assumption 1, for each culprit $x \in \mathcal{X}$ and component-wise strictly positive $\boldsymbol{p} \in \mathcal{S}_K$:*

1. *There exists a unique $\boldsymbol{\vartheta}^x \in \mathrm{Alt}_x(\boldsymbol{\theta})$ that achieves the infimum in (8).*
2. *The functions $C_x(\boldsymbol{p})$ are differentiable concave functions, with $\frac{\partial C_x(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \vartheta_i^x)$ for all $i \in [K]$.*
3. *The functions $C_x(\boldsymbol{p})$ satisfy the following partial differential equation*

$$C_x(\boldsymbol{p}) = \sum_{i \in [K]} p_i \frac{\partial C_x(\boldsymbol{p})}{\partial p_i}. \tag{9}$$

Proof. When the reward distribution follows a single-parameter exponential family, the KL divergence $d(m, m')$ is continuous in (m, m') and strictly convex in m'. A direct application of Wang et al. (2021, Theorem 4 and Lemma 7) yields a unique $\boldsymbol{\vartheta}^x$ that achieves the infimum in (8), as well as the differentiability of C_x. For the concavity, note that C_x is the infimum over a family of linear functions in $\boldsymbol{p}$. This directly implies the concavity of C_x in $\boldsymbol{p}$; see Boyd and Vandenberghe (2004, Chapter 3.2).

We say a function f is *homogeneous of degree one* if it satisfies $f(s\boldsymbol{p}) = sf(\boldsymbol{p})$ for any $s \geq 0$. The fact that $C_x(\boldsymbol{p})$ is homogeneous of degree one follows directly from its definition in (8). Combined with differentiability, the PDE characterization follows from Euler's theorem for homogeneous functions; see Apostol (1969, p. 287) for example. □

G.3 Properties of $C_x(\cdot)$ in BAI and best-k-arm identification

We provide useful properties of the functions $C_j(\boldsymbol{p})$ for BAI and the functions $C_{i,j}(\boldsymbol{p})$ for best-k identification; see Appendix A for explicit formula of these functions.

Lemma 22. *For BAI and any component-wise strictly positive $\boldsymbol{p}$, the minimizer $\boldsymbol{\vartheta}^j$ for (8) is given by*

$$\vartheta_j^j = \vartheta_{I^*}^j = \bar{\theta}_{I^*,j}(\boldsymbol{p}) \triangleq \frac{p_{I^*}\theta_{I^*} + p_j\theta_j}{p_{I^*} + p_j}, \quad \forall j \neq I^*, \quad \text{and} \quad \vartheta_k^j = 0, \quad \forall k \neq I^*, j.$$

In particular, $C_j(\boldsymbol{p})$ is concave in $\boldsymbol{p}$, strictly increasing in p_{I^*} and p_j, and

$$\frac{\partial C_j(\boldsymbol{p})}{\partial p_{I^*}} = d(\theta_{I^*}, \bar{\theta}_{I^*,j}) \quad \text{and} \quad \frac{\partial C_j(\boldsymbol{p})}{\partial p_j} = d(\theta_j, \bar{\theta}_{I^*,j}).$$

For best-k identification, the minimizer $\boldsymbol{\vartheta}^{(i,j)}$ for (8) is given by

$$\vartheta_i^{(i,j)} = \vartheta_j^{(i,j)} = \bar{\theta}_{i,j}(\boldsymbol{p}) \triangleq \frac{p_i\theta_i + p_j\theta_j}{p_i + p_j}, \quad \forall (i,j) \in \mathcal{I} \times \mathcal{I}^c, \quad \text{and} \quad \vartheta_k^{(i,j)} = 0, \quad \forall k \neq i, j.$$

In particular, $C_{i,j}(\boldsymbol{p})$ is concave in $\boldsymbol{p}$, strictly increasing in p_i and p_j, and

$$\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \bar{\theta}_{i,j}) \quad \textit{and} \quad \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j} = d(\theta_j, \bar{\theta}_{i,j}).$$

Proof. Note that C_j for BAI is equivalent to $C_{I^*,j}$ for best-k identification, where I^* is the best arm. To avoid redundancy, we focus on the functions $C_{i,j}$. To find the minimizer, we note that

$$\begin{aligned} C_{i,j}(\boldsymbol{p}) &\triangleq \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}_{i,j}} \sum_i p_i d(\theta_i, \vartheta_i) \\ &= \inf_{\vartheta_j > \vartheta_i} \{p_i d(\theta_i, \vartheta_i) + p_j d(\theta_j, \vartheta_j)\} \\ &= \inf_{\widetilde{\theta}} \left\{ p_i d(\theta_i, \widetilde{\theta}) + p_j d(\theta_j, \widetilde{\theta}) \right\} \\ &= p_i d(\theta_i, \bar{\theta}_{i,j}) + p_j d(\theta_j, \bar{\theta}_{i,j}), \quad \text{where} \quad \bar{\theta}_{i,j} \triangleq \frac{p_i \theta_i + p_j \theta_j}{p_i + p_j}. \end{aligned} \tag{36}$$

In the second line, we set $\vartheta_k = \theta_k$ for all $k \neq i, j$; the third line follows from the monotonicity of the KL divergence, and the last equality results from analytically solving the one-dimensional convex optimization.

The partial derivatives then follows directly from Lemma 1.

Alternatively, one may calculate the derivatives directly using Lemma 21 and the definition of $C_{i,j}$ in (36), as follows

$$\begin{aligned} \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} &= d(\theta_i, \bar{\theta}_{i,j}) + p_i \frac{\partial d(\theta_i, \bar{\theta})}{\partial p_i} + p_j \frac{\partial d(\theta_j, \bar{\theta})}{\partial p_i} \\ &= d(\theta_i, \bar{\theta}_{i,j}) + \frac{\partial \bar{\theta}}{\partial p_i} \cdot \left[p_i \frac{\partial d(\theta_i, \bar{\theta})}{\partial \bar{\theta}} + p_j \frac{\partial d(\theta_j, \bar{\theta})}{\partial \bar{\theta}} \right] \\ &= d(\theta_i, \bar{\theta}_{i,j}). \end{aligned}$$

The last equality follows from the first-order optimality condition for solving the convex optimization problem in the third line of (36). The calculation of $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j}$ proceeds similarly. □

G.4 Proof of Corollary 1

Assuming strictly positiveness of $\boldsymbol{p}^*$, recall from (21b) that

$$\lambda - \sum_{x \in \mathcal{X}} \mu_x \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} = 0, \quad \forall i \in [K],$$

where we used the fact that $\iota_i = 0$ due to complementary slackness. By the definition of

$$\mathcal{X}_i \triangleq \left\{ x \in \mathcal{X}(\boldsymbol{\theta}) : \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} > 0 \right\},$$

we have

$$\lambda - \sum_{x \in \mathcal{X}_i} \mu_x \frac{\partial C_x(\boldsymbol{p})}{\partial p_i} = 0, \quad \forall i \in [K]. \tag{37}$$

Notice that $\lambda = \phi = \Gamma^*_{\boldsymbol{\theta}} > 0$ from the proof of Theorem 4 and that $\frac{\partial C_x(\boldsymbol{p})}{\partial p_i} > 0$, there must exist a $x_0 \in \mathcal{X}_i$ such that $\mu_x > 0$. The complementary-slackness condition then implies that $C_{x_0}(\boldsymbol{p}^*) = \phi = \Gamma^*_{\boldsymbol{\theta}}$. Consequently,

$$\Gamma^*_{\boldsymbol{\theta}} = \min_{x \in \mathcal{X}_i} C_x(\boldsymbol{p}^*), \quad \text{for all } i \in [K].$$

H Existing algorithms in the literature

In this appendix, we briefly review existing algorithms designed for the best-k identification problem in the fixed-confidence settings. These methods serves as benchmarks for our numerical experiments.

UGapE (Gabillon et al., 2012). At each round t, the algorithm computes the upper confidence bound $U_i(t)$ and lower confidence bound $L_i(t)$ of the mean reward for each arm i:

$$\begin{aligned} U_i(t) &= \max\{q \in [\theta_{t,i}, \infty) : T_{t,i} d(\theta_{t,i}, q) \le \beta_{t,\delta}\}, \\ L_i(t) &= \min\{q \in (-\infty, \theta_{t,i}] : T_{t,i} d(\theta_{t,i}, q) \le \beta_{t,\delta}\}, \end{aligned} \tag{38}$$

where $\beta_{t,\delta}$ is the exploration rate we specify to be $\log\left((\log t + 1)/\delta\right)$. Then it computes the gap index $B_i(t) = \max^k_{i' \neq i} U_{i'}(t) - L_i(t)$ for each arm i, where $\max^k$ denotes the operator returning the k-th largest value. Based on $B_i(t)$, it evaluates a set of k arms: $J(t) = \{j : \operatorname{argmin}^j B_j(t)\}$. From $J(t)$, it computes two critical arms $u_t = \operatorname{argmax}_{i \notin J_t} U_i(t)$ and $l_t = \operatorname{argmin}_{i \in J_t} L_i(t)$, and samples the one with larger $\beta_{t,\cdot}$. The algorithm stops when $\max_{i \in J(t)} B_i(t) > 0$.

KL-LUCB (Kaufmann and Kalyanakrishnan, 2013). At each round t, the algorithm first compute the same $U_i(t)$ and $L_i(t)$ as (38). Then it samples two arms u_t and l_t:

$$u_t = \operatorname*{argmax}_{i \notin \widehat{\mathcal{I}}_t} U_i(t) \quad \text{and} \quad l_t = \operatorname*{argmin}_{i \in \widehat{\mathcal{I}}_t} L_i(t),$$

The algorithm stops when $U_{u_t}(t) > L_{l_t}(t)$.

Lazy-Mirror-Ascent (Ménard, 2019). The algorithm treats the lower bound problem as $\max_{\boldsymbol{w}} F(\boldsymbol{w})$. It uses an allocation rule based on an online lazy mirror ascent. At each round t, it tracks the allocation weights $\boldsymbol{w}(t)$ and updates it by $\widetilde{\boldsymbol{w}}(t+1) = \operatorname{argmax}_{\boldsymbol{w}} \eta_{t+1} \sum_{s=K}^{t} \boldsymbol{w} \cdot \nabla F(\widetilde{\boldsymbol{w}}(s); \boldsymbol{\mu}(s)) - \mathrm{kl}(\boldsymbol{w}, \boldsymbol{\pi})$ and $\boldsymbol{w}(t+1) = (1-\gamma_t)\widetilde{\boldsymbol{w}}(t+1) + \gamma_t \boldsymbol{\pi}$, where $\boldsymbol{\mu}(s)$ is the empirical mean estimates and $\boldsymbol{\pi}$ is the uniform distribution. There exists two tuning parameter η_t and γ_t. Based on the code by Wouter M. Koolen (https://bitbucket.org/wmkoolen/tidnabbil/src/master/) and the suggestion in Ménard (2019), we choose $\eta_t = \sqrt{\log K/t}/F^*$ and $\gamma_t = 1/(4\sqrt{t})$. Wouter M. Koolen comments that, although setting F^* relies on the knowledge of true arm means, it is critical for the performance. In appendix of Wang et al. (2021), the authors also comment that this method is very sensitive to the learning rate η_t, the current choice is only for experimental comparison and cannot be used in real-world scenarios.

m-LinGapE (Réda et al., 2021a). At each round t, the algorithm first computes the same $U_i(t)$ and $L_i(t)$ as (38). Then it computes the gap index $B_{i,j}(t) = U_i(t) - L_j(t)$ for each arm i. It chooses $J(t)$ as the set of k arms with largest empirical means. From $J(t)$, it computes two critical arms u_t and l_t:

$$u_t = \operatorname*{argmax}_{j \in J_t} \max_{i \notin J_t} B_{i,j}(t) \quad \text{and} \quad l_t = \operatorname*{argmax}_{i \notin J_t} B_{i,u_t}(t),$$

and samples one which would reduce the variance of the empirical mean difference between them. (This is the so-called "greedy selection rule" in Réda et al. (2021a).) The algorithm stops when $B_{l_t,u_t}(t) > 0$.

MisLid (Réda et al., 2021b). The algorithm is designed from the lower bound maximin problem and achieves the asymptotic optimality. It maintains a learner over the probability simplex of number of arms. At each round t, it computes a gain function and updates the learner. Then it perform tracking-rule to sample an arm. We adopt AdaHedge as the learner and use empirical gain function g_t as $\operatorname{argmin}_{\boldsymbol{\lambda} \in \mathrm{Alt}(\boldsymbol{\mu}^t)} \sum_{i=1}^K \boldsymbol{p}_i^t (\boldsymbol{\mu}_i^t - \boldsymbol{\lambda})^2$ where $\boldsymbol{p}^t$ is probability distribution returned by AdaHedge, $\boldsymbol{\mu}_t$ is the empirical mean estimate and $\mathrm{Alt}(\boldsymbol{\mu}^t)$ defines the set of bandit model returning the top-k arm set different from $(\boldsymbol{\mu}^t)$. As suggested by Réda et al. (2021b), the empirical gain function reduced the sample complexity practically compared with the optimistic gain function suggested by theory. For the tracking rule, we adopt C-Tracking in our experiments. For the stopping rule, we adopt the same GLRT statistic with threshold $\log\left((\log(t)+1)/\delta\right)$.

FWS (Wang et al., 2021). The algorithm track the lower bound maximin problem. At each round t, it performs forced exploration or Frank-Wolfe update. Due to the non-smoothness of objective function, it construct an r_t-sub-differential subspaces and then solve a zero-sum game. Note r_t is a tuning parameter. We choose $r_t = t^{-0.9}/K$ as suggested in Wang et al. (2021). For the stopping rule, we adopt the same GLRT statistic with threshold $\log\left((\log(t)+1)/\delta\right)$.

I Additional numerical experiments

In this appendix, we provide additional numerical experiments, comparing our algorithms with the following algorithms from the literature: UGapE (Gabillon et al., 2012), SAR (Bubeck et al., 2013), OCBAss (Gao and Chen, 2015) and OCBASS (Gao and Chen, 2016) in the fixed-budget setting.

We consider the best-5-arm identification problem for a Gaussian bandit with $\theta_i = 1 - 0.05 * (i-1)$ and $\sigma_i^2 = 1/4$ for $i = [20]$.

In Figure 3, we present the fixed-budget performance of our proposed `TS-KKT-IDS` algorithm, along with benchmark methods. In the subplot of Figure 3, we provide a zoomed-out illustration on a logarithmic scale, which suggests a superior convergence rate of the PICS under our algorithm.

In Figure 4, we compare the optimal allocation rate with the empirical allocation rate of the algorithms, given a budget of 10^6. We observe that both `TS-KKT-IDS` and `OCBAss` (omitted) allocate nearly exactly at the optimal rate. In this simpler case, `OCBAss` satisfies the correct sufficient

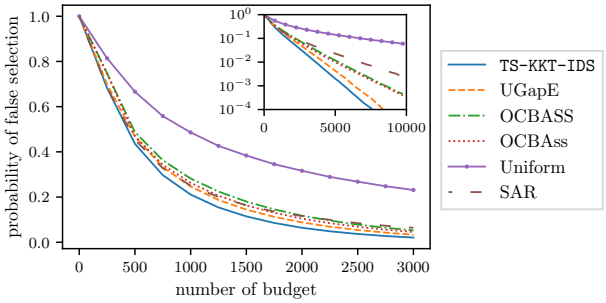


Figure 3: Fix-budget performance estimated over 10^7 independent replications.

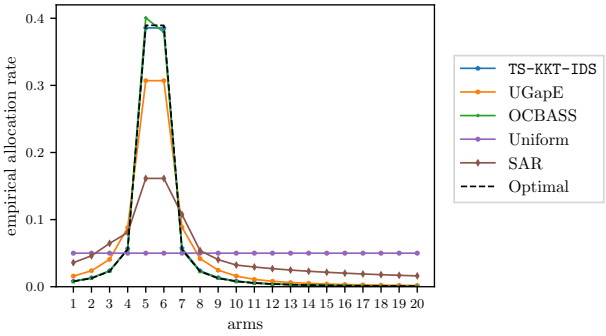


Figure 4: Compare the optimal allocation rate with the empirical allocation rate of the algorithms.

condition and thus also converges to the optimal rate. In contrast, `OCBASS`, lacking the key step of balancing the sum-of-squared allocations, does not converge to the optimal allocation rate, though it remains sufficiently close. The `UGapE` and `SAR` algorithms deviate more significantly from the optimal allocation rate, although their empirical performance in terms of PICS is still respectable.

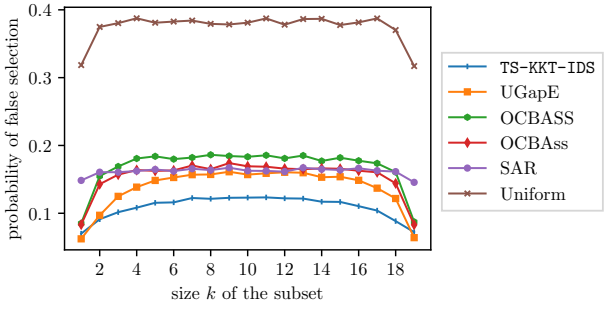


Figure 5: The probability of false selection for varying values of k with a fixed budget of 1500, estimated over 30000 independent replications.

In Figure 5, we consider the case where the number of best arms to be identified varies from $k = 1$ to 19, displaying the PICS for each k with a fixed budget of 1500 samples. We observed the our `TS-KKT-IDS` algorithm consistently outperforms the alternatives across the range of k.